



UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH

Escola Tècnica Superior d'Enginyeria
de Telecomunicació de Barcelona



Channel Knowledge Map Prediction with Deep Learning for 6G UAV-Enabled Networks

Bachelor's Degree Thesis

Submitted to the Faculty at the

Escola Tècnica Superior

d'Enginyeria de Telecomunicació de Barcelona

of the Universitat Politècnica de Catalunya

by

Guillem Moreno Garcia

In partial fulfillment

of the requirements for the

DEGREE IN TELECOMMUNICATION TECHNOLOGIES AND SERVICES ENGINEERING

Advisors: Sergi Abadal, Evgenii Vinogradov

Barcelona, June 2026

Resum

Aquest TFG estudia la predicció de mapes de coneixement de canal per a xarxes 6G amb UAVs. A partir del conjunt de dades CKM (66 ciutats reals, 16,180 mostres, mapes urbans de 513×513 píxels amb màscares LoS/NLoS i altura d'antena contínua entre 12 i 478 m) es prediuen atenuació de canal, dispersió temporal / delay spread i dispersió angular / angular spread sota un protocol estricte de validació per ciutats completes. El sistema final és un model híbrid: HARP-NET CKM, el Height-Aware Residual Prior Network for Channel Knowledge Map prediction, que combina priors interpretables calibrats amb un corrector neural residual. En els resultats finals, el seu cap intern de mescla s'avalua mitjançant la predicció determinista de la mitjana, no com una densitat predictiva calibrada.

El resultat final no és un prior pur ni una xarxa neuronal pura: HARP-NET CKM usa deep learning com a corrector multitasca condicionat per la geometria urbana, l'altura del UAV, les màscares LoS/NLoS i els mapes prior. En el split final de test de 14 ciutats no vistes, els priors congelats assoleixen 1.94 dB, 28.10 ns i 13.74 graus RMSE, mentre que HARP-NET CKM els millora fins a 1.65 dB, 26.56 ns i 11.39 graus. Això fa que els priors siguin una baseline forta per si mateixa, no només un pas intermedi: concentren la calibració per règims de densitat urbana, altura d'edificis, visibilitat i altura d'antena, i deixen al model neuronal només el residual local. El resultat és una estratègia de divideix i venceràs més que una nova llei universal de propagació.

Resumen

Este TFG estudia la predicción de mapas de conocimiento de canal para redes 6G con UAVs. A partir del conjunto de datos CKM (66 ciudades reales, 16,180 muestras, mapas urbanos de 513×513 píxeles con máscaras LoS/NLoS y altura de antena continua entre 12 y 478 m) se predicen atenuación de canal, dispersión temporal / delay spread y dispersión angular / angular spread bajo un protocolo estricto de validación por ciudades completas. El sistema final es un modelo híbrido: HARP-NET CKM, el Height-Aware Residual Prior Network for Channel Knowledge Map prediction, que combina priors interpretables calibrados con un corrector neural residual. En los resultados finales, su cabeza interna de mezcla se evalúa mediante la predicción determinista de la media, no como una densidad predictiva calibrada.

El resultado final no es un prior puro ni una red neuronal pura: HARP-NET CKM usa deep learning como corrector multitarea condicionado por la geometría urbana, la altura del UAV, las máscaras LoS/NLoS y los mapas prior. En el split final de test de 14 ciudades no vistas, los priors congelados alcanzan 1.94 dB, 28.10 ns y 13.74 grados RMSE, mientras que HARP-NET CKM los mejora hasta 1.65 dB, 26.56 ns y 11.39 grados. Esto convierte a los priors en una baseline fuerte por sí misma, no solo en un paso intermedio: concentran la calibración por regímenes de densidad urbana, altura de edificios, visibilidad y altura de antena, y dejan al modelo neuronal solo el residual local. El resultado es una estrategia de divide y vencerás más que una nueva ley universal de propagación.

Summary

This thesis studies channel knowledge map prediction for 6G UAV enabled networks. Using the CKM dataset (66 real cities, 16,180 samples, 513×513 urban maps with LoS/NLoS masks and a continuous UAV antenna height between 12 and 478 m), the project predicts channel attenuation, delay spread, and angular spread under a strict city holdout evaluation protocol, where entire cities are reserved for validation and test. The final system is HARP-NET CKM, the Height-Aware Residual Prior Network for Channel Knowledge Map prediction: a hybrid design that combines calibrated interpretable priors for channel attenuation and spreads with a neural residual corrector. In the final reported results, its internal mixture head is evaluated through the deterministic mixture mean prediction, not as a calibrated predictive density.

The final system is neither a pure prior nor a pure neural network: the final model adds deep learning as a multi task residual corrector conditioned on urban geometry, UAV height, LoS/NLoS masks, and prior maps. On the final 14-city unseen test split, the frozen priors achieve 1.94 dB, 28.10 ns, and 13.74 degree RMSE, while HARP-NET CKM improves them to 1.65 dB, 26.56 ns, and 11.39-degree RMSE. This makes the priors a strong baseline in their own right, not just an intermediate step: they concentrate the calibration over urban density, building height, visibility, and antenna height regimes, leaving the neural model to learn only the local residual. The result is a divide and conquer strategy rather than a new universal propagation law.

For all those who failed throughout their journey.

Acknowledgements

I would like to express my sincere gratitude to my supervisors, Sergi Abadal and Evgenii Vinogradov, for their guidance throughout this project. Although I worked on this thesis with a high degree of independence, their advice was extremely valuable. I am especially grateful to Evgenii Vinogradov, whose propagation-domain perspective and suggestions, such as encouraging me to study the two ray model, helped me tremendously in shaping and improving this work.

I would also like to thank ETSETB for creating the PARS programme, which allows students to begin the Master's Degree in Telecommunications Engineering before completing their Bachelor's degree. This opportunity was especially useful while working on this thesis: starting the master's before finishing the degree gave me a broader technical perspective and helped me approach the project with more maturity, depth, and ambition.

Finally, I would like to thank my family and friends for their patience and encouragement throughout the thesis.

Revision history and approval record

Revision	Date	Author(s)	Description
1.0	13/04/2026	GMG	Document creation
1.1	18/04/2026	SA, EV	Initial scope and work-plan review
1.2	22/04/2026	GMG	Scope updates after supervisor feedback
2.0	29/04/2026	GMG	First skeletal version
2.1	02/05/2026	SA, EV	Review of methodology and evaluation plan
2.2	04/05/2026	GMG	Methodology and results sections revised after review
3.0	11/05/2026	GMG	Complete technical draft
3.1	18/05/2026	GMG	Reduced version
3.2	18/05/2026	EV	First correction of the final text
3.3	26/05/2026	SA	Second correction of the final text
3.4	29/05/2026	GMG	Corrected version after supervisor feedback
3.5	04/06/2026	EV	Final text corrections and methodology review
3.6	05/06/2026	GMG	Corrected version after EV and GMG review
3.7	09/06/2026	EV	Methodology redundancy review
3.8	10/06/2026	GMG	Methodology redundancy removal and corrected version

DOCUMENT DISTRIBUTION LIST

Role	Surname(s) and Name
[Student]	Moreno Garcia, Guillem
[Project Supervisor 1]	Abadal, Sergi
[Project Supervisor 2]	Vinogradov, Evgenii

Written by:		Reviewed and approved by:	
Date	10/06/2026	Date	Pending review
Name	Moreno Garcia, Guillem	Name	Abadal, Sergi; Vinogradov, Evgenii
Position	Project Author	Position	Project Supervisors

Contents

Summary	2
Acknowledgements	5
Revision history and approval record	6
Contents	7
List of figures	10
List of tables	11
Abbreviations	13
1 Introduction	16
1.1 Work goals	16
1.2 Research questions, answers and contributions	17
1.3 Thesis outline	18
1.4 Requirements and specifications	18
1.5 Work plan	18
2 State of the Art	21
2.1 General Framework: CKM Surrogates and Evaluation Contracts	21
2.2 Channel Attenuation Prediction	22
2.2.1 Free space and two ray propagation	22
2.2.2 Empirical and semi empirical models	22
2.2.3 UAV and air to ground channel models	23
2.2.4 Spread definitions used later	23
2.2.5 Deep learning for path loss maps	24
2.2.6 Physics informed and hybrid path loss methods	25
2.3 Angular Spread Prediction	26
2.3.1 Empirical models for spread parameters	26
2.3.2 Deep learning for wideband channel statistics	27
2.4 Delay Spread Prediction	27
2.4.1 Distributional mismatch and the spike plus exponential problem	27
2.5 Distribution, Training Tools and Fair Comparison	28
2.5.1 Heteroscedastic uncertainty	28
2.5.2 Gaussian mixture model heads	28
2.5.3 Stochastic diffusion models	29
2.5.4 Multitask and joint prediction	29
2.5.5 Height conditioning for the residual model	29
2.5.6 Key benchmarks and fair comparison	30
2.5.7 Emerging techniques and implications for this thesis	32
3 Methodology	33
3.1 Dataset and split	35
3.2 General Framework and Data Flow	37

3.3	Shared support maps, visibility masks and morphology	37
3.4	Final priors: overview and literature basis	41
3.4.1	High level architecture	41
3.4.2	Prior design rules	42
3.4.3	Literature ingredients and thesis adaptations	43
3.5	Channel Attenuation Prior	44
3.5.1	LoS branch: enhanced two ray model	45
3.5.2	Calibration of the LoS two ray model	48
3.5.3	NLoS branch	51
3.5.4	Final channel attenuation prior assembly	56
3.6	Delay and Angular Spread Prior and Residual Prediction	57
3.6.1	Physical motivation for log domain modeling	57
3.6.2	Pixel wise notation for the spread prior fit	57
3.6.3	Raw spread prior in log domain	58
3.6.4	Calibrated spread prior	59
3.6.5	Delay and angular spread prior pipeline	62
3.7	HARP-NET CKM Neural Residual Model	62
3.7.1	Experimental protocol and reporting status	62
3.7.2	HARP-Net CKM Residual Head	63
3.7.3	Architecture: Shared UNet with FiLM	65
3.7.4	Composite Optimization Objective	67
4	Results	70
4.1	Standalone Frozen Prior Results	70
4.1.1	Channel attenuation prior component comparison	70
4.1.2	Channel attenuation prior summary	71
4.1.3	Delay and angular spread prior summary	71
4.1.4	Prior RMSE by transmitter height	72
4.1.5	Path loss prior by topology class	73
4.1.6	Spread priors by topology class	74
4.2	HARP-NET CKM Results Against Frozen Priors	75
4.2.1	Final test protocol and headline results	75
4.2.2	Channel attenuation result	75
4.2.3	Angular spread result	75
4.2.4	Delay spread result	76
4.2.5	Validation to test check	76
4.3	State of the Art Context and Runtime	77
4.3.1	Target vs final result	77
4.3.2	Comparison with closest state of the art metrics	77
4.4	Qualitative Analysis	80
4.5	Limitations	80
5	Sustainability Analysis and Ethical Implications	84
5.1	Sustainability matrix	84
5.2	Environmental impact	84
5.3	Economic and social impact	85

5.4	Ethical implications and SDGs	88
6	Conclusions and Future Work	90
6.1	Conclusions	90
6.1.1	What the final result means	90
6.1.2	Answers to the research questions	91
6.1.3	Main contributions	92
6.1.4	Practical implications and scope	92
6.1.5	Negative results	92
6.2	Future Directions	93
	Bibliography	94
A	Detailed Attempt Log	99
A.1	How to read the attempt numbers	99
A.2	Try-number map	100
A.3	Expanded chronological timeline	102
A.3.1	Early image to image baselines	102
A.3.2	Geometry conditioning and artifact control	102
A.3.3	Masking and evaluation contract	102
A.3.4	Prior-guided residual learning	103
A.3.5	PMHHNet and expert-routing phase	103
A.3.6	Regularization and distribution diagnosis	103
A.3.7	Distribution first modelling	103
A.3.8	Frozen calibrated priors	104
A.3.9	HARP-NET CKM	104
B	Representative HARP-NET CKM Inspection Panels	105
C	CKM Generator Interface and Reproducibility Tool	109
C.1	Public source record and checkpoint artifact	109
C.2	Interface overview	109
C.3	How the tool is run	111
C.4	Accepted inputs	111
C.5	Configurable controls	112
C.6	Image scaling and raw numeric data	112
C.7	Model-grid fitting and interpolation	112
C.8	Mask policies	113
C.9	Outputs	113
C.10	Performance controls	115
D	Declaration on Generative AI Use	116

List of Figures

1.1	Project's Gantt diagram	20
2.1	Direct and specular ground reflected paths in the two ray model.	22
3.1	High level methodology structure.	33
3.2	Detailed methodology data flow.	34
3.3	Empirical transmitter height density read directly from the CKM HDF5 database using the 3 m histogram of (3.3), linearly interpolated between bin centres for display. Most training support lies in the low and mid-altitude bands.	36
3.4	Pipeline for the hybrid path loss prior.	45
3.5	Effective ground reflection amplitude as a function of UAV height.	51
3.6	FSPL versus coherent two ray component.	51
3.7	Feature grouping for the calibrated spread prior.	60
3.8	Pipeline for the delay and angular spread priors.	62
3.9	Schematic of the final large capacity shared UNet architecture with FiLM conditioning. The backbone and decode logic are shared, while each task uses its own heads and frozen prior.	66
3.10	Detailed sinusoidal height conditioning path.	67
3.11	Detailed HARP-Net CKM residual head used by the final model.	67
3.12	Composite final loss family	69
4.1	Representative final model inspection panel.	81
B.1	HARP-NET CKM 6×3 inspection panel for an open, low-antenna validation sample in Segovia.	106
B.2	HARP-NET CKM 6×3 inspection panel for a mixed, mid-antenna validation sample in Nice.	107
B.3	HARP-NET CKM 6×3 inspection panel for a dense, high-antenna validation sample in Tunis.	108
C.1	CKM Generator Streamlit interface.	110
C.2	Generated topology and masks.	110
C.3	CKM Generator joint prediction output.	111
C.4	Generated output folder for one run.	114

List of Tables

1.1	Main CKM dataset fields used in the thesis.	19
2.1	Approximate ICASSP 2023 outdoor RMSE scale.	24
2.2	Evaluation condition comparison across representative methods.	30
3.1	Chapter 3 guide.	33
3.2	City holdout assignment used by the distribution first and final neural models.	36
3.3	External framework inputs, internally computed maps, neural model inputs and final outputs.	37
3.4	Term by term interpretation of the main frozen prior quantities.	42
3.5	Compact design rules extracted from the prior experiments.	43
3.6	Literature to design mapping used by HARP-NET CKM.	44
3.7	Term by term inventory of the LoS prior used by the final model. Unlike the NLoS branch, this is not a long pixel feature regression vector; it is a height-interpolated two ray calibration plus a radial residual table.	48
3.8	Representative smoothed deployed LoS two ray parameters by height bin.	50
3.9	Term by term inventory of the NLoS calibration vector.	54
3.10	Raw prior constants fixed before the ridge calibration in (3.70).	58
3.11	Block membership of the 23 entries in the spread calibration vector.	59
3.12	Interaction terms in the spread calibration vector.	60
3.13	Active loss settings in the HARP-Net CKM stages.	68
4.1	LoS path loss prior component comparison.	71
4.2	Final hybrid path loss prior, pixel weighted RMSE on the held out validation+test split.	71
4.3	Calibrated spread prior test results, pixel weighted RMSE.	72
4.4	Path loss prior pixel weighted RMSE by transmitter height, on the held out validation+test split.	72
4.5	Delay spread and angular spread calibrated pixel weighted RMSE by transmitter height, on the city holdout test split.	73
4.6	Hybrid path loss prior RMSE by topology class.	73
4.7	Delay spread prior RMSE by topology class on the city holdout test split.	74
4.8	Angular spread prior RMSE by topology class on the city holdout test split.	74
4.9	HARP-NET CKM test results on unseen cities.	75
4.10	HARP-NET CKM test RMSE separated by LoS and NLoS pixels. The test cities are unseen during training.	75
4.11	Validation versus test sanity check.	76
4.12	Plain language comparison with closest path loss metrics.	78
4.13	Spread side comparison with nearby state of the art.	79
4.14	Full pipeline runtime and ray tracing comparison.	80

4.15	Approximate RMSE floor from target quantisation.	82
5.1	Sustainability matrix summary for the thesis.	84
5.2	Indicative economic cost of the BT development.	86
A.1	Try-number map, early image-regression phase.	100
A.2	Try-number map, spread branches, masks, and first priors.	100
A.3	Try-number map, prior-residual and PMNet-style phase.	101
A.4	Try-number map, PMHHNet, regularization, and distribution diagnosis. .	101
A.5	Attempt map, final distribution, prior, and joint model phase.	102
B.1	Final-model appendix panel index.	105

Abbreviations

2D	two-dimensional
3D	three-dimensional
3GPP	Third Generation Partnership Project
6G	sixth-generation mobile network
A2G	air-to-ground
ABS	aerial base station
AS	angular spread
BT	bachelor thesis
cGAN	conditional generative adversarial network
CKM	channel knowledge map
CNN	convolutional neural network
COST-231	European Cooperation in Science and Technology 231
CPU	central processing unit
CUDA	Compute Unified Device Architecture
DirectML	Direct Machine Learning
DL	deep learning
DNN	deep neural network
DPM	Dominant Path Model
DS	delay spread
DS-RPP	distribution shift radio path loss prediction
EMA	exponential moving average
ETSETB	Escola Tecnica Superior d'Enginyeria de Telecomunicacio de Barcelona
FiLM	feature-wise linear modulation
FLOPs	floating-point operations
FM-RME	foundation model-empowered radio map estimation
FR3	Frequency Range 3
FSPL	free-space path loss
GAN	generative adversarial network
GMM	Gaussian mixture model
GPU	graphics processing unit
GRETST	Grau en Enginyeria de Tecnologies i Serveis de Telecomunicacio
GT	ground truth
HDF5	Hierarchical Data Format version 5
HF	high-frequency
HPC	high-performance computing
ICASSP	IEEE International Conference on Acoustics, Speech and Signal Processing

IEEE Institute of Electrical and Electronics Engineers
IRT Intelligent Ray Tracing
ITU International Telecommunication Union
JSON JavaScript Object Notation
KL Kullback-Leibler divergence
LoS line of sight
MAE mean absolute error
MATLAB Matrix Laboratory
MDN mixture density network
ML machine learning
MLP multi-layer perceptron
mmWave millimetre wave
MSE mean squared error
NLL negative log-likelihood
NLoS non-line of sight
NMSE normalized mean squared error
NPZ NumPy zipped archive
OLS ordinary least squares
PARS Programa Academic de Recorregut Successiu
PCT Parallel Computing Toolbox
PDE partial differential equation
PINN physics-informed neural network
PL path loss
PMHHNet path-loss map network with high-frequency and height awareness
PMNet path-loss map prediction network
PNG Portable Network Graphics
PSNR peak signal-to-noise ratio
RGB red-green-blue
RM3D RadioMap3DSeer
RMa rural macro
RMSE root mean squared error
RQ research question
RQ1 research question 1
RQ2 research question 2
RQ3 research question 3
RSSI received signal strength indicator
RT ray tracing
RTX NVIDIA RTX graphics-processing platform
Rx receiver
SDG Sustainable Development Goal

SOA	state of the art
SSIM	structural similarity index measure
SWA	stochastic weight averaging
TR	Technical Report
TTA	test-time augmentation
Tx	transmitter
U-Net	U-shaped convolutional encoder-decoder network
U2G	UAV-to-ground
UAV	unmanned aerial vehicle
UMa	urban macro
UMi	urban micro
USC	University of Southern California
UxNB	UAV-mounted radio access node
VQGAN	vector-quantized generative adversarial network
VRAM	video random-access memory
WiCo-PG	Wireless Channel Foundation Model for Pathloss Map Generation
WINNER	Wireless World Initiative New Radio
YAML	YAML Ain't Markup Language

Introduction

Reliable radio coverage prediction is a central requirement for future 6G systems in which UAVs may act as aerial base stations, relays, or temporary coverage nodes. In 3GPP terminology, a UAV mounted radio access node is a UxNB, defined as a radio access node on board a UAV [1]. A classical ray-tracing simulator can produce detailed channel knowledge maps, but this is too expensive to repeat every time the transmitter height, the city environment, or the deployment assumptions change. In the 6G literature, a channel knowledge map is a site-specific, location-tagged repository of channel information that can support environment-aware planning and reduce online channel-acquisition overhead [2, 3]. This thesis studies HARP-NET CKM: the Height-Aware Residual Prior Network for Channel Knowledge Map prediction. Its task is to predict dense CKM maps from a city building height map and a UAV transmitter height, while still generalizing to cities that were not seen during training. The dataset defines this transmitter as a UAV, but the same fixed centre, height conditioned task can also be read as planning for elevated radio access nodes operating at the same 7.125 GHz carrier and within the sampled height range.

The project is formulated as dense image to image regression on 513×513 pixel city maps. At a high level, the pipeline first builds physically meaningful support maps and train calibrated prior estimates, and then HARP-NET CKM corrects the remaining errors. In the CKM experiments the LoS/NLoS visibility mask is read from the dataset; for topology only deployment it can be generated geometrically from the building map and transmitter height. The predicted outputs are channel attenuation, delay spread, and angular spread. The neural network therefore focuses on residual radio map structure instead of being asked to rediscover all propagation geometry from scratch.

Terminology. The CKM HDF5 file and the training code name the dB attenuation target `path_loss`, and the implementation keeps the abbreviation PL for compatibility with those files. In the prose of this thesis, that predicted dense dB map is referred to as *channel attenuation*. The term *path loss* is kept when discussing standard propagation formulas, cited path loss models, or code and dataset names that explicitly use it.

HARP-NET CKM stands for Height-Aware Residual Prior Network for Channel Knowledge Map prediction. The name reflects its central design choice: calibrated physical/statistical priors explain the stable propagation structure, while a height conditioned neural network learns the remaining residual corrections.

1.1 Work goals

The initial technical targets were defined in physical units and were kept as the main success criteria throughout the project. The proposal also included a runtime objective, because a surrogate is only useful if it is clearly faster than generating the same maps

through ray tracing:

- channel attenuation RMSE below 5 dB;
- delay spread RMSE below 50 ns;
- angular spread RMSE below 20°;
- separate reporting for global, LoS, and NLoS regions;
- a 10 to 100× reduction in channel map generation time compared with ray tracing.

These numerical targets were set in the project proposal and kept unchanged throughout the thesis. They are meant to match the order of magnitude at which a ray tracing surrogate starts being useful for planning and coverage decisions. A few dB of channel attenuation RMSE is compatible with practical link budget margins, and tens of ns of delay spread and tens of degrees of angular spread stay well inside typical multipath spread values reported by 3GPP TR 38.901 for UMa/UMi scenarios. Chapter 4 checks the runtime target explicitly by comparing the measured final generator time with the measured MATLAB ray-tracing time on the same Barcelona geometry and hardware.

The most important methodological objective was not only to obtain a low average RMSE, but to do so under a strict city holdout split. City holdout means that entire cities, not individual pixels or tiles, are reserved for validation and testing. A test city therefore contains no samples that were seen during training. This is harder and more deployment-relevant than random pixel or random tile splitting, because the model must work on urban layouts it has never seen.

1.2 Research questions, answers and contributions

The thesis is organised around three research questions, which also define the three main scientific claims of the final pipeline:

- **RQ1.** Can a single deep learning surrogate predict per pixel channel attenuation, delay spread, and angular spread on 513×513 channel knowledge maps, under strict city holdout evaluation, with errors compatible with the thesis targets?
- **RQ2.** Which part of the prediction should be left to a calibrated analytic prior and which part should be left to the neural network, so that the hybrid model is both accurate and interpretable?
- **RQ3.** How does the continuous UAV transmitter height (12 m to 478 m) affect the prediction error, and is the same architecture able to cover the full height range without per height retraining?

In simple terms, the final answer to the three questions is positive within the fixed CKM simulation setting used in this thesis. HARP-NET CKM predicts channel attenuation, delay spread, and angular spread on cities that were not used for training. The reason it works is not that the neural network learns all propagation physics from zero. Stable effects such as distance, visibility, transmitter height, and city type are first captured by calibrated priors; the neural network then corrects the remaining local errors. The hardest cases are still low altitude and strongly blocked NLoS regions, so the results are always reported by regime instead of only as one global RMSE. Chapter 4 and the conclusions give the full technical answer.

The main contributions corresponding to these questions are:

- A reproducible CKM prediction pipeline for ground users served by a centred UAV transmitter, with the same height variable also usable for other elevated radio access nodes under the same simulation contract. The evaluation uses 513×513 maps, city holdout train/validation/test splits, valid ground receiver pixels only, and continuous UAV transmitter height conditioning.
- A train calibrated channel attenuation prior that combines enhanced two ray LoS propagation with NLoS calibration over environment features. The prior provides a transparent baseline whose exact held out city accuracy is reported in Chapter 4.
- Train calibrated delay spread and angular spread priors based on log domain ridge regression over geometry, visibility, transmitter height, and environment features.
- HARP-NET CKM, a shared neural model that receives the frozen PL, DS, and AS priors and learns bounded residual corrections for the three maps jointly.
- A distribution shape analysis showing that earlier point regression networks tended to average away difficult NLoS and spread tail cases. This motivates using mixture style residual predictions instead of a single direct regression value everywhere.

1.3 Thesis outline

Chapter 2 reviews the state of the art and the technical background needed for the thesis: classical propagation models, radio map deep learning, hybrid physics/neural methods, distribution aware prediction, and fair comparison criteria. Chapter 3 describes the final methodology: the CKM dataset, the ground receiver evaluation contract, the city holdout split, the calibrated priors, and the HARP-NET CKM residual model. The method chapter is intentionally final pipeline first; the chronological attempt trace is kept in Appendix A. Chapter 4 reports the final quantitative results, the comparison with the frozen priors and the closest published benchmarks, qualitative examples, runtime, and limitations. Chapter 5 discusses sustainability, economic impact, social impact, and ethical scope. Chapter 6 answers the research questions and presents future work. The appendices preserve the development trace, representative panels, generator documentation, and the declaration on generative AI use.

1.4 Requirements and specifications

The project uses the CKM HDF5 dataset `CKM_Dataset_270326.h5`. The CKM dataset contains 66 real city scenarios and 16180 samples. Each sample is a 513×513 pixel raster with the following fields:

The physical receiver scenario is a ground user served by the UAV transmitter. Therefore, losses and metrics are computed only on valid ground receiver pixels. Building pixels remain in the topology input because they affect propagation, but they are excluded from error accumulation because no receiver is placed inside a building pixel. The fixed receiver height used in the analytic priors is 1.5 m. The transmitter is the UAV antenna, whose height varies continuously across the dataset.

1.5 Work plan

The executed work plan can be grouped into five phases: literature and simulation contract review, dataset and split definition, calibrated prior construction, final residual

Table 1.1: *Main CKM dataset fields used in the thesis.*

Field	Shape	Type	Role
topology_map	513^2	float32	Building height in metres. Pixels with buildings are not valid receiver locations.
los_mask	513^2	uint8	Binary LoS/NLoS visibility mask.
path_loss	513^2	uint8	Ray-traced channel attenuation in dB, named path_loss in the dataset and quantized at approximately 1 dB.
delay_spread	513^2	uint16	Delay spread target, stored as integers and evaluated in ns after decoding.
angular_spread	513^2	uint8	Angular spread target, stored as integers and evaluated in degrees after decoding.
uav_height	scalar	float32	UAV transmitter height in the dataset, used as transmitter height for conditioning and physical priors.

model training, and final evaluation/writing. Figure 1.1 shows these phases at project level rather than listing all individual experiments.

The main deviation from the original proposal was methodological rather than administrative. The project moved from a generic image to image model toward a hybrid CKM generator because the dataset showed strong radial, visibility, and distributional structure that should not be left entirely to a black box neural network.

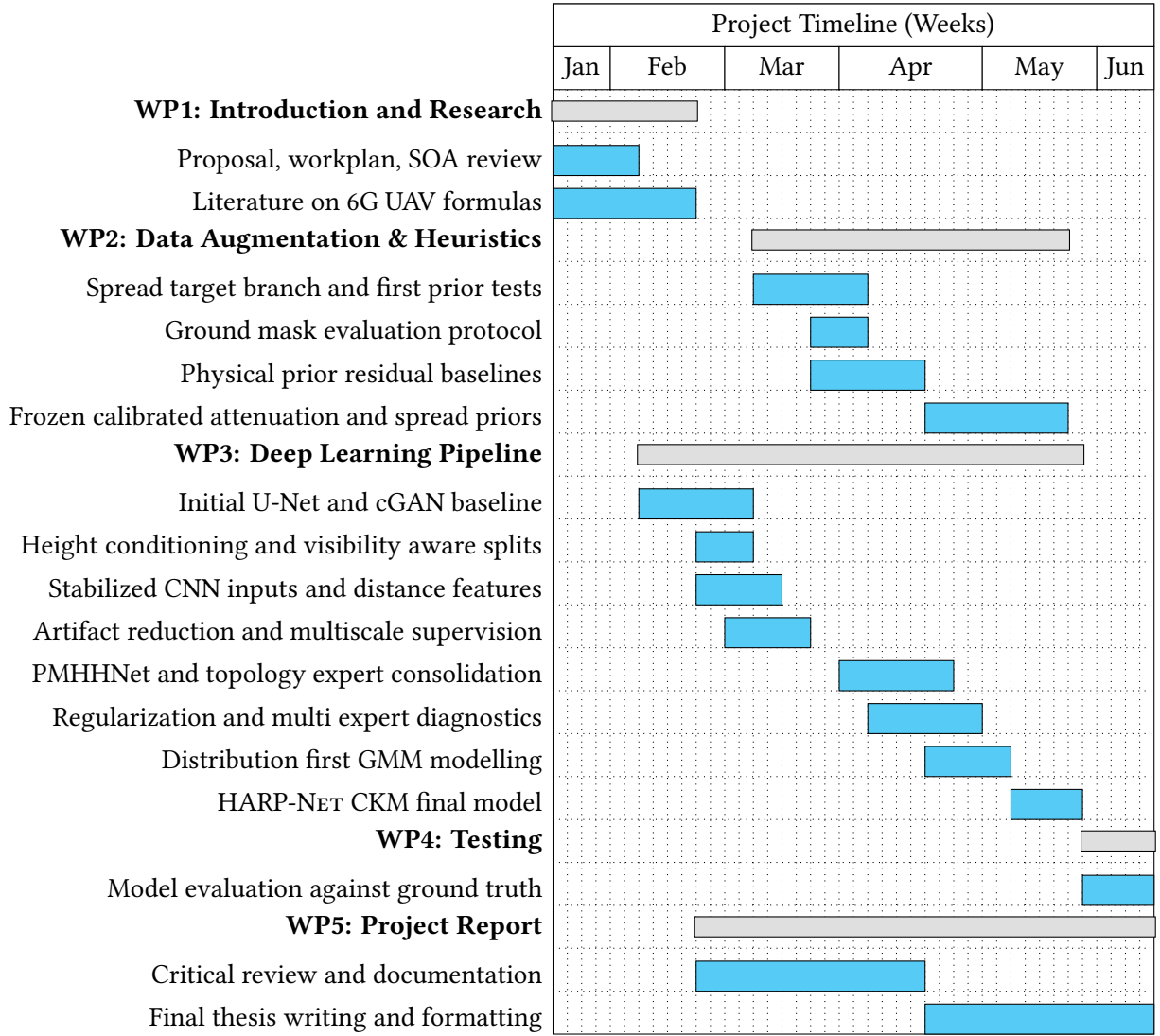


Figure 1.1: Gantt diagram of the project grouped by the main thesis phases: background review, dataset contract, prior construction, final residual model, evaluation, and writing.

State of the Art and Technical Background

This chapter gives the background needed before the method is introduced. It has four roles. First, it defines the propagation and spread quantities used in the thesis. Second, it reviews dense radio map prediction methods and their limitations. Third, it introduces the training and distribution tools used later. Fourth, it explains why external results cannot be compared as a simple leaderboard. Final thesis results are reported only in Chapter 4.

Reader map. Sections 2.1 to 2.2 introduce CKMs, path loss models and A2G propagation. The angular and delay spread sections define the two non attenuation targets. Section 2.5 collects the training tools and fair comparison rules that are needed to read the methodology and results.

2.1 General Framework: CKM Surrogates and Evaluation Contracts

The term channel knowledge map (CKM) refers to a site specific database of location tagged channel knowledge, such as channel gain, angles, or delay information, used to make future radio systems environment aware [2, 3]. This framing is more precise than a generic radio map: the map is not only an image like coverage estimate, but a queryable prior that can reduce channel state acquisition overhead, guide beamforming, support planning, and provide initial channel knowledge before online measurements are available.

Recent CKM literature also clarifies why the thesis uses a large offline ray traced dataset rather than sparse field samples. Xu and Zeng [4] analyse how CKM construction error depends on spatial sample density and on the number of neighbouring samples used for online prediction, showing the explicit cost/accuracy trade off of measurement driven CKM construction. For a thesis whose goal is to learn dense 513×513 targets under city holdout, this motivates the simulation first route: it provides full supervision everywhere, while any learned model is still best understood as a surrogate for a ray traced CKM rather than as a replacement for field calibration.

The data ecosystem has also expanded since the first path loss challenge. RadioMap3DSeer [5] and the ICASSP 2023 challenge [6] provide the closest open benchmark family for path loss image prediction, while the directive antenna dataset of Jaensch et al. [7] and CKMImageNet [8] show the broader move toward open, location specific datasets with richer channel and environment representations. The CKM dataset used in this thesis sits in the same data driven tradition, but differs in its continuous UAV height conditioning, real city holdout split, and simultaneous channel attenuation, delay spread, and angular spread targets.

2.2 Channel Attenuation Prediction

2.2.1 Free space and two ray propagation

The most fundamental path loss term is the *free space path loss* (FSPL), which depends only on carrier frequency f and Tx/Rx distance d :

$$PL_{\text{FSPL}} = 20 \log_{10}(d) + 20 \log_{10}(f) + 20 \log_{10}\left(\frac{4\pi}{c}\right). \quad (2.1)$$

FSPL is the single direct ray reference baseline in unobstructed conditions, not a lower bound on the final path loss once coherent reflections are present. In aerial scenarios a specular ground reflection can produce constructive or destructive interference periodically with distance. Let d_{2D} be the horizontal Tx/Rx distance, h_{tx} the UAV height, and h_{rx} the receiver height. The direct and reflected path lengths are

$$d_{\text{dir}} = \sqrt{d_{2D}^2 + (h_{\text{tx}} - h_{\text{rx}})^2}, \quad d_{\text{ref}} = \sqrt{d_{2D}^2 + (h_{\text{tx}} + h_{\text{rx}})^2}. \quad (2.2)$$

With wavelength λ , the coherent two ray model adds the direct and ground reflected fields before converting to dB [9, 10]:

$$E_{\text{total}} = \frac{e^{-j2\pi d_{\text{dir}}/\lambda}}{d_{\text{dir}}} + \Gamma \frac{e^{-j2\pi d_{\text{ref}}/\lambda}}{d_{\text{ref}}}, \quad (2.3)$$

where Γ is the effective ground reflection coefficient. The received field magnitude $|E_{\text{total}}|$ is then converted to a dB attenuation prediction with a fitted calibration constant. This is the physical origin of the radial rings later exploited by the path loss prior. The numerical gain of this prior is not reported here because this chapter is background. The fitted CKM result is evaluated later in Chapter 4.

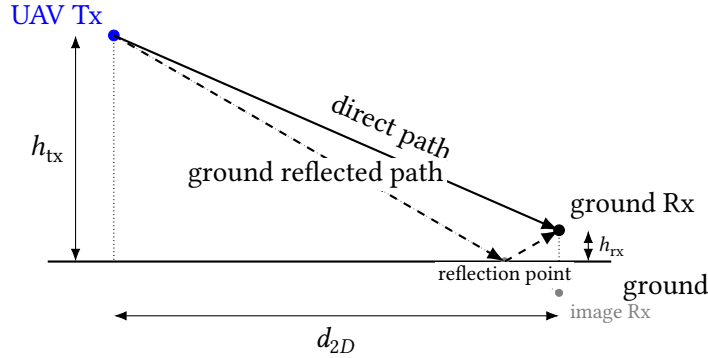


Figure 2.1: Direct and specular ground reflected paths in the two ray model.

2.2.2 Empirical and semi empirical models

Empirical models such as the Okumura-Hata and COST-231/Hata formulas extend FSPL by incorporating frequency, antenna heights, and city type correction terms [9, 11]. While computationally cheap, they are calibrated on specific frequency bands and city/environment types, limiting accuracy when the geometry departs from the fitting dataset.

The 3GPP Technical Report 38.901 [12] formalises stochastic channel models for frequencies from 500 MHz to 100 GHz for urban macro (UMa), urban micro (UMi), rural macro (RMa), and indoor scenarios. It provides separate path loss expressions for LoS and non-LoS (NLoS) conditions, breakpoint distances, and shadow fading standard deviations, which serve as reference baselines throughout this thesis.

2.2.3 UAV and air to ground channel models

Classical terrestrial channel models assume the transmitter is at rooftop height or below. Unmanned aerial vehicle (UAV) base station scenarios introduce a third dimension: the UAV can fly from near ground level to several hundred metres. Al-Hourani et al. [13] showed that the LoS probability increases monotonically with elevation angle and depends on city/environment type (suburban, urban, dense urban, high rise). The A2G modelling literature [13, 14] commonly parameterises this dependence through logistic functions of the elevation angle $\theta = \arctan((h_{\text{tx}} - h_{\text{rx}})/d_{2D})$, where h_{tx} and h_{rx} are transmit and receive heights respectively and d_{2D} is the horizontal distance. 3GPP TR 38.901 [12] is used here as standardized LoS/NLoS large scale channel model context, not as the direct source of that elevation-logistic A2G formula.

Recent work by Saboor and Vinogradov [15] makes this height dependence especially relevant for the present thesis: in simulated 26 GHz urban A2G links, LoS path loss remains close to a free space exponent, whereas NLoS path loss exponents and shadow fading evolve with ABS height and with urban layout. Vinogradov et al. [16] also propose 3D shadow projections as a fast way to generate spatially consistent A2G LoS maps without full point by point ray tracing. These results support two design choices used throughout this work: treating the LoS/NLoS mask as a physical input rather than as a cosmetic label, and conditioning every dense prediction on UAV height.

These works motivate a useful separation for the thesis methodology: LoS/NLoS visibility and continuous UAV height should be treated as physical conditioning variables, not as decorative labels appended to a generic image regression network.

2.2.4 Spread definitions used later

Beyond path loss, full channel characterisation requires second order statistics. The *root mean square delay spread* τ_{rms} measures the time domain dispersion of multipath arrivals and drives inter symbol interference in wideband systems. The *azimuth angular spread* σ_{AS} at the receiver quantifies the spatial diversity of arriving paths. Angular spread is especially relevant for antenna and array design: it affects beam width selection, beamforming robustness, and how concentrated or diffuse the useful received directions are.

For multipath components with powers P_ℓ , delays τ_ℓ , and azimuth angles ϕ_ℓ , the RMS delay spread is

$$\bar{\tau} = \frac{\sum_\ell P_\ell \tau_\ell}{\sum_\ell P_\ell}, \quad \tau_{\text{rms}} = \sqrt{\frac{\sum_\ell P_\ell (\tau_\ell - \bar{\tau})^2}{\sum_\ell P_\ell}}. \quad (2.4)$$

For angular spread, the mean angle must be computed circularly:

$$\bar{\phi} = \text{atan2}\left(\sum_\ell P_\ell \sin \phi_\ell, \sum_\ell P_\ell \cos \phi_\ell\right), \quad \sigma_{\text{AS}} = \sqrt{\frac{\sum_\ell P_\ell \text{wrap}(\phi_\ell - \bar{\phi})^2}{\sum_\ell P_\ell}}, \quad (2.5)$$

where $\text{wrap}(\cdot)$ maps angle differences to the $(-\pi, \pi]$ interval.

3GPP TR 38.901 [12] models both quantities as log normal random variables conditioned on the LoS/NLoS state and distance. WINNER II follows the same broad log domain large scale parameter convention [17]. In practice, the distributions of delay spread and angular spread are right skewed with a heavy exponential tail and often include a spike near

zero representing the dominant LoS path. A Gaussian model cannot represent this shape faithfully. This is why the methodology later checks the empirical target histograms before choosing distribution aware heads.

2.2.5 Deep learning for path loss maps

Image to image translation baselines

The radio map prediction problem can be cast as an image to image regression: given an input topology map (and optionally a transmitter location channel), the model predicts a path loss map of the same spatial extent. The pix2pix framework [18] established a general recipe for such tasks using a conditional GAN with a U-Net generator and a PatchGAN discriminator. For radio maps, however, visual sharpness is not the same as physical accuracy: the adversarial objective can sharpen spatial texture while still failing to represent the height dependent radial pattern and the heavy NLoS tails that dominate the error budget in the CKM dataset.

RadioUNet [19] is the canonical convolutional path loss map baseline before the PMNet challenge family. In its accurate map settings it reports grey level RMSE values around 0.0203–0.0384; using the paper’s 80 dB grey level conversion gives an approximate 1.6 dB–3.1 dB path loss scale. RadioGUNet [20] extends this outdoor RadioMapSeer line with group equivariant convolutions, reporting 1.304 dB on the easiest DPM setting, 1.936 dB on the IRT setting, and 1.392 dB with cars. These values are used again in Chapter 4 as scale references for the final channel attenuation result.

Limitations: These papers are useful path loss scale references, but they do not report dense delay spread or angular spread maps and do not use the CKM city holdout, continuous UAV height contract.

PMNet and the ICASSP outdoor challenge family

PMNet [21] was the winner of the first ICASSP 2023 Outdoor Pathloss Radio Map Prediction Challenge [6]. It uses a deep encoder decoder with skip connections, trained and evaluated on the RadioMap3DSeer dataset [5]. RadioMap3DSeer consists of city maps derived from urban 3D geometry with rooftop transmitters placed 3 m above the selected rooftop in the dataset description, and building interior pixels masked to zero error.

The challenge metric is RMSE in normalised image units. As an approximate dB scale reference, the RadioMap3DSeer image mapping clips path gain to the range $[PL_{thr}, PL_{max}] = [-111, -75]$ dB, yielding a 36 dB dynamic window under that interpretation. Multiplying normalised RMSE by 36 gives the rough dB scale references listed in Table 2.1.

Table 2.1: Approximate dB scale RMSE references for ICASSP 2023 outdoor challenge methods. The normalised scores are taken from the challenge family and converted with the 36 dB RadioMap3DSeer image window described in the dataset paper [6, 5].¹

Method	RMSE (norm.)	Approx. RMSE (dB)
PMNet [21]	0.0383	1.38
PPNet baseline [6]	0.0507	1.83

¹This conversion should be read as an approximate scale reference, not as a like for like physical RMSE. It uses the path gain clipping window $[PL_{thr}, PL_{max}] = [-111, -75]$ dB associated with the normalised target image, so the physical span is $-75 - (-111) = 36$ dB. Multiplying the normalised RMSE values reported in

Limitations: These values are scale references only. They are not directly comparable to CKM because RadioMap3DSeer uses a different dataset, fixed transmitter placement, a much narrower target image window, and a different building pixel convention. The CKM task later evaluated in this thesis uses unseen real city geometries, native 513×513 maps, continuous UAV altitude, and metrics over valid ground receiver pixels. The challenge includes multiple transmitter locations, but it does not test the continuous UAV height conditioning used in CKM.

Transformers, state space models, and attention based encoders

The RMTransformer [22] evaluates a hybrid MaxViT/CNN encoder–decoder on the USC/PMNet dataset rather than on the ICASSP challenge test set. It reports a normalised RMSE of 7.15×10^{-3} , compared with 1.05×10^{-2} for PMNet under a random 90/10 split on 256×256 maps. The paper normalises received power or path loss, depending on the paper convention, from -254 to 0 dBm, so the reported pixel RMSE corresponds to roughly 1.82 dB on its own scale, and the reported channel prediction error of 8.10×10^{-3} corresponds to roughly 2.06 dB. This supports the use of attention for long range urban context, but it is not inserted in Table 2.1 because it is not measured on RadioMap3DSeer or under the ICASSP challenge contract.

Recent attention and foundation model variants extend the effective receptive field of radio map predictors [23, 24]. This is useful when large scale urban context affects the prediction, but these methods normally need more data and compute than convolutional models and their reported benchmarks do not match the CKM UAV height contract.

Foundation models for radio maps

The most recent shift in the literature is towards *wireless channel foundation models* pretrained on large heterogeneous datasets before task specific fine tuning. WiCo-PG [23] uses dual VQGANs and a transformer with frequency-guided mixture-of-experts. It exploits RGB drone camera images as an auxiliary modality and achieves an NMSE of 0.012 on its constructed multimodal U2G benchmark. RadioLAM [25] targets fine grained 3D radio map estimation from ultra low sampling rates using a large generative model pipeline. FM-RME [24] frames radio map estimation as a foundation modelling problem with self supervised pretraining on simulated propagation data.

Limitations: These foundation approaches are relevant as long term background, but they require training corpora and compute budgets beyond a single CKM dataset. They are therefore cited as future context, not as methods directly compared with the final thesis model.

2.2.6 Physics informed and hybrid path loss methods

Geometry assisted and diffraction aware deep learning

Purely data driven models must implicitly learn diffraction geometry from the topology map. An alternative is to precompute a physics based channel prior and provide it as an additional input channel, reducing the learning problem to residual estimation.

Shadow projection methods [16] occupy an intermediate point between stochastic LoS probability models and full ray tracing. They use 3D building geometry to project blocked

the challenge paper [6, 5] by this factor converts them to physical dB, which gives $0.0383 \times 36 \approx 1.38$ for PMNet.

ground regions from the aerial transmitter, producing a deterministic LoS map that is spatially consistent along user trajectories. The CKM dataset already provides a ray traced LoS mask, so this thesis does not need to generate the mask itself; nevertheless, the method is directly relevant because it explains why a binary visibility channel can carry so much of the structure that the neural network otherwise has to infer from the height map alone.

Geometry assisted diffraction features [26] use virtual obstacles, multi screen knife edge propagation, and scattering aware local geometry cues to inject physically meaningful NLoS information into a deep model. This is closely related to the motivation behind the diffraction-prior experiments in this thesis, although the implementation used later is a lightweight per map channel rather than the full geometry model assisted learning framework of that paper. The limitation is that a single precomputed diffraction cue cannot replace a calibrated NLoS model when dense urban shadowing dominates.

Physics informed neural networks

The ReVeal architecture [27] operates as a physics informed neural network (PINN) by incorporating a second order PDE residual as an auxiliary loss term. This penalises predictions with inconsistent local spatial curvature under the derived propagation constraint, reducing corner transition artefacts. ReVeal achieves 1.95 dB RMSE in rural scenarios using only 30 training points by leveraging the physical constraint as a strong regulariser. A lightweight analogue for dense radio maps is a masked Laplacian penalty on the residual prediction, which penalises locally nonsmooth outputs without solving the full ReVeal operator.

Prior plus residual correction

A principled hybrid approach is to *freeze* a calibrated analytic prior and train the neural component to predict only the structured residual. This design reduces the effective learning problem from the full path loss range to a narrower residual distribution, improving data efficiency and interpretability.

Chapter 3 applies this philosophy to the three thesis targets. For the channel attenuation target, an enhanced two ray path loss model fitted per height bin provides a prior map $\hat{y}_{PL}(\mathbf{x}) = \text{TwoRay}(d, h_{tx})$, and the neural component predicts the residual $\epsilon = y - \hat{y}_{PL}$. For delay spread and angular spread, the same prior residual idea can be implemented with a ridge regression model in log space using TX centred geometric features and local morphology statistics. The important background point is the decomposition itself: a frozen train calibrated prior defines what the neural model should not have to rediscover.

2.3 Angular Spread Prediction

2.3.1 Empirical models for spread parameters

Standard channel models characterise delay spread and angular spread through log domain large scale parameters tied to propagation state and scenario. TR 38.901 [12] provides scenario specific tables and formulae for these parameters, with dependencies on variables such as distance, frequency, terminal heights, and LoS/NLoS state across UMa, UMi, and RMa scenarios. These parameterisations are limited to the morphology class and frequency band of the underlying measurement campaigns.

Log domain modelling is physically motivated: spread parameters are multiplicative

quantities in physical space, so their logarithm is more Gaussian like and additive noise assumptions are better justified. The calibrated spread prior adopts this convention by working in $\log(1 + \gamma)$ space throughout, consistent with the 3GPP empirical modelling framework.

2.3.2 Deep learning for wideband channel statistics

Work on deep learning for delay and angular spread is sparser than for channel attenuation. Most published methods either predict spread globally per sample (a single scalar) rather than per pixel, or treat it as an ancillary output of a multitask attenuation model. Per pixel dense prediction of delay spread and angular spread over 513×513 maps is, to the best of the author's knowledge, not addressed by the reviewed methods.

The closest external quantitative references for spreads are therefore scalar channel characteristic predictors rather than dense CKM-map benchmarks. Huang et al. [28] are not used in the results comparison tables because their Wireless InSite campus dataset predicts scalar A2G link characteristics from coordinates and link features, not dense CKM maps from city topologies.

Limitation for this thesis. Most reviewed angular spread work reports scalar link level angle of arrival or angle of departure statistics rather than dense CKM maps under city holdout. Chapter 4 therefore compares angular spread mainly against the frozen prior, the thesis target, and nearby papers whose tasks are related but not equivalent.

The main challenge is the spike plus exponential distribution shape of spread targets. A model trained only with mean squared error (MSE) collapses to the conditional mean, which lies in the interior of the spike and the exponential tail and is therefore a poor representative of either regime. This motivates Gaussian mixture or other distribution aware heads, especially when a log domain prior is available as an anchor before residual learning.

2.4 Delay Spread Prediction

Delay spread has the same dense map gap as angular spread, but the physical interpretation is temporal rather than directional. The target is small for many LoS dominated receivers and can become large in streets or blocks with multiple reflected and diffracted paths. This makes the quantity sensitive to rare high delay pixels, so a mean-error metric can hide the tail behaviour that matters for wideband links.

Limitation for this thesis. Most reviewed delay spread work reports scalar link level channel parameters or simulation specific statistics rather than dense 513×513 CKM maps under city holdout. Chapter 4 therefore compares delay spread mainly against the frozen prior, the thesis target, and nearby papers whose tasks are related but not equivalent.

2.4.1 Distributional mismatch and the spike plus exponential problem

The spike plus exponential shape, consisting of a Dirac like concentration near zero (the LoS dominated case) plus a heavy exponential tail (NLoS multipath), was confirmed empirically on the CKM dataset. Skewness exceeds 5 and excess kurtosis exceeds 40 for both delay and angular spread in several topology classes. A single Gaussian or even a

symmetric loss function does not fit this family.

Mixture models are the natural solution, but the required number of components depends on what is being modelled. A histogram study can select the best family for the raw target distributions, where path loss and the spread targets may need several components to match the full marginal shape. In a prior residual setting, however, the mixture head models a different quantity: the residual distribution after a frozen prior has already explained the dominant LoS, NLoS, delay spread, and angular spread structure. This distinction matters because the prior can remove much of the target side multimodality, leaving a compact mixture to represent under correction, near zero correction, and over correction modes.

2.5 Distribution, Training Tools and Fair Comparison

2.5.1 Heteroscedastic uncertainty

Kendall and Gal [29] established the framework for heteroscedastic neural networks in computer vision: predict both a mean output $\mu(x)$ and a log variance $\log \sigma^2(x)$, optimising the negative log likelihood (NLL)

$$\mathcal{L} = \frac{(\mu - y)^2}{\sigma^2} + \log \sigma^2. \quad (2.6)$$

Pixels with high aleatoric uncertainty (deep NLoS shadows, building edges) receive lower effective weight, while the network is penalised for overestimating uncertainty everywhere. In radio map prediction this formulation is useful when visually similar urban pixels can have different shadowing or multipath outcomes, because the model can express uncertainty without treating all errors as equally informative.

2.5.2 Gaussian mixture model heads

Beyond heteroscedastic Gaussians, the GMM head allows multimodal output distributions. Mixture density networks have already appeared in wireless channel modelling: Saleh et al. [30] use an MDN to predict probabilistic mmWave path loss distributions, Garcia Marti et al. [31] learn a stochastic Gaussian mixture channel model for physical layer design, and Lee et al. [32] derive conditional received power PDFs with a mixture density network. These works support the use of mixture outputs for wireless propagation, but they operate at the level of scalar or channel sample distributions rather than dense radio map generation.

A dense CKM version of a GMM head can be implemented with a two stage architecture. Stage A produces per sample GMM parameters $\{\pi_k, \mu_k, \sigma_k\}_{k=1}^K$ via a convolutional encoder conditioned on UAV height; Stage B produces per pixel component assignments $p_k(i, j)$ and within component deviations $z(i, j)$. The prediction is

$$\hat{y}(i, j) = \sum_{k=1}^K p_k(i, j) \cdot (\mu_k + z(i, j) \cdot \sigma_k). \quad (2.7)$$

This formulation avoids the mode collapse failure of MSE regression while maintaining a compact architecture. The important idea is therefore not the isolated use of a Gaussian mixture, but its role as a global distribution head that tells the dense decoder what value distribution the map should contain before the model solves the harder pixel placement problem.

2.5.3 Stochastic diffusion models

Conditional diffusion models applied to radio maps [33] generate physically plausible stochastic variations in deep shadow zones rather than predicting a single blurry mean. The iterative denoising procedure can be conditioned on topology maps, transmitter locations, and obstacle prompts. RadioDiff reports lower normalised RMSE and higher SSIM/PSNR than CNN and sequence model baselines on static and dynamic RadioMapSeer settings, but its metrics are image normalised rather than physical dB values. These models are computationally expensive at inference time and are listed as a future work horizon for this thesis.

2.5.4 Multitask and joint prediction

Training a single shared backbone across multiple output targets can improve sample efficiency and consistency between targets. Channel attenuation, delay spread, and angular spread are physically correlated: a deep shadow corresponds to high delay spread and potentially wide angular spread from diffracted paths. A joint model can therefore use the intermediate feature maps learned for attenuation to partially explain spread patterns, reducing the effective learning problem.

Chapter 3 uses this idea by sharing the topology and UAV height representation across the three targets, while allowing target specific residual heads for channel attenuation, delay spread, and angular spread. The literature point is that joint training is useful only if the shared representation is anchored by target specific priors and losses; otherwise the easiest target can dominate the common backbone.

2.5.5 Height conditioning for the residual model

Feature wise linear modulation (FiLM) [34] conditions the activations of a neural network on a scalar or vector input by applying learned affine transforms $\gamma(c) \cdot \mathbf{h} + \beta(c)$ to intermediate feature maps. Sinusoidal positional encoding of the UAV height scalar (using log spaced frequencies over the full 12 to 478 m range) is fed into FiLM layers at multiple encoder depths. This ensures that the UAV altitude continuously modulates the activation statistics rather than being handled as a discrete bin, which is critical when the effective propagation regime changes substantially with height.

Two other tools used during development, stochastic weight averaging (SWA) [35] and exponential moving average (EMA) tracking, are kept in this review because they explain some of the training choices in Appendix A. SWA averages model weights from later training checkpoints in order to select a flatter region of the loss landscape. EMA keeps a smoothed copy of the weights during optimization, so the validation checkpoint is less sensitive to short term gradient noise. Both are especially useful when validation cities are never seen during training, where a single epoch can look better or worse because the held out topology set is small.

They should nevertheless be read as training settings for optimization and checkpoint selection, not as propagation features. FiLM changes the network input conditioning because the UAV height controls intermediate activations. SWA and EMA do not add map channels, do not encode LoS/NLoS geometry, and do not change the physical variables described in Chapter 3. They only affect which trained weights are evaluated. This distinction is why the final methodology presents FiLM as part of the residual model, while SWA and EMA remain part of the training protocol.

2.5.6 Key benchmarks and fair comparison

Why headline numbers cannot be compared directly

The most widespread mistake in radio map literature reviews is comparing headline RMSE values without verifying whether the underlying tasks are equivalent. The following evaluation axes are the minimum required for a valid comparison:

1. **Split:** city holdout vs. in-distribution random split vs. same-scene split.
2. **Transmitter height:** fixed rooftop, discrete bins, or continuous UAV altitude.
3. **Building pixels:** excluded from loss, excluded from metrics, set to zero error, or included.
4. **Target quantity and units:** path loss (dB), path gain (dB), received power (dBm), delay spread (ns), or angular spread (degrees).
5. **Dynamic range and clipping:** determines the RMSE floor and ceiling independently of model quality.
6. **Inference extras and calibration source:** test time augmentation (TTA), checkpoint averaging parameters such as SWA or EMA, and whether calibration is train only, validation based, or fitted from test/field measurements.

This thesis uses a strict city holdout split, continuous UAV height, valid ground receiver metrics (building pixels excluded from loss and metrics), native 513×513 pixel maps, and no test data or field calibration after the train calibrated priors and the model under evaluation are frozen. Calibration is therefore not shown as a binary column in Table 2.2: the relevant question is not whether a method has fitted parameters, but whether it adapts to the evaluation scene using validation, test, or field measurements. Table 2.2 summarises the structural axes that can be compared compactly.

Table 2.2: *Evaluation condition comparison across representative methods.*

Method	Unseen city	Cont. UAV height	≥ 512 px	Comparable?
PMNet / ICASSP 2023 [21]	No	No	No	No
RMTransformer [22]	No	No	No	No
RadioGUNet [20]	Partial	No	No	Partial
AIRMap [36]	Partial	No	No	Partial
Gao et al. [37]	No	No	Partial	Partial
Tarhouni et al. [38]	No	No	No	Physical ref.
PathFinder [39]	Partial	No	No	Partial
This thesis task contract	Yes	Yes	Yes	Reference contract

Two consequences follow from this checklist. First, method rankings can change substantially when the split changes from random maps to unseen cities. A model that has learned the texture, street orientation, and height statistics of a known city may still fail when the same transmitter height range is evaluated on a new morphology. For this reason, this thesis treats city holdout as a core part of the task definition rather than as an optional robustness test.

Second, channel attenuation, delay spread, and angular spread should not be treated

as three copies of the same regression problem. LoS path loss often contains a strong deterministic radial component, whereas NLoS path loss depends on blocked link geometry and local morphology. Delay and angular spreads are even more distributional: many pixels are near zero, but physically important tails remain. A fair comparison therefore has to report the target, regime, masking policy, and split together; otherwise an apparently lower RMSE can simply mean that an easier quantity was measured.

Tarhouni et al.: measured suburban path loss prediction

Tarhouni et al. [38] study machine learning based path loss prediction in a sub-6 GHz suburban campus. It is useful because the errors are in physical dB and tied to a concrete environment, but it is terrestrial, lower frequency, and not a dense UAV CKM with delay- and angular spread targets. Its role here is qualitative: real environment generalisation tends to move back toward multi dB errors, which supports keeping city holdout as part of the task definition.

AIRMap: adaptive digital twin radio maps

AIRMap [36] is an adaptive U-Net autoencoder processing fixed 200×200 elevation-map tensors with variable spatial resolution (2.5–15 m/pixel, corresponding to 500 m–3 km per side) for ultra low latency digital twin synchronisation. It reports sub-4 dB RMSE against ray traced path gain maps, and a separate lightweight calibration step using 20 % of field measurements reduces the median measurement domain error to about five percent. Its strong numbers are not directly comparable to CKM because the target is path gain rather than absolute path loss, the input is a single elevation channel, and the transmitter configuration is not the continuous UAV height setting used in this thesis. Its reported ~ 4 ms inference time on an NVIDIA L40S is nevertheless a strong deployment reference: the value of a CKM surrogate is not only accuracy, but producing dense maps fast enough for planning loops.

Gao et al.: multilayer segmentation with corridor weighting

Gao et al. [37] propose a ResNet based outdoor path loss predictor with Tx depth maps, Rx depth maps, a distance map, and a weighting map that emphasises the direct Tx/Rx corridor. Using the ICASSP 2023 outdoor data, Table III of the paper reports their method at 5.59 dB RMSE, compared to 8.09 dB for PPNet, 9.56 dB for RPNet, and 8.72 dB for a twelve layer Vision Transformer baseline; Table II reports 12.19 dB RMSE on the ITU Challenge 2024 large-area subset, also ahead of the same baselines, with roughly 60% fewer FLOPs. This formalises an internal lesson of the CKM work: the important geometry is not the whole map uniformly, but the subset near the propagation corridor.

The lesson for this thesis is methodological: Tx centred distance and obstruction features are useful but are weaker than a true per receiver corridor description. A full Gao style per Rx corridor map remains future work because generating one for every receiver pixel at 513×513 resolution is substantially more expensive than using Tx centred distance and obstruction features.

PathFinder: multi transmitter domain adaptation

PathFinder [39] tackles distribution shift via disentangled feature encoding and mask guided low rank attention, reporting the lowest MSE of 0.1068 and RMSE of 0.3263 on unseen rural datasets. On the RadioMap3DSeer DS-RPP split, its normalised RMSE of 0.0331 can be approximately converted with the same 36 dB RadioMap3DSeer image

window convention, giving about 1.19 dB. The unseen rural score would be about 11.75 dB under the same window, but this is only an approximate scale reading because the paper and code report normalised PNG image units and do not state a separate rural physical target window. The explicit domain generalisation focus aligns with the CKM city holdout spirit, though PathFinder’s setup involves multi transmitter inputs and a different normalisation convention. Techniques from PathFinder such as transmitter oriented mixup augmentation are listed as medium priority future work. The same concern motivates the city holdout split used in this thesis: a CKM surrogate is useful only if it works on new urban layouts, not only on shuffled samples from the same city.

ICASSP 2025 indoor challenge

The ICASSP 2025 Indoor Pathloss Challenge [40] presents a structurally different problem, dense wall penetration NLoS effects in bounded interior spaces, but its public results are a useful warning against overinterpreting sub-2 dB outdoor challenge numbers. The official result page reports that the winning SIP2Net submission obtains a weighted average RMSE of 9.411 dB; IPP-Net reaches 9.501 dB; TerRaIn reaches 10.325 dB; and TransPathNet reaches 10.397 dB [41, 42, 43, 44]. These are not outdoor UAV results, but they show that when NLoS physics becomes harder and less directly observable from the input map, modern neural architectures again hit a roughly 9 dB to 10 dB error band. For this thesis, the lesson is diagnostic rather than competitive: harder NLoS settings favour strong priors, regime separation, and residual distribution modelling over plain direct regression.

2.5.7 Emerging techniques and implications for this thesis

Across the benchmarks reviewed here, the recurring 9 to 10 dB error band in harder NLoS heavy settings reflects the difficulty of inferring volumetric propagation from a 2D building height map. The most plausible routes beyond that wall are richer geometry and diffraction priors [26], PDE style physics losses such as ReVeal [27], sparse target domain adaptation such as RadioPiT [45], diffusion refinement [33], and larger radio map foundation models [24].

For the present thesis, the literature points to four practical design requirements. First, the comparison contract must be explicit because datasets, splits, height protocols, and target units differ widely. Second, A2G prediction requires continuous UAV height conditioning rather than a fixed transmitter height. Third, LoS/NLoS and environment type effects should be reported separately because one global RMSE hides the hardest regimes. Fourth, distribution aware outputs are useful when targets contain multiple modes or long tails. These observations motivate the methodology in Chapter 3.

Methodology

This chapter defines the final reproducible pipeline used in the thesis. Figure 3.1 gives the high level flow: the raw CKM topology map and transmitter height are converted into shared support maps and frozen calibrated priors, and HARP-NET CKM then learns bounded residual corrections for channel attenuation, delay spread and angular spread. Figure 3.2 expands the same flow into the concrete maps, priors, neural inputs and final outputs used in the experiments. Appendix A keeps the development history; this chapter keeps only the final method and reproduction details.

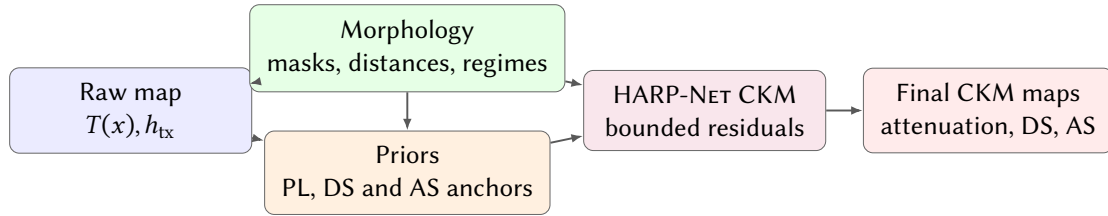


Figure 3.1: High level methodology structure. The raw building map and transmitter height are first converted into morphology maps and calibrated priors; those maps then feed HARP-NET CKM, which learns bounded residual corrections instead of predicting the radio maps from scratch.

Table 3.1: Chapter 3 guide.

Section	What it defines
Dataset and split	The HDF5 fields, city holdout split, receiver mask and height range.
General Framework and Data Flow	The complete path from raw CKM sample to frozen priors, residual model inputs and final maps.
Shared support maps, visibility masks and morphology	The support masks, distances, morphology windows and regime keys reused by every prior.
Final priors: overview and literature basis	Why the frozen priors are deterministic estimators rather than generic physical formulas.
Channel Attenuation Prior and Residual Prediction	The calibrated LoS/NLoS attenuation prior and the residual channel attenuation branch.
Delay and Angular Spread Prior and Residual Prediction	The calibrated spread prior and the residual DS/AS branches.
HARP-NET CKM Neural Residual Model	The neural network used in the final system: input channels, height conditioning, bounded residual heads and training losses.

1. External Framework Inputs

Inputs: City building/topology map $T(x)$ and UAV transmitter height h_{tx} .

Fixed dataset constants: Carrier frequency $f_c = 7.125$ GHz, receiver antenna height $h_{rx} = 1.5$ m, centred UAV.

Output: One dense 513×513 CKM sample frame.

2. Support, Geometry and Morphology Derivation (per pixel)

Inputs: Topology map $T(x) \in \mathbb{R}^{513 \times 513}$, scalar h_{tx} , and the fixed constants from Step 1.

Outputs (all 513×513 maps): Binary masks g ground receiver, m_B building, m_{LoS} visible and m_{NLoS} blocked; distances d_{2D} , direct distance d_{LoS} , reflected distance d_{ref} ; elevation θ ; morphology features and topology/height regime keys.

3. Multilinear Regression and Calculations Prior

Fitted once with training city labels. At inference it is a deterministic calculator: support, geometry and morphology maps enter; four calibrated prior maps leave.

LoS Channel Attenuation Prior

Inputs: $d_{LoS}, d_{ref}, d_{2D}, h_{tx}$; fixed f_c, h_{rx} ; selected by m_{LoS} .

Operation: FSPL + two ray term + height bias + radial residual.

Output: LoS prior map $\widehat{PL}_{LoS}(x)$ in dB.

NLoS Channel Attenuation Prior

Inputs: m_{NLoS} , raw blocked link PL, d_{2D}, h_{tx} , elevation and morphology vector $\mathbf{x}_{PL,NLoS}(x) \in \mathbb{R}^{15}$, regime key.

Fit: Training city per regime multilinear OLS coefficients.

Output: NLoS prior map $\widehat{PL}_{NLoS}(x)$ in dB.

Delay Spread Prior

Inputs (per pixel): Raw log-spread estimate $z^{(0)}, d_{2D}, \theta, h_{tx}$, LoS/NLoS state, topology class, antenna bin and morphology vector $\mathbf{x}_{SP}(x) \in \mathbb{R}^{23}$.

Fit: Regime-wise ridge coefficients, inverse $\exp(\cdot) - 1$, clipping.

Output: 513×513 delay spread prior map $\widehat{DS}_{prior}(x)$ in ns.

Angular Spread Prior

Inputs (per pixel): Raw log-spread estimate $z^{(0)}, d_{2D}, \theta, h_{tx}$, LoS/NLoS state, topology class, antenna bin and morphology vector $\mathbf{x}_{SP}(x) \in \mathbb{R}^{23}$.

Fit: Regime-wise ridge coefficients, inverse $\exp(\cdot) - 1$, clipping.

Output: 513×513 angular spread prior map $\widehat{AS}_{prior}(x)$ in degrees.

4. Deep-Learning Input Stack Assembly

Inputs: 513×513 support maps from Step 2 and prior maps from Step 3. The path loss branch also forms per pixel $\widehat{PL}_{prior}(x) = m_{LoS}(x)\widehat{PL}_{LoS}(x) + m_{NLoS}(x)\widehat{PL}_{NLoS}(x)$.

Output: Input tensor $X \in \mathbb{R}^{9 \times 513 \times 513}$ with channels $[T_{in} = Tg/90$ (geometry via masks/morphology/priors), $m_{LoS}, m_{NLoS}, g, \widehat{PL}_{prior}/180, \widehat{PL}_{LoS}/180, \widehat{PL}_{NLoS}/180, \ell_{DS}, \ell_{AS}]$, with $\ell_{DS} = \log(1 + \widehat{DS}_{prior})/\log(401)$, $\ell_{AS} = \log(1 + \widehat{AS}_{prior})/\log(91)$, plus scalar condition h_{tx} .

5. HARP-NET CKM Residual Model

Inputs: Tensor $X \in \mathbb{R}^{9 \times 513 \times 513}$ and scalar h_{tx} injected via sinusoidal FiLM conditioning.

Architecture: Shared GroupNorm U-Net encoder-decoder with task-specific LoS/NLoS residual heads; height conditioning through FiLM layers.

Train-only supervision: Per-pixel valid-ground CKM labels for channel attenuation, delay spread and angular spread.

Outputs: Per-pixel GMM parameters (p, μ, σ) , regional gates $\alpha_{k,r}$, and bounded residual maps $\Delta_{PL}, \Delta_{DS}, \Delta_{AS} \in \mathbb{R}^{513 \times 513}$.

6. Final Map Assembly and Evaluation

Inputs: Prior maps, bounded residual maps $\Delta_{PL}, \Delta_{DS}, \Delta_{AS} \in \mathbb{R}^{513 \times 513}$, LoS/NLoS masks and valid-ground mask m_{ground} .

Assembly (per pixel): Masks select LoS/NLoS point maps. PL is prior plus bounded dB residual; DS/AS mix capped native component means in $\log(1 + y)$, then invert.

Outputs: Three final dense 513×513 CKM maps: $\widehat{PL}$ (dB), $\widehat{DS}$ (ns), $\widehat{AS}$ (deg).

Evaluation: Pixel weighted RMSE and MAE over unseen city valid ground pixels.

Figure 3.2: Detailed methodology data flow from external inputs to final assembled CKM maps, showing inputs, operations and outputs of each block.

3.1 Dataset and split

The data source is CKM_Dataset_270326.h5, a project HDF5 version of the CKM radio map data associated with the public path loss and ToA radio map dataset of Yapar et al. [5]. At dataset level, it contains 16180 ray traced urban samples from 66 real city scenarios. The fields used by the final pipeline are the building topology map, the CKM target stored as path_loss, the LoS/NLoS support mask, the UAV transmitter height uav_height, delay_spread, and angular_spread. The rest of this section moves from that dataset level view to the per sample receiver grid and city split.

Each sample is a complete 513×513 pixel CKM. The spatial resolution is $1 \text{ m} \times 1 \text{ m}$ per pixel, so each map covers a $513 \text{ m} \times 513 \text{ m}$ square. The UAV transmitter is horizontally fixed at the centre of the map, while its antenna height h_{tx} varies from sample to sample and is treated as a primary conditioning variable. The dataset labels this transmitter as a UAV, but the mathematical setting is an elevated 7.125 GHz transmitter with sampled heights above roughly 10 m, so the same surrogate can describe UAVs or other elevated radio access nodes under the same simulation assumptions. Every non building pixel is an active receiver location, with the receiver antenna placed exactly at 1.5 m above ground. Building pixels are not receiver positions and are excluded from the training loss and reported metrics rather than being counted as easy background pixels. The model never receives a random crop as an independent training city. Instead, the split is performed at city level. For the distribution first and final neural models the split uses split_seed = 42, with 15% validation and 15% test samples after city assignment. This avoids the most common optimistic error in spatial radio prediction: training on one part of a city and testing on another visually similar part of the same city.

The transmitter height samples are not uniformly distributed. They are produced by the dataset generator through the affine beta mixture $h_{\text{design}} = 10 + 490X$, where

$$X \sim 0.75 \text{Beta}(3, 23) + 0.25 \text{Beta}(4, 4). \quad (3.1)$$

For $a, b > 0$, $\text{Beta}(a, b)$ denotes the distribution with density

$$f_{a,b}(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}, \quad 0 \leq x \leq 1, \quad (3.2)$$

and density zero outside $[0, 1]$. Thus the generator first draws a normalized height variable from the mixture density $0.75 f_{3,23}(x) + 0.25 f_{4,4}(x)$, then maps it to metres. However, the finite HDF5 database is the relevant distribution for training and reporting. Reading the stored uav_height field for all 16180 samples gives an empirical range of 11.61 m to 477.54 m, usually rounded in the text as about 12 m to 478 m; 477.54 m is the largest sampled HDF5 value, not a hard endpoint of the generator distribution. This is statistically normal: under $10 + 490X$, $P(h_{\text{tx}} > 478 \text{ m}) = 3.19 \times 10^{-5}$ per draw, so only 0.52 such samples are expected in 16180 draws and the probability of drawing none is about 0.60. A reproducible formulaic version of the realized database distribution is the 3 m empirical density. Let $\Delta h = 3 \text{ m}$, $B_k = [10 + 3k, 13 + 3k)$, and let h_i be the stored height of sample i . Then

$$\begin{aligned} \hat{f}_{\Delta h}(h) &= d_k, \quad h \in B_k, \\ d_k &= \frac{n_k}{N \Delta h}, \quad n_k = \sum_{i=1}^N \mathbf{1}\{h_i \in B_k\}, \quad N = 16180. \end{aligned} \quad (3.3)$$

When the density is visualized, the plotted curve linearly interpolates between the bin centre values (c_k, d_k) , with $c_k = 11.5 + 3k$. Figure 3.3 shows this realized transmitter height density directly from the stored HDF5 samples.

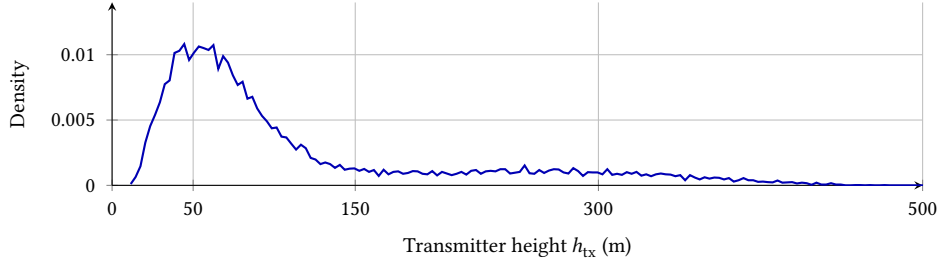


Figure 3.3: Empirical transmitter height density read directly from the CKM HDF5 database using the 3 m histogram of (3.3), linearly interpolated between bin centres for display. Most training support lies in the low and mid-altitude bands.

The all-database height counts are 3972 samples below 50 m, 8456 from 50 m to 150 m, 2490 from 150 m to 300 m, and 1262 above 300 m. In the actual training cities, the same bins contain 2676, 5654, 1644, and only 866 samples, respectively. This matters for interpreting the height binned results: the higher altitude bins contain fewer training examples, so their lower RMSE later in the thesis is evidence that those cases are physically easier, not that they are better represented in the training set.

The prior artifacts use the same city holdout split as the final model: `val_ratio = 0.15`, `test_ratio = 0.15`, and `split_seed = 42`. The path loss prior and the spread priors are fitted on the 10840 training samples. Standalone prior audits can then use the full held out validation and test pool of 5340 samples, while final model against prior comparisons use the 2590-sample final test split. This keeps training cities, validation cities, and final test cities separated throughout the prior and residual model evaluations.

Table 3.2: City holdout assignment used by the distribution first and final neural models.

Split	Samples	Cities	City names
Train	10840	37	Abidjan, Abu Dhabi, Alexandria, Antigua Guatemala, Athens, Bangalore, Bangkok, Bath, Berlin, Bratislava, Brugge, Buenos Aires, Canberra, Chefchaouen, Dalian, Hanoi, Ho Chi Minh City, Hyderabad, Kinshasa, Leuven, Ljubljana, Luanda, Lyon, Manaus, Minsk, Nagpur, Paris, Rio de Janeiro, Santiago, Shanghai, Siem Reap, St. Louis, Thane, Toledo, Valparaiso, Venice, Vienna.
Validation	2750	15	Bilbao, Chiang Mai, Cleveland, Dhaka, Dubrovnik, Fortaleza, Marrakesh, Mexico City, Munich, Nazareth, Nice, Salvador, Segovia, Tunis, Victoria.
Test	2590	14	Barcelona, Beijing, Carcassonne, Copenhagen, Halifax, Jaipur, Johannesburg, Key West, Kuala Lumpur, Osaka, Phuket, Pune, Surat, Vancouver.

3.2 General Framework and Data Flow

Table 3.3: *External framework inputs, internally computed maps, neural model inputs and final outputs.*

Pipeline block	Maps / variables
External inputs to the full framework	Building/topology map $T(x)$ and UAV transmitter height h_{tx} , stored as <code>uav_height</code> . The carrier frequency $f_c = 7.125$ GHz and receiver height $h_{rx} = 1.5$ m are fixed dataset constants, not external inputs.
Internally computed support, geometry and morphology maps	Ground/building masks, LoS/NLoS masks, horizontal distance, direct LoS distance, reflected distance, elevation angle, local morphology features and topology/height regime keys.
Multilinear regression and calculation prior outputs	Combined path loss prior, path loss LoS prior, path loss NLoS prior, delay spread prior, angular spread prior.
Inputs to HARP-NET CKM	Nine map channels in the final model order: topology, LoS mask, NLoS mask, ground mask, combined path loss prior, path loss LoS prior, path loss NLoS prior, delay spread prior and angular spread prior; plus scalar transmitter-height FiLM conditioning.
Final outputs	Channel attenuation map, delay spread map, angular spread map.

The full reader map has already been introduced in Figure 3.2. Sections 3.3 to 3.7 define the final method at the level needed to reproduce the reported pipeline: which maps are fitted, which maps are kept fixed, which channels enter HARP-NET CKM, how the deterministic outputs are assembled, and how the final maps are evaluated.

Therefore, the full framework has two external inputs and three final outputs. The neural residual model receives nine spatial map channels plus a separate scalar transmitter height conditioning path because the framework has already computed visibility masks, receiver masks and fixed prior maps. This is the main distinction between the deployable framework and the internal neural model: the user provides only a city height map and the transmitter height, while the pipeline constructs the remaining maps before inference.

The rest of this chapter therefore describes the selected final method rather than the chronological path used to discover it. Appendix A keeps the development evidence and discarded routes; this chapter keeps only the final constraints that survive into the reported system: no building pixel credit, no random city leakage, explicit transmitter height conditioning, separate LoS/NLoS support, training city regression/calculation priors, and bounded residual correction.

3.3 Shared support maps, visibility masks and morphology

Before any prior is evaluated, the raw map is converted into a common set of deterministic support maps. This section defines those maps once, because the LoS attenuation prior, the NLoS attenuation prior, the two spread priors and the final neural residual model all use the same shared geometry.

Let $H = W = 513$, let $x = (i, j)$ denote a pixel with $i, j \in \{0, \dots, 512\}$, and let the

transmitter be at $x_{\text{tx}} = (i_{\text{tx}}, j_{\text{tx}}) = (256, 256)$. This is the zero based convention used by the implementation. The topology matrix is:

$$\mathbf{T} \in \mathbb{R}_{\geq 0}^{H \times W}, \quad T_{ij} = \text{building height at pixel } (i, j) \text{ in metres.} \quad (3.4)$$

The pixel pitch is $\Delta = 1$ m. The receiver height is fixed for the whole thesis:

$$h_{\text{rx}} = 1.5 \text{ m.} \quad (3.5)$$

The transmitter height h_{tx} changes with the sample and is the only scalar radio geometry input besides the topology map.

The ground receiver mask and building occupancy mask are:

$$g_{ij} = \mathbf{1}[T_{ij} \leq \epsilon_g], \quad \epsilon_g = 10^{-6} \text{ m,} \quad (3.6)$$

$$m_{\text{B},ij} = 1 - g_{ij}. \quad (3.7)$$

Equivalently, $T(x)$ denotes the building height stored at pixel x , $g(x) = 1$ means that x is a ground receiver pixel, and $m_{\text{B}}(x) = 1$ means that x is occupied by a building. The HDF5 maps use exact zero for ground, while CKMGenerator uses the small tolerance ϵ_g so that image or array inputs with numerical noise are handled safely. The valid receiver set is:

$$\Omega_g = \{(i, j) : g_{ij} = 1\}. \quad (3.8)$$

Every pixel in Ω_g is a ground receiver.

For every pixel $x = (i, j)$, the horizontal distance from the centered transmitter is:

$$d_{2D}(x) = \Delta \sqrt{(i - i_{\text{tx}})^2 + (j - j_{\text{tx}})^2}. \quad (3.9)$$

The direct LoS path length is:

$$d_{\text{LoS}}(x) = \sqrt{d_{2D}(x)^2 + (h_{\text{tx}} - h_{\text{rx}})^2}. \quad (3.10)$$

The only other 3D path length used by the final LoS prior is the image source ground reflection distance:

$$d_{\text{ref}}(x) = \sqrt{d_{2D}(x)^2 + (h_{\text{tx}} + h_{\text{rx}})^2}. \quad (3.11)$$

The notation d_{ref} is reserved for the reflected two ray path.

The elevation angle is:

$$\theta(x) = \arctan 2(h_{\text{tx}} - h_{\text{rx}}, \max(d_{2D}(x), 1)), \quad (3.12)$$

with the $\max(\cdot, 1)$ term preventing division by zero below the UAV. The normalized angle maps are:

$$\theta_{\text{norm}}(x) = \text{clip}\left(\frac{2\theta(x)}{\pi}, 0, 1\right), \quad (3.13)$$

$$\theta_{\text{inv}}(x) = 1 - \theta_{\text{norm}}(x). \quad (3.14)$$

This is the angle divided by 90° , written directly in radians. Therefore θ_{norm} is high for near overhead links, while θ_{inv} is high for grazing links that pass through more urban structure.

The LoS mask is treated as a support map, not as a target radio map. In the HDF5 experiments the stored CKM mask is used directly and then restricted to ground receivers:

$$m_{\text{LoS}}(x) = g(x) \mathbf{1}[m_{\text{LoS},\text{HDF5}}(x) > 0.5], \quad (3.15)$$

$$m_{\text{NLoS}}(x) = g(x) (1 - \mathbf{1}[m_{\text{LoS},\text{HDF5}}(x) > 0.5]). \quad (3.16)$$

This ground multiplication is essential: $1 - m_{\text{LoS}, \text{HDF5}}$ alone would incorrectly mark building pixels as NLoS receivers. When CKMGenerator receives an input that has no provided LoS mask, it generates one from the topology and transmitter height by ray casting. In words, for each candidate ground receiver CKMGenerator walks along the horizontal segment from the centered UAV to that receiver, skips the transmitter and receiver cells themselves, and checks whether any sampled building reaches or crosses the height of the straight aerial link. If that happens the receiver is generated as NLoS; otherwise it is generated as LoS. Building pixels remain outside both masks because both generated masks are multiplied by $g(x)$.

For a candidate ground receiver $x = (i, j)$, define:

$$L_x = \max\left(1, \left\lceil \frac{\max(|i - i_{\text{tx}}|, |j - j_{\text{tx}}|)}{s} \right\rceil\right), \quad s = 1 \text{ pixel} \quad (3.17)$$

for the default generator setting. The implementation enforces $s \geq 0.25$ pixel if the user requests a finer step. For $\ell = 1, \dots, L_x - 1$, the sampled point on the horizontal ray is:

$$t_{\ell, x} = \frac{\ell}{L_x}, \quad (3.18)$$

$$q_{\ell, x} = (\text{round}(i_{\text{tx}} + (i - i_{\text{tx}})t_{\ell, x}), \text{round}(j_{\text{tx}} + (j - j_{\text{tx}})t_{\ell, x})). \quad (3.19)$$

The height of the ideal straight segment from the UAV to the receiver is:

$$z_{\ell, x} = h_{\text{tx}} + (h_{\text{rx}} - h_{\text{tx}})t_{\ell, x}. \quad (3.20)$$

With default clearance $c_{\text{clear}} = 0$, the generated obstruction flag is:

$$o(x) = \mathbf{1}[\exists \ell \in \{1, \dots, L_x - 1\} : q_{\ell, x} \notin \{x_{\text{tx}}, x\} \text{ and } T(q_{\ell, x}) + c_{\text{clear}} \geq z_{\ell, x}]. \quad (3.21)$$

The generated masks are then:

$$m_{\text{LoS}, \text{gen}}(x) = g(x) [1 - o(x)], \quad (3.22)$$

$$m_{\text{NLoS}, \text{gen}}(x) = g(x) o(x). \quad (3.23)$$

CKMGenerator uses the provided mask in automatic mode when it exists; otherwise it uses (3.22). The final experiments in this thesis use the stored HDF5 mask; the generated mask is only fallback deployment logic.

The local morphology features are also computed before any prior is evaluated. The 15×15 windows capture near receiver texture, while the 41×41 windows capture a wider urban block context. In the code these two window sizes are fixed constants. For $k \in \{15, 41\}$, let BoxMean_k be a reflected padding $k \times k$ average:

$$\text{BoxMean}_k(f)(i, j) = \frac{1}{k^2} \sum_{u=i-p}^{i+p} \sum_{v=j-p}^{j+p} f_{\text{ref}}(u, v), \quad p = \left\lfloor \frac{k}{2} \right\rfloor, \quad (3.24)$$

where f_{ref} is f reflected at the map boundary. The final prior bundle uses:

$$\delta_k(x) = \text{BoxMean}_k(m_{\text{B}})(x), \quad (3.25)$$

$$h_k(x) = \frac{\text{BoxMean}_k(T \odot m_{\text{B}})(x)}{H_{\text{norm}}}, \quad H_{\text{norm}} = 90 \text{ m}, \quad (3.26)$$

$$n_k(x) = \text{BoxMean}_k(m_{\text{NLoS}})(x). \quad (3.27)$$

δ_k is local building coverage, h_k is density weighted local building height, and n_k is the local fraction of the window that is both ground and NLoS. For the spread prior, three

additional 41×41 morphology terms are used:

$$b_{41}(x) = \frac{\text{BoxMean}_{41}(\max(T - h_{\text{rx}}, 0) \odot m_B)(x)}{H_{\text{norm}}}, \quad (3.28)$$

$$c_{41}(x) = \frac{\text{BoxMean}_{41}(\max(h_{\text{tx}} - T, 0) \odot m_B)(x)}{H_{\text{norm}}}, \quad (3.29)$$

$$t_{41}(x) = \text{BoxMean}_{41}(\mathbf{1}[T > h_{\text{tx}}] \odot m_B)(x). \quad (3.30)$$

Here b_{41} measures local blocker depth above the receiver, c_{41} measures the UAV height margin over nearby buildings, and t_{41} measures the local area fraction occupied by buildings taller than the UAV. These are area weighted 41×41 summaries over the full window: ground cells contribute zero because of m_B . Thus c_{41} is both a clearance and local building presence feature, and t_{41} is not a fraction conditioned on the cells already being buildings.

Two topology regime systems are used because the calibrated attenuation prior and the spread prior were fitted with different levels of granularity. Both are computed once per map from:

$$\bar{\delta}_{\text{map}} = \frac{1}{HW} \sum_x m_B(x), \quad (3.31)$$

$$\bar{h}_{\text{map}} = \frac{\sum_x T(x)m_B(x)}{\max(\sum_x m_B(x), 1)}. \quad (3.32)$$

The three class path loss regime is:

$$\text{class}_{\text{PL}} = \begin{cases} \text{dense_highrise}, & \bar{\delta}_{\text{map}} \geq 0.2549 \text{ or } \bar{h}_{\text{map}} \geq 15.95 \text{ m}, \\ \text{open_lowrise}, & \bar{\delta}_{\text{map}} \leq 0.1957 \text{ and } \bar{h}_{\text{map}} \leq 10.91 \text{ m}, \\ \text{mixed_midrise}, & \text{otherwise.} \end{cases} \quad (3.33)$$

The six class spread regime is:

$$\text{class}_{\text{SP}} = \begin{cases} \text{open_sparse_lowrise}, & \bar{\delta}_{\text{map}} \leq 0.12, \bar{h}_{\text{map}} \leq 12 \text{ m}, \\ \text{open_sparse_vertical}, & \bar{\delta}_{\text{map}} \leq 0.12, \bar{h}_{\text{map}} > 12 \text{ m}, \\ \text{dense_block_midrise}, & \bar{\delta}_{\text{map}} \geq 0.28, \bar{h}_{\text{map}} \leq 28 \text{ m}, \\ \text{dense_block_highrise}, & \bar{\delta}_{\text{map}} \geq 0.28, \bar{h}_{\text{map}} > 28 \text{ m}, \\ \text{mixed_compact_lowrise}, & 0.12 < \bar{\delta}_{\text{map}} < 0.28, \bar{h}_{\text{map}} \leq 12 \text{ m}, \\ \text{mixed_compact_midrise}, & 0.12 < \bar{\delta}_{\text{map}} < 0.28, \bar{h}_{\text{map}} > 12 \text{ m}. \end{cases} \quad (3.34)$$

The shared calibration height key is:

$$\text{ant_bin}(h_{\text{tx}}) = \begin{cases} \text{low_ant}, & h_{\text{tx}} \leq 58.12 \text{ m}, \\ \text{mid_ant}, & 58.12 \text{ m} < h_{\text{tx}} \leq 103.85 \text{ m}, \\ \text{high_ant}, & h_{\text{tx}} > 103.85 \text{ m}. \end{cases} \quad (3.35)$$

The normalized scalar height feature used in the spread vector is:

$$h_{\text{norm}} = \text{clip}\left(\frac{\log(1 + \max(h_{\text{tx}} - h_{\text{rx}}, 0))}{\log(401)}, 0, 1\right). \quad (3.36)$$

The denominator $\log(401)$ corresponds to a 400 m transmitter minus receiver height difference after the $1 + \cdot$ transform. This is a modelling cap, not the maximum HDF5

height: the database reaches about 478 m, but the high altitude tail contains too few training samples after city and regime splits to estimate a separate response reliably. Heights above this scale are therefore saturated at $h_{\text{norm}} = 1$. The feature is a scalar for each sample and is broadcast to every valid receiver row when the spread design matrix is built; it is not a local morphology map.

3.4 Final priors: overview and literature basis

This section provides the entry point for the ML calibrated physical path loss and spread priors and the HARP-NET CKM residual architecture. The propagation ingredients follow classical FSPL, two ray, empirical NLoS and large scale parameter models [9, 10, 11, 12, 17, 14], while the residual network inherits dense prediction, FiLM conditioning and distribution aware regression ideas [46, 34, 29, 30]. The detailed mathematical derivations and implementation level fitting notes are included in this methodology chapter, so the final pipeline can be read without leaving the method section.

3.4.1 High level architecture

The central design decision in the final model is *not* to learn path loss, delay spread, and angular spread from scratch. Instead:

- the path loss prior provides a frozen prior for `path_loss`,
- the spread prior module provides frozen priors for delay spread (`delay_spread`) and angular spread (`angular_spread`),
- HARP-NET CKM learns bounded corrections around those priors.

The deterministic or interpretable structure is kept in the prior, and the deep model is used only for the residual part that the prior cannot explain. Figure 3.2 illustrates the overall pipeline.

This prior phase is not presented as a new universal closed form propagation law. Its value is more practical: it turns physics, morphology, visibility, and height dependent calibration into a strong auditable baseline. The final model then follows a split of responsibility: calibrated priors remove the stable component of the targets, and the neural residual model focuses on the local errors that remain.

In this chapter, a prior is a deterministic frozen estimator map computed before the neural residual correction. Three such estimators are used:

- **LoS path loss prior.** This is the clearest physical branch: $PL_{\text{LoS}} = \text{FSPL} + C_{2\text{ray}} + \beta_{\text{LoS}} + r$. FSPL uses the 3D direct Tx–Rx distance; $C_{2\text{ray}}$ comes from a coherent direct plus ground reflected field sum [9, 10]; β_{LoS} is a fitted height dependent dB offset; and r is a smoothed radial residual table. The UAV is always centred horizontally, so the radial coordinate is unambiguous, but h_{tx} varies by sample: $\rho(h_{\text{tx}})$, $\phi(h_{\text{tx}})$, $\beta_{\text{LoS}}(h_{\text{tx}})$ and $r(d_{2D}, h_{\text{tx}})$ are fitted in 5 m height bins, smoothed, and linearly interpolated at inference. It is not a generic free space formula and it is not a learned linear regressor.
- **NLoS path loss prior.** This starts from the raw blocked link propagation estimate $PL^{(0)}$, which includes $PL_{\text{NLoS,raw}}$ on obstructed pixels, then calibrates it with local morphology: building density, building height, NLoS support, the elevation dependent shadowing proxy σ_{shadow} , elevation angle, and interactions. It uses

antenna height regimes (low_ant, mid_ant, high_ant) because the obstruction geometry changes qualitatively when the centred UAV rises above the local skyline, consistent with A2G elevation and scenario modelling practice [11, 12, 13, 14]. It is deliberately separate from the LoS branch because blocked pixels do not follow the same two ray mechanism.

- **Delay and angular spread priors.** These are fitted in $\log(1 + y)$ space. The raw estimate captures propagation state, distance, elevation, and local morphology; the final ridge calibration adds the 23-feature vector of Sec. 3.6. That vector includes continuous normalized UAV height, density weighted transmitter clearance over nearby building pixels, and the local window area fraction occupied by buildings taller than the transmitter. This follows the log domain convention used for large scale spread statistics and recent A2G spread predictors [12, 17, 47, 28]. This is why the spread prior can fit the near zero LoS mass and the NLoS tail without letting the largest tail values dominate the whole calibration, unlike a native space global average.

The notation is kept distinct across blocks. The symbol $\rho(h_{\text{tx}})$ is reserved for the LoS reflection amplitude. Local building density is written as δ_k , and the LoS dB bias is written as $\beta_{\text{LoS}}(h_{\text{tx}})$. The spread symbols b_{topology} and b_{41} remain local to the spread prior: the first is a log domain topology offset and the second is a blocker depth feature.

Table 3.4: Term by term interpretation of the main frozen prior quantities.

Term	Prior block	Interpretation
d_{LoS}	LoS path loss	3D direct Tx–Rx distance. It sets the FSPL baseline and uses the fixed $h_{\text{rx}} = 1.5$ m receiver height.
d_{ref}	LoS path loss	Image source reflected distance. It is the length of the single ground reflected path used in the coherent two ray correction.
$C_{2\text{ray}}$	LoS path loss	Direct plus reflected field correction in dB. Negative values mean constructive gain over FSPL; positive values mean destructive fading.
$\rho(h_{\text{tx}})$	LoS path loss	Height interpolated effective reflection amplitude. It is fitted from CKM rather than assumed from one fixed ground material.
$\phi(h_{\text{tx}})$	LoS path loss	Height interpolated effective reflection phase, in radians. It absorbs the phase offset of the simplified one reflection geometry.
$\beta_{\text{LoS}}(h_{\text{tx}})$	LoS path loss	Height dependent dB bias remaining after the coherent two ray fit.
$r(d_{2D}, h_{\text{tx}})$	LoS path loss	Smoothed radial residual table. It corrects distance dependent errors that remain after FSPL, two ray interference, and height bias.
$\text{PL}^{(0)}, \text{PL}_{\text{NLoS,raw}}$	NLoS path loss	Raw hybrid formula map and its obstructed pixel branch before map morphology calibration.
δ_k	NLoS PL and spreads	Fraction of building pixels in a $k \times k$ neighbourhood; a local density proxy.
h_k	NLoS PL and spreads	Local building height summary in a $k \times k$ neighbourhood; a proxy for nearby reflectors and blockers.
n_k	NLoS PL and spreads	Local NLoS support fraction; indicates how much of the neighbourhood is geometrically blocked.
σ_{shadow}	NLoS path loss	Elevation dependent shadowing proxy used by the NLoS linear calibration.
$\theta, \theta_{\text{norm}}, \theta_{\text{inv}}$	PL and spreads	Elevation angle terms. Low elevation links interact with more buildings, so θ_{inv} is especially useful for spread tails.
b_{41}, c_{41}, t_{41}	Spreads	Blocker depth, transmitter clearance, and local taller building fraction; they separate open overhead links from obstructed street canyon like links.

3.4.2 Prior design rules

The main prior design rule is that a useful prior is not just a physical looking equation. It is a complete estimator: the formula, fitted calibration, feature units, LoS/NLoS mask, topology features, and evaluation mask must be identical between fitting and inference.

The final prior is strong because it keeps those pieces self contained and auditable under the same city holdout split used for the final model.

The distinction matters most for NLoS. A calibration fitted for one physical formula cannot reliably correct another; local height and density features must have the same units when fitted and when applied; and regime assignment should come from the sample geometry rather than only from a broad city label. Once those constraints were enforced, the prior could separate deterministic large scale structure from the genuinely local residual that the final model later learned.

This also explains why the final network is residual and distribution aware. Blocked NLoS and spread targets are multimodal and heavy tailed, so a single point estimate without a strong anchor tends to average incompatible regimes. HARP-NET CKM therefore keeps the priors as frozen anchors and lets the neural head model bounded residual structure and uncertainty without turning the main method into an unconstrained direct regressor.

Table 3.5: Compact design rules extracted from the prior experiments.

Rule	Effect on the final pipeline
Fit and apply the same formula	The calibration corrects the actual runtime prior, not a neighbouring variant.
Keep units explicit	Distances, heights, topology values, and normalized features have one shared definition across fitting, validation, and test.
Use the metric mask during fitting	The prior is optimized for valid receiver pixels rather than for building cells that are excluded from the final metric.
Hard separate LoS and NLoS	The two regimes keep different physics, feature sets, and calibration parameters when the dataset provides a reliable mask.
Let literature define axes, not all coefficients	Distance, height, frequency, and elevation come from propagation theory; effective reflection and local morphology corrections are fitted from CKM.
Learn bounded residuals after the prior	The final model is allowed to correct local errors without overwriting the calibrated large scale structure.
Treat calibration artifacts as part of the model	Large JSON and coefficient artifacts are not bookkeeping only; they store the height, density, visibility, and morphology-specific decisions that make the prior a competitive baseline.

3.4.3 Literature ingredients and thesis adaptations

The final pipeline is a hybrid between classical propagation models, distribution aware regression, and dataset-specific calibration. Table 3.6 keeps the mapping explicit; the detailed equations and fitting procedures follow in this chapter, covering the attenuation prior, spread prior, and HARP-NET CKM residual model.

Table 3.6: Literature to design mapping used by HARP-NET CKM.

Block	Literature basis	CKM adaptation	New role in this thesis
LoS path loss prior	FSPL and coherent two ray propagation [9, 10].	Effective complex reflection, height bin interpolation, and radial residual fitting on CKM LoS pixels.	Reconstructs the radial interference pattern and supplies the frozen LoS anchor.
NLoS path loss calibration	COST-231/Hata style attenuation, A2G elevation envelopes, and LoS/NLoS separation [11, 12, 13, 14].	Map derived density, height, NLoS support, elevation, and shadowing features with per regime calibration.	Turns a broad empirical curve into a local, topology aware NLoS prior.
Delay and angular spread priors	Log normal spread statistics from 3GPP/WINNER style channel models and UAV elevation trends [12, 17, 47, 28].	Log domain ridge regression with local morphology features and a sparse regime fallback hierarchy.	Provides stable spread anchors without pretending that spreads are smooth deterministic channel attenuation maps.
Final residual model	U-Net style dense prediction, FiLM conditioning, probabilistic regression, and mixture density ideas [18, 46, 34, 29, 30, 31].	Prior channels, explicit masks, residual caps, anchor gates, and a shared multitask trunk.	Learns only the bounded local correction and uncertainty left after the calibrated priors.

3.5 Channel Attenuation Prior and Residual Prediction

The CKM dataset stores the attenuation target under the code name `path_loss`. The final attenuation prior is therefore built from physical path loss formulas, but its dense output is reported in the thesis as channel attenuation. The prior coefficients and calibration tables are fixed before neural training and have two branches.

The LoS branch starts from free space path loss, adds a coherent two ray ground reflection correction, and then applies height dependent bias and radial residual calibration. This captures the ring like structure visible around the centred transmitter. The NLoS branch starts from a blocked link propagation estimate and calibrates it with local morphology, obstruction and elevation features. Both branches are fitted only on training cities. At inference, the explicit LoS/NLoS support mask selects the branch at each ground receiver pixel, producing the fixed attenuation prior map used by HARP-NET CKM.

Chapter 4 reports the quantitative prior audit. The full equations, feature inventory and fitting details are included below.

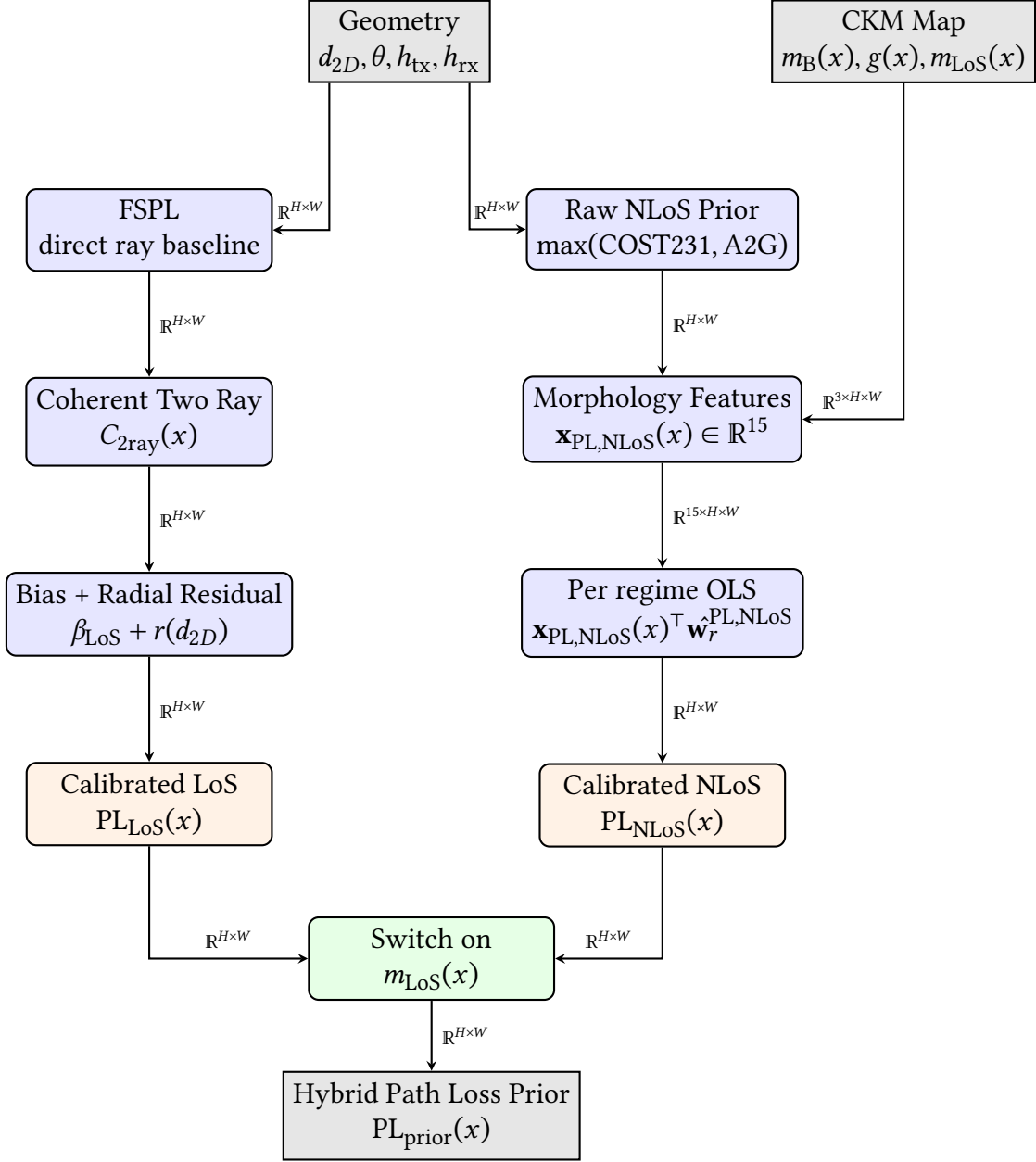


Figure 3.4: Pipeline for the hybrid path loss prior. The LoS branch uses an explicit two ray interference model, while the NLoS branch calibrates a raw prior against map derived morphology features using per regime OLS. The two branches are hard switched by the LoS mask to produce the final frozen prior.

3.5.1 LoS branch: enhanced two ray model

The LoS prior is not a pure FSPL prior. FSPL is kept because it is the direct ray normalization term of the coherent two ray model: it gives the path loss that would be observed if only the direct ray existed. The deployed LoS branch then adds the fitted ground reflection interference term, a height dependent bias, and a smoothed radial residual profile. This differs from the NLoS branch, which uses a longer pixel feature vector and a per regime linear calibration.

FSPL baseline used by the LoS prior

The shared distances d_{LoS} and d_{ref} are defined once in Sec. 3.3. The LoS branch uses the standard free space path loss expression as the direct ray baseline:

$$\text{FSPL}(x) = 32.45 + 20 \log_{10} \left(\frac{d_{\text{LoS}}(x)}{1000} \right) + 20 \log_{10}(f_{\text{MHz}}) \quad (3.37)$$

Enhanced two ray correction

In the two ray model, the received field is the superposition of the direct path and a ground reflected path. Let $\lambda = c/f$ be the wavelength. The complex sum of the two waves at the receiver is:

$$E_{\text{total}} \propto \frac{e^{-j2\pi d_{\text{LoS}}/\lambda}}{d_{\text{LoS}}} + \Gamma \frac{e^{-j2\pi d_{\text{ref}}/\lambda}}{d_{\text{ref}}} \quad (3.38)$$

where Γ is the complex ground reflection coefficient. In the standard textbook derivation, Γ is computed from the Fresnel equations for the ground material. Here, it is replaced by an *effective* complex parameter:

$$\Gamma(h_{\text{tx}}) = \rho(h_{\text{tx}}) e^{-j\phi(h_{\text{tx}})} \quad (3.39)$$

where $\rho \in [0, 1.5]$ is the effective amplitude used during the calibration search and $\phi \in [-\pi, \pi]$ is the fitted phase offset, in radians, applied to the reflected field relative to the direct field. The search range is only a robustness allowance for a fitted CKM parameter, not a claim that a passive ground reflection coefficient exceeds unity; the deployed smoothed values reported later remain below one. Both parameters depend on the UAV height bin.

The two ray path loss correction term in dB is:

$$C_{2\text{ray}}(x) = -20 \log_{10} \left| 1 + \rho(h_{\text{tx}}) \frac{d_{\text{LoS}}(x)}{d_{\text{ref}}(x)} e^{-j \left(2\pi \frac{d_{\text{ref}}(x) - d_{\text{LoS}}(x)}{\lambda} + \phi(h_{\text{tx}}) \right)} \right| \quad (3.40)$$

Term by term interpretation.

- $\rho(h_{\text{tx}})$: amplitude of the ground reflection. A value of $\rho = 0$ recovers pure FSPL. The fitted values are $\rho \approx 0.46$ to 0.56 for heights 10 to 100 m, consistent with a moderate ground reflection coefficient.
- $d_{\text{LoS}}(x)/d_{\text{ref}}(x)$: path amplitude ratio. This is always ≤ 1 since the reflected path is longer; it corrects for the $1/d$ amplitude decay of the reflected ray relative to the direct ray.
- $2\pi(d_{\text{ref}}(x) - d_{\text{LoS}}(x))/\lambda$: the phase difference accumulated due to the extra path length of the reflected wave.
- $\phi(h_{\text{tx}})$: an additional phase offset capturing phase shifts upon reflection (e.g., phase shift at the boundary). The fitted values are $\phi \approx 0$ for all height bins, meaning the fitted extra phase offset is near zero under this sign convention.

- $-20 \log_{10} |\cdot|$: the total field amplitude at the receiver, normalized to the direct path field. When the two waves add in phase, the field amplitude is > 1 , giving a *negative* correction (the signal is *stronger* than FSPL alone); when they cancel, the correction is *positive* (fading loss).

This spatial interference pattern creates the characteristic ring like behavior in the LoS path loss maps: concentric rings of constructive and destructive interference around the transmitter.

Effective two ray calibration

Equation (3.40) keeps the standard coherent two ray field sum: one direct ray and one ground reflected ray are added before conversion to dB. The adaptation in this thesis is not an extra propagation term inside that field sum, but the fitted effective reflection coefficient $\Gamma(h_{\text{tx}}) = \rho(h_{\text{tx}})e^{-j\phi(h_{\text{tx}})}$. Thus ρ and ϕ are calibration parameters fitted from training city LoS pixels in 5 m UAV height bins, not material Fresnel constants from a single flat ground plane [9, 48]. The search grid allowed $\rho \leq 1.5$ for robustness, but the smoothed deployed values remain below unity. The additional terms $\beta_{\text{LoS}}(h_{\text{tx}})$ and $r(d_{2D}, h_{\text{tx}})$ are post fit calibration terms outside the literature two ray formula.

Calibrated LoS prior

The full calibrated LoS prior is:

$$\text{PL}_{\text{LoS}}(x) = \text{FSPL}(x) + C_{2\text{ray}}(x) + \beta_{\text{LoS}}(h_{\text{tx}}) + r(d_{2D}(x), h_{\text{tx}}) \quad (3.41)$$

where:

- $\beta_{\text{LoS}}(h_{\text{tx}})$ is a height dependent bias (dB): a constant correction that shifts the overall two ray prediction for a given UAV height.
- $r(d_{2D}(x), h_{\text{tx}})$ is a radial residual correction: a smooth function of horizontal distance (and height bin) that captures systematic over-/under-prediction of the two ray model as a function of range. It is obtained by smoothing the empirical mean residual in each distance bin after the two ray fit.

Equation (3.41) is applied as a frozen calibrated LoS prior and then clipped to the [20, 180] dB physical path loss range used by the final model. Table 3.7 summarizes the calibrated LoS terms; the shared distance and angle definitions are not repeated here.

Stored / computed term	Mathematical role	Interpretation
$\text{FSPL}(d_{\text{LoS}}, f)$	Direct ray baseline.	Gives the geometric free space loss before any reflected path is added. It is a component of the two ray prior, not the final LoS model by itself.
$\rho(h_{\text{tx}})$	Height interpolated reflected ray amplitude.	Controls how strongly the ground reflected wave contributes relative to the direct wave.
$\phi(h_{\text{tx}})$	Height interpolated phase offset.	Captures the effective reflection phase after fitting to CKM instead of assuming a fixed Fresnel coefficient.
$C_{2\text{ray}}(x)$	Coherent interference correction.	Converts the complex direct plus reflected field sum into an oscillatory dB correction around FSPL.
$\beta_{\text{LoS}}(h_{\text{tx}})$	Height interpolated bias.	Removes a remaining per height offset after the coherent two ray fit.
$r(d_{2D}, h_{\text{tx}})$	Smoothed radial residual profile, clipped to ± 2.5 dB.	Corrects systematic distance dependent residuals not captured by the two ray equation.
$\text{clip}_{[20,180]}$	Output guard.	Keeps the frozen prior inside the physical path loss range used by the final model.

Table 3.7: Term by term inventory of the LoS prior used by the final model. Unlike the NLoS branch, this is not a long pixel feature regression vector; it is a height-interpolated two ray calibration plus a radial residual table.

3.5.2 Calibration of the LoS two ray model

Data preparation

All training samples are used. For each sample, valid LoS pixels are selected as those satisfying:

1. topology label = 0 (ground pixel),
2. $m_{\text{LoS}}(x) > 0$ (marked as LoS),
3. the measured path loss is finite and ≥ 20 dB.

If a sample has more than 2 500 valid LoS pixels, a random subsample of 2 500 pixels is drawn (with a fixed seed). Samples are then grouped by UAV height into height bins of 5 m width:

$$h_{\text{bin}} = 5 \cdot \text{round}\left(\frac{h_{\text{tx}}}{5}\right) \quad (3.42)$$

Each bin is further subsampled to a maximum of 8 000 pixels total.

Grid search for (ρ, ϕ)

For each height bin, (ρ, ϕ) are found by grid search:

$$\phi_{\text{grid}} = \left\{ -\pi + \frac{2\pi q}{N_\phi} : q = 0, 1, \dots, N_\phi - 1 \right\}, \quad N_\phi = 48 \quad (3.43)$$

$$\rho_{\text{grid}} = \{0.00, 0.05, 0.10, \dots, 1.50\} \quad (3.44)$$

For each pair $(\hat{\rho}, \hat{\phi}) \in \rho_{\text{grid}} \times \phi_{\text{grid}}$, and for each pixel x_i in the height bin, define the two ray prediction without bias:

$$\hat{p}_i(\hat{\rho}, \hat{\phi}) = \text{FSPL}(x_i) - 20 \log_{10} \left| 1 + \hat{\rho} \frac{d_{\text{LoS}}(x_i)}{d_{\text{ref}}(x_i)} e^{-j \left(2\pi \frac{d_{\text{ref}}(x_i) - d_{\text{LoS}}(x_i)}{\lambda} + \hat{\phi} \right)} \right| \quad (3.45)$$

The optimal LoS bias for each $(\hat{\rho}, \hat{\phi})$ pair is solved analytically as the empirical mean of residuals:

$$\hat{\beta}_{\text{LoS}}(\hat{\rho}, \hat{\phi}) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{p}_i(\hat{\rho}, \hat{\phi})) \quad (3.46)$$

where y_i is the measured path loss at pixel x_i and N is the number of pixels in the bin. The biased prediction is $\hat{p}_i + \hat{\beta}_{\text{LoS}}$, and the RMSE for the pair is:

$$\text{RMSE}(\hat{\rho}, \hat{\phi}) = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{p}_i(\hat{\rho}, \hat{\phi}) - \hat{\beta}_{\text{LoS}}(\hat{\rho}, \hat{\phi}))^2} \quad (3.47)$$

The best pair is:

$$(\rho^*, \phi^*) = \arg \min_{(\hat{\rho}, \hat{\phi})} \text{RMSE}(\hat{\rho}, \hat{\phi}) \quad (3.48)$$

Post fit Gaussian smoothing across height bins

After fitting all bins independently, a Gaussian-weighted smooth is applied to stabilize sparse high altitude bins. Let $\{h_k\}$ be the fitted height bin centers, $\{(\rho_k^*, \phi_k^*, \beta_{\text{LoS},k}^*)\}$ the per bin fitted values, and $\{N_k\}$ the pixel counts used in fitting. The implementation weights each bin by $\sqrt{N_k}$ rather than by N_k itself, so very populated bins stabilize the curve without completely dominating neighboring sparse bins. The smoothed values are:

$$\tilde{\rho}(h) = \frac{\sum_k w_k(h) \sqrt{N_k} \rho_k^*}{\sum_k w_k(h) \sqrt{N_k}}, \quad w_k(h) = \exp\left(-\frac{(h - h_k)^2}{2\sigma_h^2}\right), \quad \sigma_h = 10 \text{ m} \quad (3.49)$$

and similarly for ϕ and β_{LoS} . The implementation also stores a suspicious-bin mask for bins where both ρ_k^* and $\beta_{\text{LoS},k}^*$ differ noticeably from the smoothed global trend, typically high altitude bins with few samples. That mask is diagnostic metadata: the runtime calibration arrays used by the final residual model are the smoothed ρ , ϕ , and β_{LoS} curves themselves, not a mixture of raw values for ordinary bins and smoothed values only for flagged ones. At inference time the UAV height is not forced to be exactly one of the fitted 5 m bin centers. The implementation linearly interpolates $\rho(h_{\text{tx}})$, $\phi(h_{\text{tx}})$, $\beta_{\text{LoS}}(h_{\text{tx}})$, and the radial residual profile between the two nearest height bins, and clamps to the

nearest endpoint outside the fitted range. This keeps the prior continuous with respect to UAV height instead of introducing discrete jumps every 5 m.

Radial residual correction

After fitting the two ray model, a residual correction is computed separately per height bin and integer radius:

$$r(r_{\text{px}}, h_{\text{tx}}) = \text{clip}\left(\text{GaussSmooth}_{\sigma_r}\left(\mathbb{E}_{h\text{-bin}, r_{\text{px}}}\left[\text{clip}(y - \hat{y}_{2\text{ray}}, -2.5, +2.5)\right]\right), -2.5, +2.5\right) \quad (3.50)$$

where r_{px} is the rounded 2D pixel radius from the transmitter, $\sigma_r = 1.5$ pixels is the Gaussian smoothing radius, and the ± 2.5 dB clip is applied both to individual residual samples during accumulation and to the interpolated profile used at inference. Radius bins with no samples in a height bin fall back to the global radial profile before smoothing. At inference, the profile is linearly interpolated across the stored height bins, then indexed by the pixel-radius map and added to the coherent two ray prediction.

Smoothed deployed parameter values

Table 3.8 shows representative LoS two ray parameters after the post fit Gaussian smoothing across height bins. These are the deployed calibration values used by the prior at inference time, not the unsmoothed per bin grid-search optima.

Table 3.8: Representative smoothed deployed LoS two ray parameters by height bin.

h_{tx} (m)	$\tilde{\rho}$	$\tilde{\phi}$ (rad)	$\tilde{\beta}_{\text{LoS}}$ (dB)
10	0.5605	0.000	-0.037
20	0.5242	0.000	-0.042
30	0.4950	0.000	-0.042
50	0.4632	0.000	-0.047
80	0.4958	0.000	-0.013
100	0.4926	0.000	-0.011
140	0.5308	0.000	-0.028
460	0.8966	-0.030	+1.432

The dominant deployed pattern is $\tilde{\rho} \approx 0.46$ to 0.56 with $\tilde{\phi} \approx 0$ for heights 10 to 100 m, meaning a moderate-amplitude ground reflection with a near zero fitted extra phase offset. Biases are near zero (< 0.05 dB) for these bins. At very high altitudes, the smoothed deployed $\tilde{\rho}$ and $\tilde{\beta}_{\text{LoS}}$ increase because training data is sparser; those bins are stabilized by Gaussian smoothing before deployment.

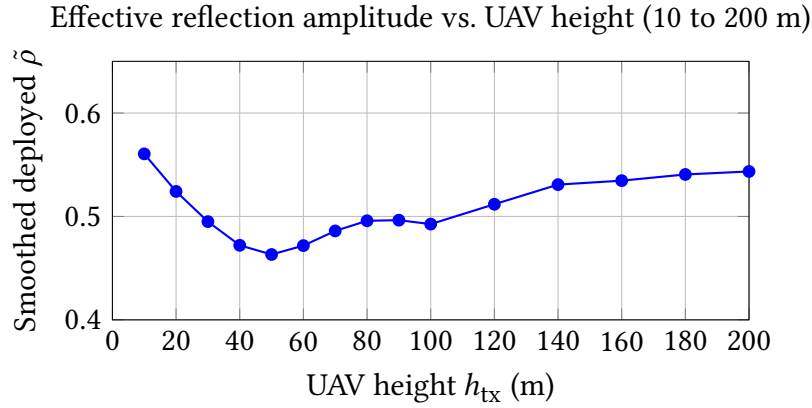


Figure 3.5: Effective ground reflection amplitude $\tilde{\rho}$ as a function of UAV height after post fit smoothing across 5 m height bins. Values are stable in the range 0.46 to 0.56 for 10 to 200 m, then rise at the sparsely sampled very-high altitude bins used by the calibration.

FSPL vs. coherent two ray component ($h_{tx} = 60$ m, $f = 7.125$ GHz)

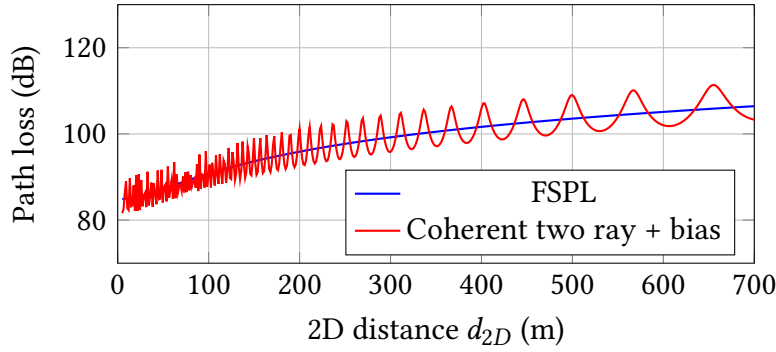


Figure 3.6: Schematic comparison of FSPL and the calibrated coherent two ray component for a 60 m UAV at 7.125 GHz ($h_{tx} = 1.5$ m), zoomed to the 700 m near field that contains the full 513×513 map diagonal. The plotted red curve uses the code loaded 60 m calibration values before adding the radial residual profile, so it shows the constructive/destructive interference term rather than the full final LoS prior.

3.5.3 NLoS branch

The NLoS path loss branch used in the final model is not a pure literature formula. It is a hybrid engineering model built from five calculation blocks.

Raw NLoS propagation prior

Two coarse physical estimates are combined.

COST-231 term. The COST-231 / Hata extension for urban macro cells gives the empirical large scale path loss expression used below [11]. It is included because it is one of the standard references for urban NLoS macrocell loss, even though the CKM scenario is outside its original frequency and base station height range. None of the COST-231 coefficients is therefore treated as a physically transferred absolute predictor at 7.125 GHz; the equation is only a log distance and height sensitive basis that the train only NLoS calibration can bend:

$$\begin{aligned} \text{PL}_{\text{COST231}}(x) = & 46.3 + 33.9 \log_{10}(f_{\text{MHz}}) - 13.82 \log_{10}(h_{\text{tx}}) - a(h_{\text{rx}}) \\ & + (44.9 - 6.55 \log_{10}(h_{\text{tx}})) \log_{10}(d_{\text{km}}) + 3 \end{aligned} \quad (3.51)$$

where $d_{\text{km}} = \max(d_{2D}(x)/1000, 0.001)$ and the implemented mobile antenna correction is the COST-231 small/medium city correction:

$$a(h_{\text{rx}}) = (1.1 \log_{10}(f_{\text{MHz}}) - 0.7) h_{\text{rx}} - (1.56 \log_{10}(f_{\text{MHz}}) - 0.8) \quad (3.52)$$

(3.51) should therefore be read as a calibrated *feature basis*, not as a trusted absolute predictor or as a claim that COST-231 itself transfers to this 7.125 GHz UAV geometry. It contributes the expected log distance growth, transmitter height dependence and receiver height correction of an urban empirical model; the train only calibration fit then adapts that shape to the ray tracing setting used here.

Elevation-aware UAV envelope. The A2G NLoS envelope is used as a conservative elevation dependent shape term for urban air to ground path loss. UAV channel models motivate elevation angle as an important modelling axis because obstruction and shadowing statistics vary with elevation [14, 12]. The particular constants below, however, are not used as a transferred monotonic law. With the positive exponential term, the raw envelope can be larger near overhead; it is therefore only one conservative basis branch that is later combined with COST-231 through a maximum and corrected by the training city per regime NLoS calibration.

$$\text{PL}_{\text{A2G,NLoS}}(x) = \lambda_0(h_{\text{tx}}, f) + \beta_{\text{NLoS}} + A_{\text{NLoS}} \exp\left(-\frac{90 - \theta^\circ(x)}{\tau_{\text{NLoS}}}\right) \quad (3.53)$$

where:

$$\lambda_0(h_{\text{tx}}, f) = 20 \log_{10}\left(\frac{4\pi h_{\text{tx}} f}{c}\right) \quad (3.54)$$

The fixed A2G-envelope constants used by the implementation are: $\beta_{\text{NLoS}} = -16.16$ dB, $A_{\text{NLoS}} = 12.0436$, $\tau_{\text{NLoS}} = 7.52$ deg. These A2G constants are implementation constants in the path loss prior code rather than entries in the calibration JSON files; the JSON artifacts store the fitted LoS two ray curves and the per regime NLoS calibration coefficients. The exponential term is kept only as a conservative elevation envelope: near overhead links approach $\lambda_0 + \beta_{\text{NLoS}} + A_{\text{NLoS}}$, while grazing links approach $\lambda_0 + \beta_{\text{NLoS}}$. The maximum operation and the later per regime calibration decide how much of this branch remains active in the final prior.

The raw NLoS prior is then the conservative (higher) of the two estimates:

$$\text{PL}_{\text{NLoS,raw}}(x) = \max(\text{PL}_{\text{COST231}}(x), \text{PL}_{\text{A2G,NLoS}}(x)) \quad (3.55)$$

The maximum in (3.55) is deliberately pessimistic. If the empirical COST-231 curve predicts stronger attenuation than the A2G envelope, the model keeps the urban macro value; if the elevation aware envelope predicts stronger attenuation, the model keeps the A2G value. This prevents the raw prior from becoming unrealistically optimistic in NLoS streets where neither source model is fully valid on its own.

Raw formula map for NLoS calibration

The NLoS construction above defines only the obstructed branch. The calibration feature $PL^{(0)}(x)$ is the raw formula map used as an anchor for the NLoS regressor. It is not the final channel attenuation prior. The implementation forms it by hard switching between an auxiliary LoS formula helper and the conservative raw NLoS branch:

$$PL^{(0)}(x) = m_{LoS}(x) PL_{LoS,aux}(x) + (1 - m_{LoS}(x)) PL_{NLoS,raw}(x) \quad (3.56)$$

The auxiliary term $PL_{LoS,aux}$ is only a legacy formula helper used to keep $PL^{(0)}$ defined over the whole map. In the final frozen path, LoS output pixels come directly from the separate enhanced two ray prior PL_{LoS} of (3.41); the raw formula map $PL^{(0)}$ in (3.56) is used only as one input feature for the calibrated NLoS branch. The LoS/NLoS mask used here is the explicit CKM mask provided with the dataset and treated as an available side channel. A deployment version would need to obtain an equivalent visibility mask from deterministic geometry or another inexpensive pre-processing step; replacing the supplied mask is not evaluated in this thesis.

Map derived morphology features

The raw hybrid prior $PL^{(0)}(x)$ defined in (3.56) is the scalar formula input to the calibrated NLoS branch. The map derived features below do not define a competing path loss model; they give the linear calibration the local context that the raw formula cannot see: building density, building height, nearby NLoS support, elevation and simple interactions. From $PL^{(0)}(x)$ and the shared topology maps, the following 15 element vector is computed. This follows the same intuition as urban NLoS models based on building rows and street level descriptors [49, 50], and as path profile surrogates that use map derived profile statistics [51], but all variables here are computed directly from the CKM grid and fitted only on training city pixels.

$$\mathbf{x}_{PL,NLoS}(x) = \begin{bmatrix} PL^{(0)}(x)^2 \\ PL^{(0)}(x) \\ \log(1 + d_{2D}(x)) \\ \delta_{15}(x) \\ \delta_{41}(x) \\ h_{15}(x) \\ h_{41}(x) \\ \delta_{41}(x) \log(1 + d_{2D}(x)) \\ n_{15}(x) \\ n_{41}(x) \\ n_{41}(x) \log(1 + d_{2D}(x)) \\ \sigma_{shadow}(x) \\ \theta_{norm}(x) \\ n_{41}(x) \theta_{norm}(x) \\ 1 \end{bmatrix} \quad (3.57)$$

The lowercase bold notation is intentional: $\mathbf{x}_{\text{PL,NLoS}}(x)$ is a per pixel feature vector. When vectors from many pixels in one regime are stacked for calibration, they form the uppercase design matrix $\mathbf{X}_r$. For compact table notation, let $\ell_d(x) = \log(1 + d_{2D}(x))$. Shared support maps such as d_{2D} , θ_{norm} , δ_k , h_k , and n_k are defined once in Sec. 3.3; Table 3.9 only states how they enter the NLoS calibration.

Term	Mathematical role	Physical interpretation
$\text{PL}^{(0)}(x)^2$	Quadratic basis term.	Lets the linear calibration bend the raw path loss curve when the raw model is systematically curved.
$\text{PL}^{(0)}(x)$	Linear raw prior term.	Keeps the calibrated model anchored to the physical prior instead of relearning path loss from scratch.
$\log(1 + d_{2D}(x))$	Smooth distance basis.	Mirrors log distance propagation laws used in classical models [9, 11].
$\delta_{15}(x), \delta_{41}(x)$	Local building density averages.	Count how much nearby urban structure can block, reflect or diffract the signal.
$h_{15}(x), h_{41}(x)$	Density weighted local height averages.	Distinguish low dense blocks from taller blocks with stronger obstruction.
$\delta_{41}(x)\ell_d(x)$	Density-distance interaction.	Allows distance slope to increase in dense areas.
$n_{15}(x), n_{41}(x)$	Local NLoS support fractions.	Measure how much nearby ground is geometrically shielded from the UAV.
$n_{41}(x)\ell_d(x)$	NLoS distance interaction.	Allows blocked streets to accumulate loss differently with range.
$\sigma_{\text{shadow}}(x)$	Elevation dependent shadowing proxy.	Encodes the stronger low elevation variability reported in A2G channel models [14, 12].
$\theta_{\text{norm}}(x)$	Normalized elevation angle.	Separates grazing links from near overhead links.
$n_{41}(x)\theta_{\text{norm}}(x)$	NLoS elevation interaction.	Allows the NLoS correction to depend on whether shielding occurs at low or high elevation.
1	Bias term.	Lets each regime learn a constant offset after all physical terms are included.

Table 3.9: Term by term inventory of the NLoS calibration vector.

Shadowing proxy. The only additional scalar not already defined in the shared support map section is the shadowing proxy used by the NLoS calibration:

$$\sigma_{\text{LoS}}(x) = \alpha_{\text{LoS}} (90 - \theta^\circ(x))^{\mu_{\text{LoS}}} \quad (3.58)$$

$$\sigma_{\text{NLoS}}(x) = \alpha_{\text{NLoS}} (90 - \theta^\circ(x))^{\mu_{\text{NLoS}}} \quad (3.59)$$

with fitted values $\alpha_{\text{LoS}} = 0.0272$, $\mu_{\text{LoS}} = 0.7475$, $\alpha_{\text{NLoS}} = 2.3197$, and $\mu_{\text{NLoS}} = 0.2361$. The blended proxy is:

$$\sigma_{\text{shadow}}(x) = m_{\text{LoS}}(x) \sigma_{\text{LoS}}(x) + (1 - m_{\text{LoS}}(x)) \sigma_{\text{NLoS}}(x) \quad (3.60)$$

This acts as a spatially varying uncertainty proxy rather than a measured shadow fading standard deviation. Higher values occur at low elevation angles, where propagation conditions are more difficult; the LoS/NLoS blend lets the same feature have a different scale in the two visibility regimes.

The interaction terms in Table 3.9 are retained because they let the calibration change the distance or elevation slope in dense or blocked regions. Their coefficients should be interpreted together with the full fitted vector, not as independent physical constants.

Per regime linear calibration

The regime r is identified by three keys:

- *Coarse topology*: class_{PL} from (3.33),
- *Propagation state*: LoS or NLoS,
- *Antenna height bin*: $\text{ant_bin}(h_{\text{tx}})$ from (3.35).

This gives up to $3 \times 2 \times 3 = 18$ regimes.

The stored calibration for this legacy mixed-prior map still contains both LoS and NLoS regime coefficients. However, in the final frozen path loss assembly only the NLoS pixels from this calibrated map are used, because LoS pixels are replaced by the separate enhanced two ray prior. The regime labels are map level labels, not extra per pixel features. Their derivation from $\bar{\delta}_{\text{map}}$, $\bar{h}_{\text{map}}$ and h_{tx} is given once in Sec. 3.3 so that the path loss and spread priors use the same height bin convention.

The calibrated prediction for regime r is:

$$\widehat{\text{PL}}_r(x) = \mathbf{x}_{\text{PL,NLoS}}(x)^\top \mathbf{w}_r^{\text{PL,NLoS}} = \sum_{j=1}^{15} x_j(x) w_{r,j}^{\text{PL,NLoS}} \quad (3.61)$$

How $\mathbf{w}_r^{\text{PL,NLoS}}$ is fitted. Across all training samples whose regime label equals r , a linear regression is run:

$$\hat{\mathbf{w}}_r^{\text{PL,NLoS}} = \arg \min_{\mathbf{w} \in \mathbb{R}^{15}} \sum_{x \in \mathcal{S}_r} (y(x) - \mathbf{x}_{\text{PL,NLoS}}(x)^\top \mathbf{w})^2 \quad (3.62)$$

where $\mathcal{S}_r$ is the set of all valid pixels in training samples belonging to regime r . This ordinary least squares problem has the closed form solution, when $\mathbf{X}_r^\top \mathbf{X}_r$ is nonsingular:

$$\hat{\mathbf{w}}_r^{\text{PL,NLoS}} = (\mathbf{X}_r^\top \mathbf{X}_r)^{-1} \mathbf{X}_r^\top \mathbf{y}_r \quad (3.63)$$

where $\mathbf{X}_r$ stacks the feature vectors and $\mathbf{y}_r$ stacks the measured path loss values. The runtime code does not refit this model; it loads the stored coefficient vectors from the NLoS regime calibration artifact. At inference, if the exact antenna height key is absent, the implementation only tries alternative antenna bins in the order current, mid, low, high while keeping the same path loss prior city type and LoS/NLoS label. It does not use the broader cross-topology fallback hierarchy of the spread prior calibration.

The terms in (3.63) are standard linear least squares quantities. The matrix $\mathbf{X}_r^\top \mathbf{X}_r$ is the *Gram matrix*; its diagonal entries measure the energy of each feature inside regime r , and its off-diagonal entries measure correlations between features. The vector $\mathbf{X}_r^\top \mathbf{y}_r$ measures how strongly each feature correlates with measured path loss. The inverse maps those feature target correlations into the fitted coefficient vector $\hat{\mathbf{w}}_r^{\text{PL,NLoS}}$. This is the classical ordinary least squares solution, but applied separately per propagation regime rather than globally. Because the feature vector intentionally contains correlated terms, the important object is the stored least squares coefficient vector; if a normal equation matrix is rank deficient or ill conditioned, (3.63) should be understood as the full-rank notation for the corresponding least squares or pseudoinverse solution, not as a requirement that every calibration block be perfectly invertible.

An example fitted coefficient vector for the `mixed_midrise|NLoS|high_ant` regime is:

$$\mathbf{w}_{\text{mixed|NLoS|high}}^{\text{PL,NLoS}} = \begin{bmatrix} -0.000338, & -0.0266, & 8.233, & -1.452, & -11.659 \\ & 3.508, & -5.611, & 1.904, & 0.926, & 12.084 \\ -1.628, & -7.487, & -12.665, & -10.959, & 121.480 \end{bmatrix}^\top \quad (3.64)$$

The large bias weight and the sizeable morphology, elevation and raw prior weights should be interpreted together as one fitted calibration vector, not as independent physical constants. Their magnitudes mainly reflect the scale and correlation of the 15 input features in this NLoS regime. This makes the path loss prior an interpretable calibrated surrogate rather than a pure first-principles propagation formula: the two ray and A2G terms set the shape, while per regime fitting adapts the coefficients to the CKM simulator and to the dataset's stored masks.

NLoS clipping

$$\text{PL}_{\text{NLoS}}(x) = \text{clip}(\mathbf{x}_{\text{PL,NLoS}}(x)^\top \mathbf{w}_r^{\text{PL,NLoS}}, 20, 180) \quad (3.65)$$

Hard clipping to $[20, 180]$ dB prevents the linear model from producing physically unreasonable values in the obstructed branch.

3.5.4 Final channel attenuation prior assembly

The calibrated LoS and NLoS branches are assembled only after both branches have been defined. The final frozen channel attenuation prior switches with the CKM LoS mask:

$$\text{PL}_{\text{prior}}(x) = \begin{cases} \text{PL}_{\text{LoS}}(x), & m_{\text{LoS}}(x) > 0.5 \\ \text{PL}_{\text{NLoS}}(x), & \text{otherwise} \end{cases} \quad (3.66)$$

The quantitative audit of this prior is reported in Chapter 4. This assembly is the attenuation input later supplied to HARP-NET CKM.

3.6 Delay and Angular Spread Prior and Residual Prediction

Angular spread uses the shared spread prior described below. Its final prediction is the fixed angular spread prior plus the bounded residual correction predicted by HARP-NET CKM. The target is reported in degrees and is evaluated only on valid ground receiver pixels.

Delay spread uses the same shared spread prior machinery, but it is evaluated in nanoseconds and is more sensitive to rare high delay pixels. The model therefore uses the prior as a stable baseline and learns only a bounded correction.

For both spread targets, the prior is fitted in $\log(1 + y)$ space because the maps are nonnegative, strongly skewed, and contain near zero LoS mass plus heavier NLoS tails. The raw spread estimate combines propagation state, distance, elevation and local morphology. A train only ridge calibration then specializes this estimate by metric, topology class, LoS/NLoS state and transmitter height bin, with fallback keys for sparse regimes. The calibrated prior is inverse transformed to native units before it is compared with ray-traced ground truth or passed to HARP-NET CKM.

The full spread feature vector, ridge objective and fallback structure are given below.

3.6.1 Physical motivation for log domain modeling

Delay spread (DS) and angular spread (AS) are nonnegative, often zero inflated, and heavy tailed. Their empirical distributions are commonly modeled in the logarithmic domain in standard channel models [12, 17]. For a spread variable y , define:

$$z(x) = \log(1 + y(x)), \quad \hat{y}(x) = \exp(\hat{z}(x)) - 1 \quad (3.67)$$

For two pixels with spread values in original units y_a and y_b , their difference in transformed space is

$$z_a - z_b = \log\left(\frac{1 + y_a}{1 + y_b}\right). \quad (3.68)$$

Thus, a fixed error in z corresponds approximately to a fixed multiplicative error in $1 + y$. This is more appropriate than least squares in the original measured units because large NLoS tails would otherwise dominate the fit and the near zero LoS mass would be ignored. Throughout this spread section, original units means nanoseconds for delay spread and degrees for angular spread, as stored in the CKM targets. The calibrated spread prior is still a deterministic log domain ridge estimator; it is not claiming to estimate the full spike plus tail probability distribution.

3.6.2 Pixel wise notation for the spread prior fit

Let n index a valid ground pixel from a training map. For a selected metric $m \in \{\text{DS}, \text{AS}\}$, define the measured target $y_{m,n}$, the transformed target $z_{m,n} = \log(1 + y_{m,n})$, and the raw log prior value $z_{m,n}^{(0)}$. The calibrated prior is linear in a 23-dimensional feature vector:

$$\hat{z}_{m,n} = \mathbf{x}_{m,n}^\top \mathbf{w}_{m,r(n)}, \quad \mathbf{x}_{m,n} \in \mathbb{R}^{23}, \quad \mathbf{w}_{m,r} \in \mathbb{R}^{23}. \quad (3.69)$$

The regime $r(n)$ is a discrete key containing the metric, topology class, LoS/NLoS state and UAV height bin. In other words, the spread prior does not fit one global line through

all pixels. It fits a family of small linear models, each one specialized to a propagation regime.

3.6.3 Raw spread prior in log domain

$$z^{(0)}(x) = \log\left(1 + b_{\text{LoS/NLoS}}^{\text{orig}}(x)\right) + b_{\text{topology}} + a_1 \log(1 + d_{2D}) + a_2 \theta_{\text{inv}} + a_3 \delta_{41} + a_4 h_{41} + a_5 n_{41} + a_6 n_{41} \theta_{\text{inv}}. \quad (3.70)$$

where $\theta_{\text{inv}}(x) = 1 - \theta_{\text{norm}}(x)$ (high value at low elevation), and $b_{\text{LoS/NLoS}}^{\text{orig}}$ is the LoS or NLoS anchor in original measured units. The implementation first converts this anchor to $\log(1 + b)$, then adds the remaining log domain terms. After (3.70), the raw log prior is clipped to $[0, 8]$, inverse transformed with $\exp(\cdot) - 1$, and clipped to the metric upper bound. For reproducibility, the $z^{(0)}(x)$ entries used later in $\mathbf{x}_{\text{SP}}$ denote the log value recovered from this clipped raw prior in original units, i.e., $\log(1 + \text{raw_prior}_{\text{clipped}}(x))$, not an unconstrained copy of (3.70).

The raw prior is not a 3GPP formula copied directly into CKM. Instead, it takes the 3GPP/WINNER lesson that spread statistics should be separated by propagation state and represented in log space [12, 17], then adds CKM specific map morphology terms. UAV measurement studies also motivate the elevation angle dependence: low elevation links usually interact with more rooftops and street canyons than near overhead links [52, 53]. The development repository documents this spread module as a transparent non neural baseline whose purpose is to measure how much of DS/AS can already be explained by geometry, morphology and lightweight calibration [54]. In those notes, physical intuition means the qualitative ordering imposed before calibration: which variables should increase spread and which propagation regimes should start higher. That ordering has four sources. First, 3GPP and WINNER model delay and angular spreads as log domain large scale parameters. Second, UAV A2G measurements show that spread statistics vary with link distance, flight height and elevation angle. Third, LoS links are dominated by the direct ray whereas NLoS links collect delayed and angularly separated reflected or diffracted components. Fourth, local building density, height and NLoS support are map proxies for available scatterers and blockers. The raw constants encode these expected signs and relative magnitudes, while the per regime ridge stage is the part fitted from training city pixels. The exact constants used by the implementation are the entries of METRIC_SPECS and TOPOLOGY_PRIOR_BIAS_LOG; Table 3.10 is the single inventory of those fixed constants. They are not the fitted ridge weights $\mathbf{w}_{m,r}$, which are stored separately in the calibration file.

Table 3.10: Raw prior constants fixed before the ridge calibration in (3.70).

Quantity	Delay spread	Angular spread	Meaning
$b_{\text{LoS}}^{\text{orig}}$	4.0 ns	3.0°	LoS anchor in original units, converted to $\log(1 + b)$.
$b_{\text{NLoS}}^{\text{orig}}$	11.0 ns	7.0°	NLoS anchor in original units, converted to $\log(1 + b)$.
$a_1, \dots, a_6$	0.11, 0.62, 0.48, 0.32, 0.86, 0.60	0.05, 0.52, 0.60, 0.38, 0.58, 0.35	Fixed weights multiplying $\log(1 + d_{2D})$, θ_{inv} , δ_{41} , h_{41} , n_{41} , and $n_{41} \theta_{\text{inv}}$, respectively.
b_{topology}	0.00, 0.18, 0.06, 0.22, 0.24, 0.34	0.00, 0.09, 0.05, 0.16, 0.18, 0.24	Log domain topology offsets in the six class order open sparse low rise, open sparse vertical, mixed compact low rise, mixed compact mid rise, dense block mid rise, dense block high rise.

Every term in (3.70) has a physical interpretation:

- $b_{\text{LoS/NLoS}}^{\text{orig}}$ is the base spread level for the propagation state, in original measured units. NLoS starts higher because blocked paths arrive after additional reflections and diffractions.
- b_{topology} is a morphology offset. Dense and high rise topologies receive larger offsets because they contain more reflectors.
- $\log(1 + d_{2D})$ gives a smooth range dependence. Longer links have more opportunity to encounter delayed or angularly separated paths.
- θ_{inv} increases at low elevation angle, where the UAV to ground ray intersects more urban structure before reaching the receiver.
- δ_{41} , h_{41} , and n_{41} describe local building density, mean building height, and NLoS support in a 41×41 window. These are direct map derived proxies for multipath richness.
- $n_{41}\theta_{\text{inv}}$ is an interaction term. It activates when low elevation geometry and local obstruction occur together.

3.6.4 Calibrated spread prior

Spread feature vector and design matrix

$$\mathbf{x}_{\text{SP}}(x) = \begin{bmatrix} z^{(0)}(x)^2, z^{(0)}(x), \log(1 + d_{2D}), \theta_{\text{norm}}, \theta_{\text{inv}} \\ h_{\text{norm}}, h_{\text{norm}}^2, \delta_{15}, \delta_{41}, h_{15}, h_{41} \\ n_{15}, n_{41}, n_{41} \log(1 + d_{2D}), \delta_{41}\theta_{\text{inv}}, b_{41}, 1 \\ c_{41}, t_{41}, \theta_{\text{norm}}\delta_{41}, c_{41}\theta_{\text{inv}}, t_{41}\delta_{41}, t_{41}n_{41} \end{bmatrix}^{\top} \quad (3.71)$$

For reading, the vector in (3.71) can be grouped into five conceptual blocks: the raw prior block $\mathbf{x}_{\text{prior}}$, the range/elevation/height block $\mathbf{x}_{\text{geom}}$, the local building block $\mathbf{x}_{\text{morph}}$, one bias term, and the interaction block $\mathbf{x}_{\text{int}}$. The displayed order follows the implementation, so these blocks are not all contiguous in the column vector. Table 3.11 states the membership explicitly.

Block	Entries from $\mathbf{x}_{\text{SP}}(x)$	Purpose
$\mathbf{x}_{\text{prior}}$	$z^{(0)}(x)^2, z^{(0)}(x)$	Raw prior anchor and curvature.
$\mathbf{x}_{\text{geom}}$	$\log(1 + d_{2D}), \theta_{\text{norm}}, \theta_{\text{inv}}, h_{\text{norm}}, h_{\text{norm}}^2$	Range, elevation and UAV height.
$\mathbf{x}_{\text{morph}}$	$\delta_{15}, \delta_{41}, h_{15}, h_{41}, n_{15}, n_{41}, b_{41}, c_{41}, t_{41}$	Local urban morphology and blockage.
Bias	1	Regime offset.
$\mathbf{x}_{\text{int}}$	$n_{41} \log(1 + d_{2D}), \delta_{41}\theta_{\text{inv}}, \theta_{\text{norm}}\delta_{41}, c_{41}\theta_{\text{inv}}, t_{41}\delta_{41}, t_{41}n_{41}$	Joint geometry and morphology effects.

Table 3.11: Block membership of the 23 entries in the spread calibration vector.

The direct entries are not redefined here: distances, angles, height bins and local morphology terms are the shared support quantities of Sec. 3.3. The useful distinction is that h_{norm} is a broadcast transmitter height scalar, whereas b_{41} , c_{41} and t_{41} are deterministic 41×41 map summaries. The $z^{(0)}$ and $(z^{(0)})^2$ entries anchor the fit to the raw log prior and let the calibration bend it quadratically, while the constant entry 1 gives the selected regime an additive offset. Some entries are intentionally correlated: for instance, $z^{(0)}(x)^2$

is determined by $z^{(0)}(x)$, and $t_{41}(x)n_{41}(x)$ is tied to both taller building and NLoS support features. Individual fitted coefficients should therefore not be read as isolated physical constants; they are a calibrated basis.

Only the interaction block needs an extra inventory because those terms combine two physical effects at once:

Term	Mathematical role	Physical interpretation
$n_{41} \log(1 + d_{2D})$	NLoS distance interaction.	Lets range act differently in blocked streets.
$\delta_{41}\theta_{\text{inv}}$	Density elevation interaction.	Activates dense, low elevation morphology.
$\theta_{\text{norm}}\delta_{41}$	Overhead density interaction.	Allows dense areas to change under high elevation.
$c_{41}\theta_{\text{inv}}$	Clearance elevation interaction.	Couples rooftop clearance with grazing geometry.
$t_{41}\delta_{41}$	Tall building density interaction.	Highlights dense blocks with buildings above the UAV.
$t_{41}n_{41}$	Tall building NLoS interaction.	Activates where tall buildings and shielding coincide.

Table 3.12: Interaction terms in the spread calibration vector.

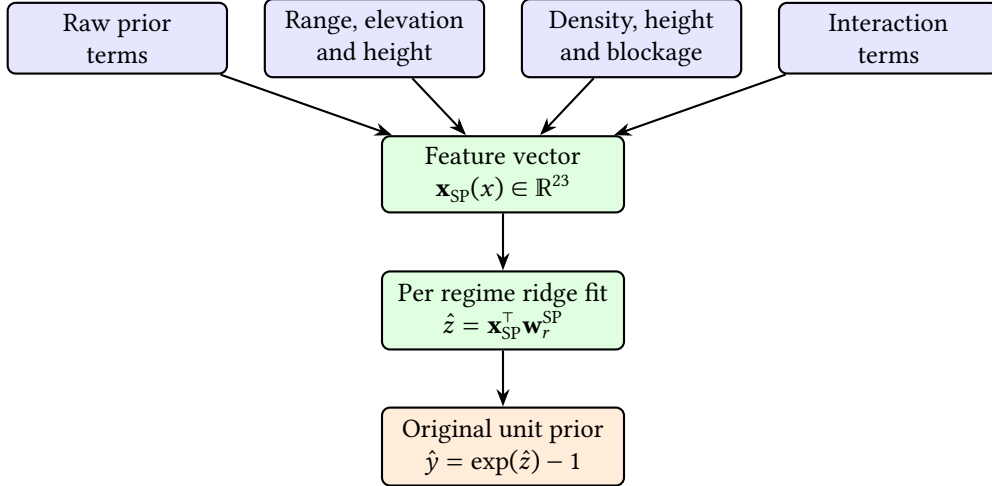


Figure 3.7: Feature grouping used by the calibrated spread prior. The calibrated model is linear in $\mathbf{x}_{\text{SP}}$, but the features include nonlinear transforms and interactions selected from propagation intuition.

The spread vector reuses the shared support features defined in Sec. 3.3: h_{norm} , δ_k , h_k , n_k , b_{41} , c_{41} , t_{41} , θ_{norm} , and θ_{inv} . Their definitions are not repeated here. The six topology labels of (3.34) are map level routing keys for the ridge models, not extra per pixel morphology features.

Ridge calibration by regime

The regime r for the spread prior is keyed by metric (DS or AS), class_{SP} as defined in (3.34), propagation state (LoS or NLoS), and ant_bin as defined in (3.35). This gives up to $2 \times 6 \times 2 \times 3 = 72$ regimes.

Valid pixels are subsampled at rate $p = 0.02$ (2%). The ridge regression:

$$\hat{\mathbf{w}}_r^{\text{SP}} = (\mathbf{X}_r^\top \mathbf{X}_r + \lambda \mathbf{D}^\top \mathbf{D})^{-1} \mathbf{X}_r^\top \mathbf{z}_r, \quad \lambda = 10^{-3} \quad (3.72)$$

Here $\mathbf{X}_r \in \mathbb{R}^{N_r \times 23}$ stacks one row $\mathbf{x}_{\text{SP},n}^\top$ per sampled pixel in regime r , and $\mathbf{z}_r \in \mathbb{R}^{N_r}$ stacks the corresponding transformed targets. The coefficient vector $\hat{\mathbf{w}}_r^{\text{SP}}$ minimizes the

regularized least squares problem

$$\hat{\mathbf{w}}_r^{\text{SP}} = \arg \min_{\mathbf{w} \in \mathbb{R}^{23}} \|\mathbf{X}_r \mathbf{w} - \mathbf{z}_r\|_2^2 + \lambda \|\mathbf{D} \mathbf{w}\|_2^2, \quad (3.73)$$

where $\mathbf{D} = \text{diag}(d_1, \dots, d_{23})$, with $d_{23} = 0$ and $d_i = 1$ for $i \neq 23$. Thus, the final interaction term $t_{41}n_{41}$, which is the last entry of (3.71), is left unpenalized by the ridge term in the frozen calibration used by the final model. This is an implementation detail of the stored artifact, not a separate physical assumption. The regularizer is small, but it prevents unstable coefficients when a regime contains few valid pixels or when two morphology features are strongly correlated.

Term by term, $\mathbf{X}_r^\top \mathbf{X}_r$ is the feature correlation matrix for regime r , $\lambda \mathbf{D}^\top \mathbf{D}$ adds a small diagonal stabilizer to every coefficient except the final $t_{41}n_{41}$ coefficient, and $\mathbf{X}_r^\top \mathbf{z}_r$ is the feature target correlation vector in log spread space. This is ridge regression: it is the ordinary least squares fit plus a penalty that discourages very large coefficients. The penalty is important here because many morphology features are related; for example, δ_{41} , h_{41} , b_{41} and t_{41} all increase in dense high rise areas.

Sufficient statistics are accumulated incrementally:

$$\mathbf{A}_r += \mathbf{X}_{r,s}^\top \mathbf{X}_{r,s} \quad \forall s \quad (3.74)$$

$$\mathbf{v}_r += \mathbf{X}_{r,s}^\top \mathbf{z}_{r,s} \quad \forall s \quad (3.75)$$

so the full design matrix is never materialised in memory. This is important in practice because the calibration artifacts are already large even after compression into fitted coefficients and profiles. The path loss LoS calibration JSON reused by the final model is approximately 200 000 lines, and the spread calibration must be rerun separately for delay spread and angular spread whenever the feature set changes.

Fallback hierarchy. During fitting, each sampled pixel updates the exact accumulator and the four broader fallback accumulators listed below. A coefficient vector is written only for accumulators with at least $4 \times 23 = 92$ sampled pixels. During inference, the implementation tries the same keys in this order and uses the first one present:

1. exact regime: (metric, topology class, LoS/NLoS, antenna bin),
2. same topology class + LoS/NLoS + *all* antenna heights,
3. same topology class + *all* LoS states + all antenna heights,
4. global topology + LoS/NLoS + all antenna heights,
5. fully global fallback: (metric, global, all_los, all_ant), where all_los is the implementation key for all visibility states, not LoS only pixels.

Inverse transform and clipping

$$\hat{y}(x) = \exp(\hat{z}(x)) - 1, \quad \hat{z}(x) = \mathbf{x}_{\text{SP}}(x)^\top \hat{\mathbf{w}}_r^{\text{SP}} \quad (3.76)$$

Regime dependent clipping is applied after the inverse transform:

$$\hat{y}_{\text{clip}}(x) = \min(y_{\text{max},r}, \max(0, \exp(\hat{z}(x)) - 1)), \quad (3.77)$$

with delay spread clipped to $[0, 400]$ ns and angular spread clipped to $[0, 15^\circ]$ in LoS or $[0, 90^\circ]$ in NLoS. The asymmetric angular limit is deliberate: LoS angular spread is dominated by the direct path and is physically closer to a near zero spike plus small reflections, whereas NLoS angular spread can occupy a much wider support.

3.6.5 Delay and angular spread prior pipeline

The topology class RMSE tables for the calibrated spread priors are reported in Chapter 4. This subsection keeps the final pipeline diagram and the method level data flow used by both spread modalities.

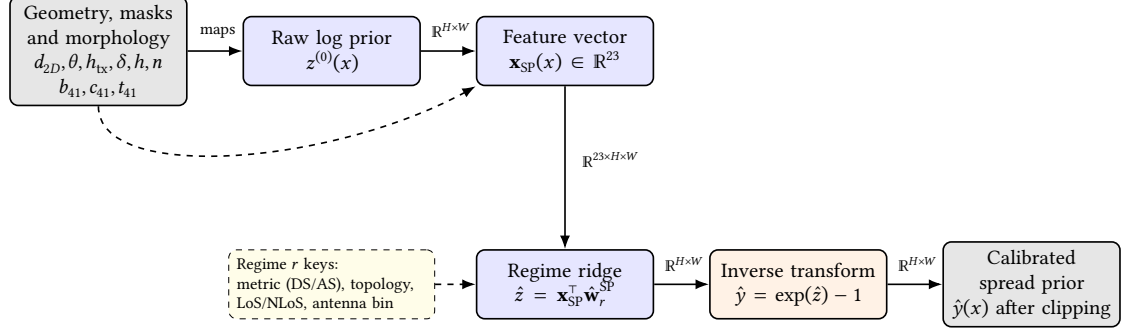


Figure 3.8: Pipeline for the delay and angular spread priors. Both spread modalities share this identical linear architecture, predicting expected spread in the log domain through ridge calibration by regime over 23 spatial and geometric features.

3.7 HARP-NET CKM Neural Residual Model

HARP-NET CKM receives nine spatial channels plus a separate scalar transmitter height conditioning path. The channels are the normalized topology map, LoS mask, NLoS mask, ground mask, combined attenuation prior, LoS attenuation prior, calibrated NLoS attenuation branch, delay spread prior and angular spread prior. The scalar height is encoded with sinusoidal features and injected through FiLM layers rather than copied as a constant image.

The network is a shared GroupNorm U-Net with task specific residual heads. For each target and visibility region, the head predicts bounded residual components around the fixed prior baseline. The probabilistic parameterization is kept because it represents residual tails during training, but the reported maps are deterministic: channel attenuation is read in native dB space, while delay spread and angular spread are predicted in $\log(1+y)$ model space and then mapped back to ns or degrees. The explicit LoS/NLoS masks select the final regional branch at each valid ground receiver pixel.

Training uses the same city holdout split, receiver mask and pixel weighted RMSE rule defined in Chapter 4. The selected final stage is RMSE/MAE oriented, with small anchor and prior guard terms so that the residual model improves the prior baselines without freely overwriting them. The full architecture diagram, residual assembly equations and loss family are included below.

3.7.1 Experimental protocol and reporting status

HARP-NET CKM follows the same strict split philosophy as the recent distribution first models. The split is a city holdout split with `split_seed = 42`, `val_ratio = 0.15`, and `test_ratio = 0.15`. Metrics and losses are computed only on valid ground pixels, using the target specific no data masks and the explicit LoS/NLoS masks. The model input contains nine spatial channels:

1. normalized topology,
2. LoS mask,

3. NLoS mask,
4. ground mask,
5. combined path loss prior,
6. LoS path loss prior,
7. calibrated NLoS path loss branch map,
8. delay spread prior,
9. angular spread prior.

The seventh channel is named `path_loss_nlos_prior` in the code, but it should be read carefully. It is the calibrated mixed/NLoS path loss map computed by the per regime path loss calibration, not a map that has been zeroed outside NLoS pixels. In the final combined path loss prior and in the final output selection, only the explicit LoS/NLoS masks decide which regional branch is used.

The three mask channels are passed as binary maps. The topology channel is masked to ground pixels and divided by 90 m; the three path loss prior channels are divided by 180 dB; and the delay and angular spread priors are passed as $\log(1 + \text{prior}) / \log(1 + 400)$ and $\log(1 + \text{prior}) / \log(1 + 90)$, respectively. These scaling constants only affect the neural network input representation; the residual targets and reported metrics remain in the native units.

The scalar UAV height is not appended as a constant image channel. It is encoded with a sinusoidal embedding and injected into the network through FiLM conditioning. The selected final configuration uses a shared encoder decoder with `base_width = 128`, `cond_dim = 160`, three GMM components per task and region, α -gate bias -2 , and fixed residual caps specified separately for path loss, delay spread and angular spread.

The selected final model reported in Chapter 4 uses these frozen path loss and spread priors as anchors. This section therefore documents both the frozen prior inputs and the residual architecture used by the final evaluation.

3.7.2 HARP-NET CKM Residual Head

The final model introduces a joint multitask residual mixture head architecture that emits a *Gaussian Mixture Model (GMM)*-parameterized residual head at each pixel, rather than only a direct scalar output. For each task $k \in \{\text{PL}, \text{DS}, \text{AS}\}$, region $r \in \{\text{LoS}, \text{NLoS}\}$, component $j \in \{1, 2, 3\}$, and valid pixel x , the model predicts a mixture weight, a mean and a standard deviation. Channel attenuation is modeled in native dB space, while delay spread and angular spread are modeled in transformed $\log(1 + y)$ space inside the probabilistic head and converted back to native units for deterministic metrics. In the selected final fine tuning stage the same GMM parameterization is kept, but the probabilistic losses are disabled; therefore the final reported claim is based on the deterministic mixture mean map rather than on calibrated predictive densities.

The native residual caps depend on task and region:

$$\delta_{\max,k,r} = \begin{cases} 7 \text{ dB}, & k = \text{PL}, r = \text{LoS}, \\ 25 \text{ dB}, & k = \text{PL}, r = \text{NLoS}, \\ 20 \text{ ns}, & k = \text{DS}, r = \text{LoS}, \\ 20 \text{ ns}, & k = \text{DS}, r = \text{NLoS}, \\ 17^\circ, & k = \text{AS}, r = \text{LoS}, \\ 15^\circ, & k = \text{AS}, r = \text{NLoS}. \end{cases} \quad (3.78)$$

The bounded native residual is assembled from a global map level offset and a local pixel level correction:

$$\Delta_{k,r,j}(x) = \text{clip}\left(\delta_{k,r,j}^{(g)} + \delta_{k,r}^{(\ell)}(x), -\delta_{\max,k,r}, \delta_{\max,k,r}\right). \quad (3.79)$$

The mean of each mixture component is then anchored to the frozen prior:

$$\mu_{k,r,j}^{\text{native}}(x) = y_k^{\text{prior}}(x) + \alpha_{k,r}(x)\Delta_{k,r,j}(x), \quad (3.80)$$

where:

- $y_k^{\text{prior}}(x)$ is the per task frozen prior selected inside the shared output loop, then broadcast to the LoS and NLoS residual branches,
- $\delta_{k,r,j}^{(g)}$ is a map level residual offset for component j ,
- $\delta_{k,r}^{(\ell)}(x)$ is a per pixel residual refinement shared across components,
- $\alpha_{k,r}(x) \in (0, 1)$ is a learned spatial gate, initialized with bias -2 , so the model starts close to the prior,
- the residual is bounded to $\pm\delta_{\max,k,r}$ to prevent the DNN from deviating catastrophically from the prior.

For the spread tasks, the native anchored mean is mapped into transformed space before the Gaussian likelihood:

$$\mu_{k,r,j}(x) = \begin{cases} \mu_{k,r,j}^{\text{native}}(x), & k = \text{PL}, \\ \log(1 + \max\{\mu_{k,r,j}^{\text{native}}(x), 0\}), & k \in \{\text{DS}, \text{AS}\}. \end{cases} \quad (3.81)$$

The DNN also predicts per pixel mixture probabilities $p_{k,r,j}(x)$ and combines global and local standard deviation terms into the component standard deviation:

$$\sigma_{k,r,j}(x) = \sqrt{\left(\sigma_{k,r,j}^{(g)}\right)^2 + \left(\tilde{\sigma}_{k,r}^{(\ell)}(x)\right)^2}. \quad (3.82)$$

The Gaussian variance used by the density is therefore $\sigma_{k,r,j}^2(x)$. Both standard deviation terms are constrained by sigmoid transforms to lie between $\sigma_{\min} = 0.05$ and $\sigma_{\max} = 3.0$. These bounds stabilize the probabilistic head and prevent the NLL term, when it is enabled, from collapsing to unrealistically tiny uncertainty or exploding to a uselessly flat density.

The local component logits are transformed with a per pixel softmax, so $\sum_j p_{k,r,j}(x) = 1$. The global mixture weights $\pi_{k,r,j}^{(g)}$ emitted by the global branch belong to the map level distributional branch used when map level KL style terms are active; they are not multiplied into the deterministic point map in (3.83). This distinction is important for

the selected final checkpoint, where the distributional losses are disabled and RMSE is computed from the local weighted point prediction.

The architecture supports a full GMM Negative Log Likelihood (NLL), and the preceding large capacity residual stage used nonzero probabilistic weights. The selected final model keeps the same GMM head, but the last residual fine tuning stage is RMSE/MAE dominant: its NLL, KL, moment and outlier budget weights are set to zero in the resolved YAML configuration. The deterministic point prediction used for RMSE evaluation is the mixture mean in the modelled space:

$$\hat{z}_{k,r}(x) = \sum_{j=1}^3 p_{k,r,j}(x) \mu_{k,r,j}(x), \quad \hat{y}_{k,r}(x) = \begin{cases} \hat{z}_{k,r}(x), & k = \text{PL}, \\ \exp(\hat{z}_{k,r}(x)) - 1, & k \in \{\text{DS}, \text{AS}\}. \end{cases} \quad (3.83)$$

For path loss, this is also the native space mixture mean because the modelled space is dB. For delay spread and angular spread, the mixture is averaged in $\log(1 + y)$ space and then mapped back with $\exp(\cdot) - 1$. Therefore the reported point map should be read as the deterministic prediction used for RMSE, not as the exact native space expectation of the probabilistic mixture. The final task map selects $\hat{y}_{k,\text{LoS}}$ or $\hat{y}_{k,\text{NLoS}}$ using the explicit CKM LoS/NLoS masks. This prevents a visually plausible NLoS correction from leaking into pixels whose geometric path is LoS, and conversely prevents LoS ring structure from being used as an explanation for blocked pixels.

The training loss also includes a prior preservation term:

$$\mathcal{L}_{\text{guard},k} = \mathbb{E}_{x \in \Omega_k} \left[\max \left(0, (\hat{y}_k(x) - y_k(x))^2 - \left(y_k^{\text{prior}}(x) - y_k(x) \right)^2 \right) \right] \quad (3.84)$$

This *pixel wise* prior guard penalizes the model explicitly on any pixel where its squared error exceeds the prior’s squared error. The guard is not used alone; it is part of the composite optimization objective in Sec. 3.7.4.

3.7.3 Architecture: Shared UNet with FiLM

The final model processes the map inputs through a shared multitask convolutional encoder decoder. In the final large capacity configuration, scalar UAV height is encoded into a condition vector $\mathbf{c} \in \mathbb{R}^{160}$ and injected dynamically into the convolutional feature maps at multiple scales via *Feature wise Linear Modulation (FiLM)* layers.

For a given convolutional feature map $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, a linear projection of the height condition $\mathbf{c}$ outputs a scale vector $\mathbf{g} \in \mathbb{R}^C$ and a shift vector $\mathbf{b} \in \mathbb{R}^C$. The feature map is then modulated via an affine transformation:

$$\mathbf{X}' = \mathbf{X} \odot (1 + \mathbf{g}) + \mathbf{b} \quad (3.85)$$

where $\mathbf{g}$ and $\mathbf{b}$ are broadcasted spatially across the height and width. The FiLM mechanism and injection points are unchanged between the wide and large capacity variants; only the width of the feature maps changes, so the FiLM projection emits $2C$ parameters for the larger channel counts.

The word “joint” in this model refers to shared representation learning and joint optimization, not to an autoregressive or cross output decoder. Path loss, delay spread and angular spread all pass through the same encoder, the same height conditioning path,

the same decoder feature map and the same HARP-NET CKM residual assembly rule. After these shared features are computed, however, the implementation iterates over the three tasks and applies task specific global and local heads. Therefore PL, DS and AS can exchange information through the common spatial representation, but the final residual parameters, mixture weights and frozen prior are selected independently for each task.

The architecture separates global and local effects with two task specific heads. The **global head** average pools the deepest bottleneck feature map, fuses it with the height condition, and predicts map level distributional quantities: auxiliary mixture weights $\pi_{k,r,j}^{(g)}$, offsets $\delta_{k,r,j}^{(g)}$, and standard deviations $\sigma_{k,r,j}^{(g)}$. The **local head** applies 1×1 convolutions to the decoder output and predicts spatial fields $p_{k,r,j}(x)$, $\delta_{k,r}^{(\ell)}(x)$, $\alpha_{k,r}(x)$, and $\tilde{\sigma}_{k,r}(x)$.

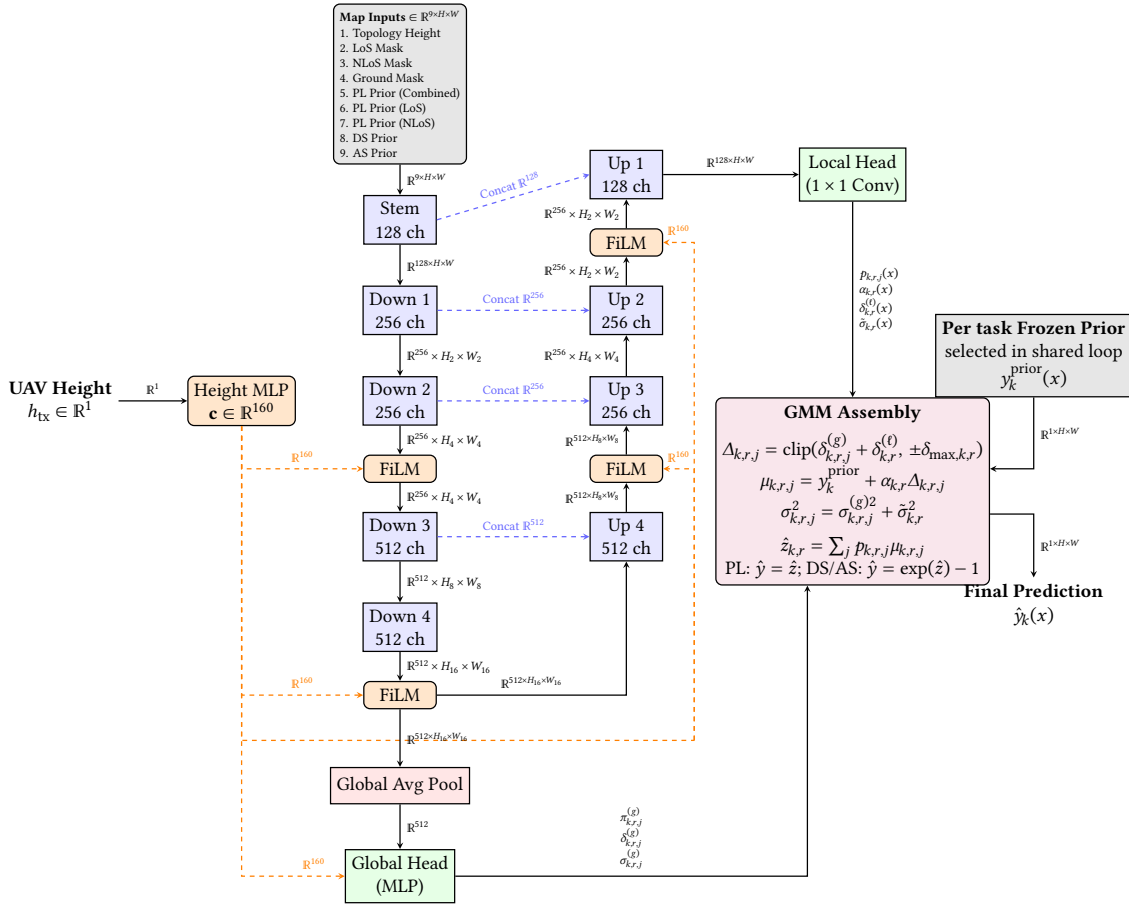


Figure 3.9: Schematic of the final large capacity shared UNet architecture with FiLM conditioning. The backbone and decode logic are shared, while each task uses its own heads and frozen prior.

Detailed sinusoidal height embedding and FiLM path. The scalar UAV height h_{tx} is first converted into a 32-dimensional sinusoidal embedding:

$$f_i = \exp\left(\log 12 + \frac{i-1}{15}(\log 478 - \log 12)\right), \quad i = 1, \dots, 16 \quad (3.86)$$

$$\mathbf{e}(h_{tx}) = \left[\sin\left(\frac{h_{tx}}{f_1}\right), \dots, \sin\left(\frac{h_{tx}}{f_{16}}\right), \cos\left(\frac{h_{tx}}{f_1}\right), \dots, \cos\left(\frac{h_{tx}}{f_{16}}\right) \right] \in \mathbb{R}^{32} \quad (3.87)$$

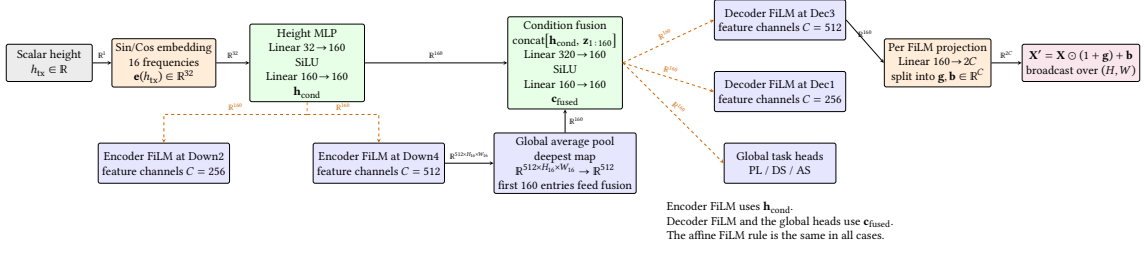


Figure 3.10: Detailed sinusoidal height conditioning path.

Detailed HARP-Net CKM residual head. For each task k and for each region $r \in \{\text{LoS}, \text{NLoS}\}$, the model predicts a 3-component mixture. The global head outputs $\pi_{k,r,j}^{(g)}$, $\delta_{k,r,j}^{(g)}$ and $\sigma_{k,r,j}^{(g)}$, while the local head outputs per pixel maps $p_{k,r,j}^{(\ell)}(x)$, $\delta_{k,r}^{(\ell)}(x)$, $\alpha_{k,r}(x)$, and $\tilde{\sigma}_{k,r}(x)$. After softmax over components, $p_{k,r,j}^{(\ell)}(x)$ becomes the local probability $p_{k,r,j}(x)$ used in the regional model space mean. The auxiliary global weights $\pi_{k,r,j}^{(g)}$ are retained for the distributional training branch and diagnostics, but the selected RMSE/MAE dominant checkpoint uses the local probabilities for the deterministic output map.

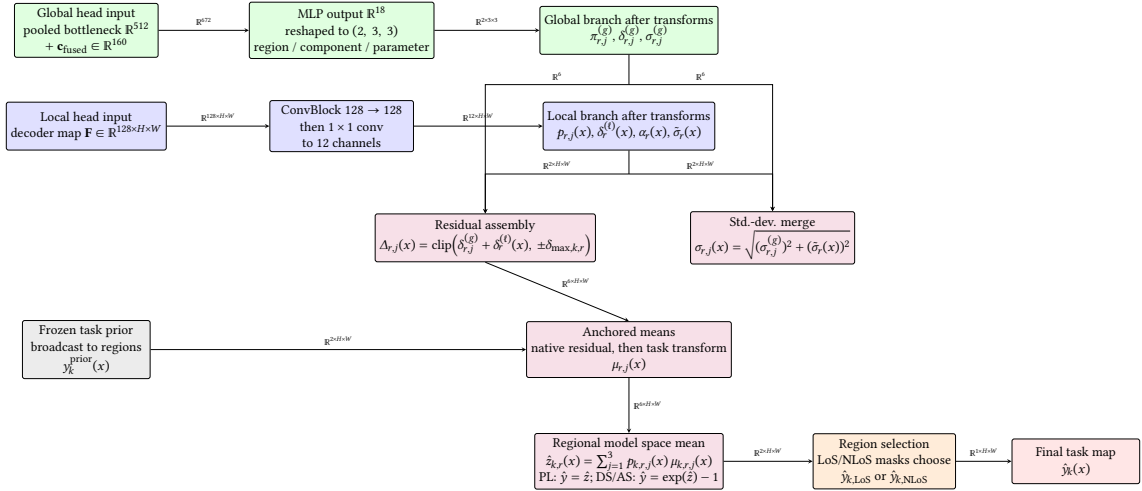


Figure 3.11: Detailed HARP-NET CKM residual head used by the final model.

3.7.4 Composite Optimization Objective

The implementation defines a composite loss that can train both the probabilistic GMM head and the deterministic residual map. For each task, let Ω_k be the valid ground pixel set, let $q_k(z | x)$ be the predicted mixture density in the modelled space, with $z = y$ for path loss and $z = \log(1 + y)$ for delay spread and angular spread, and let $\hat{y}_k(x)$ be the native space point prediction used for RMSE. The total objective is:

$$\begin{aligned} \mathcal{L} = \sum_{k \in \{\text{PL}, \text{DS}, \text{AS}\}} w_k & \left[\lambda_{\text{NLL}} \mathcal{L}_{\text{NLL},k} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL},k} + \lambda_{\text{mom}} \mathcal{L}_{\text{mom},k} \right. \\ & + \lambda_{\text{anchor}} \mathcal{L}_{\text{anchor},k} + \lambda_{\text{guard}} \mathcal{L}_{\text{guard},k} + \lambda_{\text{out}} \mathcal{L}_{\text{out},k} \\ & \left. + \lambda_{\text{RMSE}} \mathcal{L}_{\text{RMSE},k} + \lambda_{\text{MAE}} \mathcal{L}_{\text{MAE},k} \right]. \end{aligned} \quad (3.88)$$

Table 3.13 gives the resolved weights for both the large capacity residual stage and the

Table 3.13: Active loss settings in the HARP-NET CKM stages.

Term group	Large capacity residual stage	Final residual fine tuning stage	Interpretation
NLL, KL, moment and outlier budget	1.00 / 0.25 / 0.10 / 0.02	0 / 0 / 0 / 0	Active in the preceding large capacity stage, then disabled in the selected deterministic fine tuning stage.
RMSE / MAE	1.50 / 0.10	1.00 / 0.05	Supervised pixel losses in native output units; the final fine tuning stage makes them the dominant active terms.
Anchor / prior guard	0.05 / 0.10	0.01 / 0.02	Keep residual corrections small and penalize model states that become worse than the frozen priors, with lighter regularization in the fine tuning stage.
Task weights	PL 1.00, DS 1.00, AS 1.00	PL 1.00, DS 0.25, AS 0.25	Equal target weighting before the final fine tuning stage; then path loss becomes the main optimization target while spread supervision remains in the joint model.

final residual fine tuning stage; the equations below define the terms used in that table.

Map negative log likelihood. The likelihood is computed separately in LoS and NLoS regions, then averaged over the nonempty regions. For a transformed target $z_k(x)$,

$$q_{k,r}(z_k(x) | x) = \sum_{j=1}^3 p_{k,r,j}(x) \mathcal{N}(z_k(x); \mu_{k,r,j}(x), \sigma_{k,r,j}^2(x)), \quad (3.89)$$

and the regional NLL is

$$\mathcal{L}_{\text{NLL},k,r} = -\frac{1}{|\Omega_{k,r}|} \sum_{x \in \Omega_{k,r}} \log q_{k,r}(z_k(x) | x). \quad (3.90)$$

The implemented version evaluates (3.90) using $\log \sum \exp$ for numerical stability.

Distributional KL term. The global branch predicts residual components $\delta_{k,r,j}^{(g)}$. To keep those components aligned with the empirical residual distribution of the current batch, residuals $\epsilon_k(x) = y_k(x) - y_k^{\text{prior}}(x)$ are softly binned into an empirical probability vector $\mathbf{u}_{k,r}$. The global mixture offsets define a predicted residual histogram $\mathbf{v}_{k,r}$. The loss is

$$\mathcal{L}_{\text{KL},k} = \frac{1}{2} \sum_{r \in \{\text{LoS}, \text{NLoS}\}} \sum_{b=1}^B u_{k,r,b} \log \frac{u_{k,r,b}}{v_{k,r,b}}, \quad B = 64. \quad (3.91)$$

This term does not force a perfect pixel wise correction. It only encourages the map level mixture to place probability mass where the batch residuals actually live.

Moment matching. The point prediction is also constrained to have approximately the same first two moments as the target map:

$$\mathcal{L}_{\text{mom},k} = (\bar{\hat{y}}_k - \bar{y}_k)^2 + (s_{\hat{y},k} - s_{y,k})^2, \quad (3.92)$$

where bars denote masked means over Ω_k and s denotes masked standard deviation. When active, this term prevents a low NLL solution from matching local density while drifting in map average level. In the final fine tune it is defined by the code but has zero weight.

Anchor and guard terms. The anchor penalty keeps the correction small:

$$\mathcal{L}_{\text{anchor},k} = \mathbb{E} \left[\alpha_k(x)^2 + 0.25 \left(\delta_k^{(g)} \right)^2 + 0.50 \left(\delta_k^{(\ell)}(x) \right)^2 \right]. \quad (3.93)$$

The guard term is (3.84). Together, these two terms encode the final design philosophy: the DNN is allowed to improve the prior, but it must pay a penalty for large or harmful deviations. In (3.93), the residual variables are the same tensors used by the implementation: native dB offsets for path loss, and the corresponding $\log(1 + y)$ -space offsets for delay and angular spread after the native residual caps have been applied. The guard term, by contrast, is evaluated in the native units of each task against the frozen prior map.

Outlier budget and deterministic losses. When the outlier budget term is active, the implementation uses the last local mixture component as a proxy for broad/outlier usage. This is an indexing convention rather than a hard architectural constraint: the model is not forced to make that component the broadest one. If the mean usage of this proxy exceeds $\tau = 0.60$, a hinge penalty is applied:

$$\mathcal{L}_{\text{out},k} = \max \left(0, \frac{1}{|\Omega_k|} \sum_{x \in \Omega_k} p_{k,\text{out}}(x) - \tau \right). \quad (3.94)$$

The RMSE and MAE terms are conventional masked deterministic losses:

$$\mathcal{L}_{\text{RMSE},k} = \sqrt{\frac{1}{|\Omega_k|} \sum_{x \in \Omega_k} (\hat{y}_k(x) - y_k(x))^2}, \quad \mathcal{L}_{\text{MAE},k} = \frac{1}{|\Omega_k|} \sum_{x \in \Omega_k} |\hat{y}_k(x) - y_k(x)|. \quad (3.95)$$

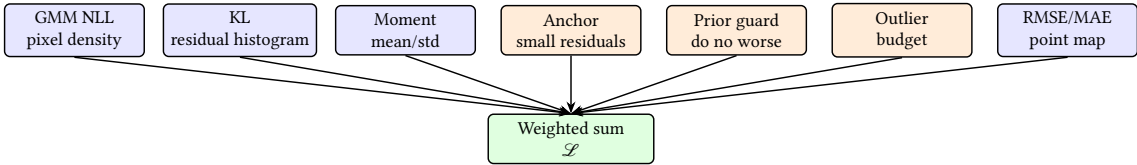


Figure 3.12: Composite final loss family. In the final residual fine tuning stage, the selected final model keeps the GMM head and uses RMSE/MAE, anchor and guard penalties; NLL, KL, moment and outlier budget weights are set to zero. Earlier residual stages used nonzero probabilistic weights.

Results

This chapter is organized in two blocks. First, it reports the frozen prior results on their own: the channel attenuation prior, the delay spread prior, and the angular spread prior. Second, it reports the final HARP-NET CKM residual model against those frozen priors on the unseen test cities. The standalone path loss prior audit uses the city holdout validation+test pool: 5340 samples from 29 unseen cities. The selected final model is then evaluated on the final 2590-sample, 14-city test split. Two path loss prior numbers therefore appear in this chapter: 1.9246 dB is the standalone calibrated hybrid prior on the held out validation+test pool, whereas 1.9383 dB is the frozen path loss prior baseline used by HARP-NET CKM on the final 14-city test split. All final model versus prior comparisons use the latter.

All RMSE values in this chapter are pixel weighted over valid ground receivers:

$$\text{RMSE} = \sqrt{\frac{\sum_{s,i,j} m_s(i,j) (\hat{y}_s(i,j) - y_s(i,j))^2}{\sum_{s,i,j} m_s(i,j)}}. \quad (4.1)$$

Here $m_s(i,j)$ is the valid receiver mask for sample s . This is stricter than averaging per sample RMSEs because larger valid maps contribute proportionally to their number of receiver pixels. The same rule is used for channel attenuation, delay spread and angular spread. For the two spread targets, every non building pixel is also a receiver pixel in the CKM ground truth, so the reported spread RMSE compares the dense prediction with the dense ray traced ground truth over all valid ground receivers.

The prior is not a weak reference included only for comparison. It is one of the central results of the thesis: a strong, train calibrated baseline that absorbs the stable part of the map before the residual model is asked to solve the harder local errors. Appendix A keeps the development history and negative attempts; this chapter focuses on the frozen priors and final test evaluation.

4.1 Standalone Frozen Prior Results

The final path loss prior is the clearest evidence that the prior is a strong path loss baseline in its own right. The enhanced two ray LoS model reduced LoS RMSE from roughly 3.95 dB with FSPL to 1.75 dB. The final hybrid prior reached 1.9246 dB overall under the pixel weighted metric on the held out validation+test pool.

4.1.1 Channel attenuation prior component comparison

The LoS branch of the final path loss prior is compared through three increasingly physical component variants fitted on the same city holdout split, so the gain from each modelling choice can be isolated (Table 4.1).

Two lessons follow. First, adding a purely radial correction on top of FSPL only buys about

Table 4.1: LoS path loss prior component comparison: three priors evaluated on the same held out validation+test pool (pixel weighted RMSE).

Prior	LoS RMSE (dB)	Gain over previous
FSPL only	3.9540	N/A
FSPL + radial residual	3.7086	−0.25
Enhanced two ray	1.7516	−1.96

0.25 dB. The residuals that a non coherent model can absorb are small compared to the structured deviation from FSPL. Second, switching to the *enhanced* two ray model with a fitted effective reflection coefficient (ρ, ϕ) per 5 m transmitter height bin removes almost 2 dB of additional error. This is exactly the magnitude expected when the dataset contains a visible ground reflection interference pattern, because the radial-residual model cannot reproduce the oscillatory ring structure with a purely monotonic function of distance. The two ray model is therefore not just a better fit numerically; it is a physically motivated explanation of the dominant LoS residual shape in CKM, and it is why the final model uses the two ray output directly as a frozen anchor rather than re-learning it from pixels.

4.1.2 Channel attenuation prior summary

Split comparability note. The final priors and HARP-NET CKM use the same city_holdout split with `val_ratio = 0.15`, `test_ratio = 0.15`, and `split_seed = 42`. The standalone path loss prior audit uses the training cities and the full validation+test held out pool, giving 1.9246 dB over 5340 held out samples. The final model comparison remains test only, so the frozen path loss prior baseline in Table 4.9 is 1.9383 dB.

The calibrated attenuation prior is the strongest non neural baseline in the thesis. It is split into LoS and NLoS branches because those regimes have different error mechanisms, but the construction details belong to Chapter 3. In this chapter the prior is used as a frozen baseline for the final test comparison.

The final reporting separates LoS and NLoS wherever the target definition allows it. A single overall number can hide whether the model is improving direct visibility pixels, blocked pixels, or only the easier majority regime.

Table 4.2: Final hybrid path loss prior, pixel weighted RMSE on the held out validation+test split.

Region	RMSE	Unit
LoS	1.7516	dB
NLoS	3.3967	dB
Overall	1.9246	dB

4.1.3 Delay and angular spread prior summary

The final spread priors showed that a large part of the spread targets can also be explained by geometry, building layout, and calibration without a neural network. The fitted calibration dictionary contains 114 case keys, but this number is not one simple Cartesian product. For each spread metric, the exact grid has 6 topology classes $\times$ 2 LoS/NLoS states $\times$ 3 antenna height bins, or 36 exact keys. The implementation then adds fallback calibrations for sparse cases: 12 antenna collapsed topology and LoS/NLoS keys, 6

topology wide keys, 2 global LoS/NLoS keys, and 1 fully global key. This gives 57 keys per spread metric and 114 keys across delay spread and angular spread. Equivalently, the dictionary contains 72 exact DS/AS cases plus 42 fallback cases. The point of this dictionary is not novelty by formula alone, but coverage: the baseline is strong because it stores many small regime specific decisions about density, building height, visibility, and antenna height instead of forcing one global estimator to explain all pixels. The calibrated spread prior module applies this idea separately to delay spread and angular spread. The raw spread estimate is only the first feature; the calibrated model then uses a 23-term vector with raw prior curvature, range, elevation angle, transmitter height, local building density at two scales, local height at two scales, LoS/NLoS support at two scales, blocker depth, transmitter clearance, taller building fraction, and interaction terms. Both spread targets are fitted in $\log(1 + y)$ space, so the model can handle the LoS spike near zero and the long NLoS tail with one calibrated framework while still reporting metrics in native ns and degrees.

Table 4.3: *Calibrated spread prior test results, pixel weighted RMSE.*

Metric	Overall	LoS	NLoS
Delay spread (ns)	28.10	27.49	29.62
Angular spread (°)	13.74	15.28	8.66

4.1.4 Prior RMSE by transmitter height

The CKM dataset covers stored transmitter heights from about 12 m to almost 478 m, while the generator distribution has support up to 500 m. The absence of samples above 478 m is statistically normal for this finite draw, because the expected count above that height is only 0.52. Therefore the conditional error at a single aggregate RMSE can hide strong altitude-dependent structure. Tables 4.4 and 4.5 therefore break the path loss prior and the delay spread and angular spread priors into four transmitter height bins, using the same pixel weighted aggregation as the headline tables.

Table 4.4: *Path loss prior pixel weighted RMSE by transmitter height, on the held out validation+test split.*

Transmitter height bin	Overall	LoS	NLoS	Samples
12 to 50	2.180	1.797	3.756	1211
50 to 150	1.936	1.785	3.228	2634
150 to 300	1.680	1.666	2.259	786
300 to 500	1.627	1.619	2.255	362

The altitude dependence is monotonic and physically intuitive. Higher UAV transmitters see more of the ground as unobstructed LoS and encounter fewer deep shadow pockets, so both the NLoS pixel fraction and the worst-case tails shrink. Path loss NLoS RMSE drops from 3.76 dB at low altitudes to 2.26 dB above 300 m, and delay spread NLoS RMSE collapses from 35.13 ns to 2.76 ns in the same comparison. The opposite is true for the absolute number of samples: the dataset is dominated by the 50 – 150 m band, so the global aggregate RMSE reflects that regime more than the higher-altitude extremes. Together with the empirical HDF5 histogram of (3.3) and the counts reported in Chapter 3, this

Table 4.5: Delay spread and angular spread calibrated pixel weighted RMSE by transmitter height, on the city holdout test split.

Transmitter height bin	Metric	Overall	LoS	NLoS
12 to 50	Delay spread (ns)	39.90	43.04	35.13
50 to 150	Delay spread (ns)	27.17	26.72	28.29
150 to 300	Delay spread (ns)	8.70	8.79	8.13
300 to 500	Delay spread (ns)	3.70	3.80	2.76
12 to 50	Angular spread (°)	15.69	18.81	9.93
50 to 150	Angular spread (°)	14.19	15.89	8.41
150 to 300	Angular spread (°)	10.39	11.07	4.49
300 to 500	Angular spread (°)	8.85	9.34	2.85

means the high altitude rows are better despite having much less training support, not because they are over represented in the dataset. A practical consequence for the final residual model is that the neural residual head has less to correct at high altitudes, where the path loss prior is already close to the quantisation floor. The remaining headroom lives in the low altitude, NLoS heavy bin, which is exactly where a distribution aware residual can matter.

4.1.5 Path loss prior by topology class

The headline path loss prior number (1.92 dB overall RMSE on the held out validation+test split, see Table 4.2) hides the same building layout structure captured by the topology aware breakdown. The per sample evaluation records were aggregated again by classifying each evaluated sample into the six topology classes with the same final routing rule. The pixel weighted RMSE per topology class is reported in Table 4.6.

Table 4.6: Hybrid path loss prior, pixel weighted RMSE on the held out validation+test split, broken down by topology class. The overall row reproduces the RMSE values in Table 4.2; the sample count is the sum of samples assigned to the six topology classes. The aggregate prior evaluator contains 4994 topology assigned samples.

Topology class	LoS	NLoS	Overall	Samples
open sparse lowrise	1.39	2.92	1.43	895
open sparse vertical	1.43	3.91	1.55	370
mixed compact lowrise	1.81	2.91	1.90	1240
mixed compact midrise	1.91	3.84	2.21	1060
dense block midrise	2.19	3.20	2.40	1325
dense block highrise	2.44	4.09	2.97	104
Overall	1.75	3.40	1.92	4994

The LoS dependency on building layout is monotonic. The two ray prior fits its own ground reflection ring against an LoS marginal whose width grows with city density and building height, so the LoS RMSE of the analytic model rises from 1.39 dB in the open sparse lowrise class to 2.44 dB in the dense block highrise class. The NLoS branch is less ordered by building layout: the best NLoS topology class is *mixed compact lowrise* at 2.91 dB, while the worst is *dense block highrise* at 4.09 dB. The dataset is also unbalanced

in this respect: dense block highrise contributes only 104 of the 4994 topology assigned evaluation samples, so its high RMSE has a relatively small impact on the overall pixel weighted aggregate but is the regime most likely to limit deployment in dense urban environments.

This topology breakdown also makes the central methodological claim of the thesis quantitative. The train calibrated prior is not only strong on average, but also auditable by city type and LoS/NLoS case. The longer comparison with the appendix diagnostic models is kept in Appendix A.

4.1.6 Spread priors by topology class

The delay spread and angular spread priors are also topology dependent. These tables use the same city holdout test split as Table 4.3 and keep the dense pixel weighted metric separated by visibility regime.

Table 4.7: Delay spread prior RMSE by topology class on the city holdout test split.

Topology class	Overall	LoS	NLoS
open sparse lowrise	16.88	17.01	15.43
open sparse vertical	40.84	40.44	44.08
mixed compact lowrise	17.27	16.65	18.91
mixed compact midrise	37.88	38.26	37.25
dense block midrise	26.90	29.00	25.02
dense block highrise	43.52	40.16	45.08
Overall	28.10	27.49	29.62

Table 4.8: Angular spread prior RMSE by topology class on the city holdout test split.

Topology class	Overall	LoS	NLoS
open sparse lowrise	10.01	10.30	6.33
open sparse vertical	12.70	12.98	9.97
mixed compact lowrise	14.37	15.99	8.17
mixed compact midrise	15.26	17.85	9.57
dense block midrise	15.53	21.20	8.17
dense block highrise	14.64	21.33	9.78
Overall	13.74	15.28	8.66

The spread priors show a different topology pattern from channel attenuation. Delay spread is hardest in open sparse vertical and dense block highrise settings, while angular spread is dominated by the LoS rows of denser classes. This supports the final design choice to keep separate residual outputs for the three targets instead of treating all CKM maps as one generic image regression problem.

4.2 HARP-NET CKM Results Against Frozen Priors

4.2.1 Final test protocol and headline results

HARP-NET CKM freezes the attenuation and spread priors and learns a bounded residual correction for the three radio maps. It is tested on the city holdout test split with 2590 samples from 14 cities that were not seen in training. The model should be read as a two stage final pipeline, not only as the last fine tuning loss: the large capacity probabilistic residual stage keeps NLL, KL, moment matching and outlier budget terms active, while the reported checkpoint is the state selected on validation after the deterministic RMSE/MAE dominant fine tuning stage. The exact weights of both stages are listed in Table 3.13. The final claim is therefore an RMSE improvement over the frozen priors rather than a calibrated probabilistic prediction claim. The pixel weighted RMSE rule in (4.1) is used for the model and for the frozen priors. Therefore `model_rmse` in the evaluation JSON is target specific: it is the RMSE for channel attenuation, delay spread, or angular spread individually, not a sum of the three RMSE values. The three targets are kept separate because their units are incompatible. Delay spread and angular spread are evaluated exactly like channel attenuation: for each valid ground receiver pixel, the predicted value is compared with the ray-traced CKM ground truth value, and the RMSE is accumulated over all valid ground receivers.

Table 4.9: HARP-NET CKM test results on unseen cities, compared with the frozen priors. Negative Δ means that the model improves the prior.

Target	Model RMSE	Prior RMSE	Δ RMSE	Model MAE	Prior MAE	Δ MAE	Unit
Attenuation	1.6519	1.9383	-0.2865	0.9298	1.2319	-0.3020	dB
Delay spread	26.5570	28.1023	-1.5453	7.9897	7.3033	+0.6864	ns
Angular spread	11.3854	13.7416	-2.3562	5.0088	4.9890	+0.0198	°

Table 4.10: HARP-NET CKM test RMSE separated by LoS and NLoS pixels. The test cities are unseen during training.

Target	Scope	Model RMSE	Prior RMSE	Δ RMSE
Attenuation	LoS	1.4675	1.7519	-0.2844
Attenuation	NLoS	3.1482	3.5143	-0.3661
Delay spread	LoS	25.4419	27.4851	-2.0432
Delay spread	NLoS	29.2045	29.6157	-0.4112
Angular spread	LoS	12.3510	15.2818	-2.9308
Angular spread	NLoS	8.4424	8.6614	-0.2191

4.2.2 Channel attenuation result

Channel attenuation is the cleanest final result because both RMSE and MAE improve over the frozen prior on unseen test cities. Overall RMSE drops from 1.9383 dB to 1.6519 dB, while MAE drops from 1.2319 dB to 0.9298 dB. The LoS and NLoS rows in Table 4.10 show that the improvement is not coming from only one visibility case: LoS improves by 0.2844 dB RMSE and NLoS by 0.3661 dB.

4.2.3 Angular spread result

For angular spread, the frozen prior already meets the original project target, but HARP-NET CKM still reduces RMSE on unseen test cities. Overall RMSE drops from 13.7416° to 11.3854°. The MAE is almost unchanged (+0.0198°), so this is a modest residual correction

rather than a uniform improvement at every pixel. The clearest gain appears in LoS pixels, from 15.2818° to 12.3510° , while NLoS improves only slightly, from 8.6614° to 8.4424° .

No reviewed external method reports the same dense CKM city holdout angular spread map task. One or two papers have similar link level channel statistics goals, but not the same dense map input, transmitter height protocol, or held out city evaluation. For that reason, the angular spread result is compared mainly against the frozen prior and the original thesis target.

4.2.4 Delay spread result

Delay spread also improves in RMSE, although it is the only output where RMSE and MAE move in opposite directions. Overall RMSE drops from 28.1023 ns to 26.5570 ns, while MAE increases from 7.3033 ns to 7.9897 ns. This should be read as a tail focused gain: suppressing the largest delay errors can improve RMSE while many small local shifts slightly increase MAE. The LoS RMSE improves by 2.0432 ns; the NLoS RMSE improves only by 0.4112 ns.

As with angular spread, no reviewed paper is directly equal to the dense CKM city holdout delay spread map task. The nearest references predict scalar channel characteristics or use different simulation contracts, so the main comparison is the frozen delay spread prior plus the original 50 ns project target.

4.2.5 Validation to test check

The validation metrics are not the final test claim. They evaluate the same selected checkpoint on the validation cities. The same selected model is then evaluated on the separate test cities, so the table checks validation to test drift without changing checkpoint or metric. Table 4.11 reports the two scopes side by side.

Table 4.11: Validation versus test sanity check for the same selected final model. The validation row evaluates the selected checkpoint on validation cities; the test row is the final unseen city generalisation result.

Split	Samples	PL RMSE	DS RMSE	DS prior RMSE	AS RMSE	Meaning
Validation	2750	1.6157	22.1061	23.4494	11.3782	Same selected model state evaluated on validation cities.
Test	2590	1.6519	26.5570	28.1023	11.3854	Separate 14-city holdout; this is the thesis generalisation number.

The larger delay spread gap between validation and test is mainly a city mix effect, not a change of model or metric. The frozen delay spread prior also worsens from 23.45 ns RMSE on validation to 28.10 ns on test, so the test cities contain harder delay spread distributions before the neural residual is applied. In particular, Kuala Lumpur contributes only 13.2% of the valid delay spread test pixels but 36.2% of the test delay spread squared error, with 44.00 ns model RMSE and 46.93 ns prior RMSE. This is why the final test delay spread RMSE is higher than the validation row, while still improving over the prior.

Together, the three target sections show the main generalisation result: the test cities are completely unseen, and the residual model reduces RMSE for channel attenuation, angular spread and delay spread simultaneously.

4.3 State of the Art Context and Runtime

4.3.1 Target vs final result

The selected final model clears the original numerical targets on the unseen city test split: 1.65 dB channel attenuation against the 5 dB target, 26.56 ns delay spread against the 50 ns target, and 11.39° angular spread against the 20° target. The same target check also holds on validation and on the all city aggregation; even the worst unseen-test subexpert for each output remains below its target (2.66 dB, 36.89 ns, and 14.06°). On the all city subexpert audit, the corresponding worst cases are 2.45 dB, 42.24 ns, and 13.97° , still below the same targets.

4.3.2 Comparison with closest state of the art metrics

The state of the art numbers cannot be used as a direct leaderboard because their datasets, height assumptions, city splits, and target definitions differ. Still, Table 4.12 places the final test result next to the closest available metrics from Section 2.5.7. Channel attenuation has several external dB scale references, but no external paper found here reports the same dense CKM city holdout contract for delay spread and angular spread. The table is therefore limited to path loss like external rows plus the final model reference row; scalar channel parameter papers are kept out of this results table. In the table, PL means channel attenuation RMSE and lower RMSE is better. The better/worse ratios should be read cautiously because the rows use different data, different inputs, or different evaluation contracts. Earlier internal baselines are summarized in Appendix A instead of repeated here.

Table 4.13 adds the missing spread side comparison. It is intentionally not a leaderboard: the closest external papers mostly predict scalar or link level channel characteristics, and the angular quantities are often angle of arrival or angle of departure spreads rather than the dense angular spread map target used here. This thesis predicts dense 513×513 delay spread and angular spread maps under city holdout, so the table is useful as scope context rather than as a ranking.

This comparison supports a cautious interpretation. The final channel attenuation number is not claimed to beat every published benchmark in absolute terms; PMNet’s reported ICASSP score is lower on a different contract, and RadioGUNet is lower on the easiest RadioMapSeer setting. The channel attenuation result is nevertheless in the same broad outdoor simulated-map band as RadioUNet and RadioGUNet, below the PathFinder unseen rural value under the approximate RM3D scaling, numerically well below Gao et al.’s recent corridor weighted segmentation numbers, close to ReVeal’s real outdoor sparse measurement scale, and slightly below the converted RMTransformer value after applying that paper’s own 254 dB normalisation window. For spreads, Table 4.13 makes the external reading explicit instead of omitting the comparison. The thesis claim is not that the spread maps beat external scalar predictors; it is that the model reports dense delay spread and angular spread RMSE on unseen CKM cities, improves the closest same protocol spread baselines, and keeps the stricter ground mask, LoS/NLoS, and continuous height protocol visible.

Inference time sanity check. The secondary runtime objective was checked on the Barcelona geometry supplied with the MATLAB ray tracing script. The benchmark

Table 4.12: Plain language comparison with the closest available path loss side metrics. No scalar indoor or link level spread predictor is included as a table row, because those numbers are not dense CKM-style map benchmarks.

Reference	What it reports	Verdict for our final model	Why the verdict is limited
RadioUNet [19]	1.6 dB–3.1 dB equivalent channel attenuation RMSE in accurate map settings	Our final PL RMSE lies inside the reported 1.6 dB–3.1 dB dB scale band.	2D channel attenuation maps only; no transmitter height conditioning or spread targets.
RadioGUNet [20]	1.304 dB DPM, 1.936 dB IRT, and 1.392 dB DPM with cars on RadioMapSeer	Our final PL RMSE is close to the reported RadioMapSeer dB scale range, while remaining above its best DPM number.	Outdoor simulated terrestrial maps; held out test layouts, but no transmitter height or spread targets.
PMNet / ICASSP 2023 winner [21, 6]	approx. 1.38 dB converted from the RadioMap3DSeer normalized score	Final PL is close but higher under a different benchmark contract; use only as scale context, not as a direct ranking.	Not directly comparable: ICASSP/RadioMap3DSeer testing differs from our unseen CKM-city split, continuous transmitter height conditioning, and joint PL/DS/AS target contract.
Gao et al. [37]	5.59 dB on ICASSP 2023; 12.19 dB on ITU Challenge 2024, with about 60% fewer FLOPs than PPNNet	Our final PL RMSE is about 3.4× lower than the ICASSP number and 7.4× lower than the ITU number.	Recent outdoor PL-map model with Tx/Rx depth and corridor weighting, but no CKM variable height UAV, no joint PL/DS/AS output, and different split/features.
ReVeal PINN [27]	1.95 dB outdoor rural/suburban radio environment RMSE with sparse measurements	Final PL is in the same dB scale range as ReVeal’s sparse measurement result.	Real outdoor measurements, but sparse RSSI radio environment mapping rather than dense transmitter height conditioned CKM generation.
AIRMap [36]	path gain error reported below 4 dB against ray tracing	Our final PL RMSE is below 2 dB, but an exact ratio is not recoverable from the reported path gain threshold.	Reports path gain, not the same path loss calibration.
RMTransformer [22]	7.15×10^{-3} normalised pixel RMSE, approximately 1.82 dB with its 254 dB USC/PMNet scaling window; 8.10×10^{-3} channel prediction error, approximately 2.06 dB	Our final PL RMSE is slightly below the converted dB scale values under that paper’s normalisation window.	Recent dense path loss radio map reference, but on 256 px USC/PMNet maps with a random 90/10 split, no CKM city holdout, no continuous transmitter height, and no spread targets.
PathFinder [39]	0.0331 normalized RMSE on RM3D DS-RPP, approximately 1.19 dB with the 36 dB RM3D image window convention; 0.3263 on unseen rural, approximately 11.75 dB only under the same assumption	Our final PL RMSE is above the converted DS-RPP value and far below the rural value under the same approximate 36 dB assumption.	The DS-RPP conversion follows RM3D scaling; the rural conversion is only an approximate scale check because the paper reports normalised image-space RMSE.
Final model	1.6519 dB PL , 26.5570 ns DS , 11.3854° AS on unseen CKM cities	Reference row for the ratios above.	Only row here reporting all three dense CKM targets under the final city holdout protocol.

Table 4.13: *Spread side comparison with nearby state of the art channel parameter prediction papers. Rows with external papers are scope comparisons, not direct rankings against the dense CKM map task.*

Reference	Spread side output and reported scale	Similarity to this work	Main limitation of the comparison
Yang et al. [47]	A2G mmWave delay spread prediction with random forest and k -nearest neighbours; reported lower RMSE than a lognormal contrast model.	Closest early ML reference for A2G delay spread prediction.	Scalar link level DS only; no angular spread output, no dense maps, and no CKM city holdout evaluation.
Huang et al. [28]	Transformer UAV channel predictor; merged data RMSE of 4.577 ns for RMS DS and 1.549° , 0.375° , 0.864° , 0.074° for RMS azimuth/elevation angle of arrival and departure quantities.	Recent UAV mmWave model predicting delay spread and AoA/AoD angular statistics.	Predicts per link channel characteristics from coordinates/frequency/location tags, not dense 513×513 AS/DS maps from building height images.
Mi et al. [55]	Regression PointNet from indoor 60 GHz point clouds; predicts RMS DS, azimuth angular spread, path loss and K -factor; reports average angular spread RMSE of 7.39° .	Measurement based DL reference that includes both DS and AS.	Indoor point cloud measurement task with mostly link level samples, not outdoor UAV CKM generation or unseen city dense map testing.
Final model	26.5570 ns DS RMSE and 11.3854° AS RMSE on unseen CKM cities.	Only row here with dense PL/DS/AS maps and the final ground mask city holdout contract.	Numbers should be compared primarily against the same protocol frozen spread priors and project targets.

uses the same local laptop as the final generator: an AMD Ryzen 5 5600H CPU and an RTX 3050 Ti Laptop GPU with 4 GiB of VRAM. The ray tracing run was configured at 1 m receiver spacing, so each non building pixel of the 513×513 area is represented by a receiver at 1.5 m. Two Barcelona layouts and five transmitter heights per layout give ten full resolution CKM samples. The local MATLAB installation does not include Parallel Computing Toolbox, so the ray tracer was run on CPU with the same propagation settings as the supplied script: one reflection, no diffraction, 7.125 GHz, and extraction of channel attenuation, delay spread, and angular spread from the ray tracing paths.

The surrogate timing uses the same ten Barcelona topologies and antenna heights. It includes the complete non-RT generation path after the checkpoint is loaded once: geometric LoS/NLoS mask generation from the height map, the final ML calibrated path loss prior, the final ML calibrated delay spread and angular spread priors, and mixed-precision inference with HARP-NET CKM, including the default uncompressed .npz numerical archive export. The excluded costs are the one time checkpoint load, the one time CUDA warm up, and optional PNG exports for masks or rendered maps.

The ten ray traced samples contain 156,088 and 115,825 valid receiver pixels in the two Barcelona layouts, respectively. The measured average is 99.89 s per full resolution sample for ray tracing and 0.88 s per sample for the complete HARP-NET CKM generator, including masks, priors, predictions, and the default numerical .npz archive export, but not optional PNG exports. The resulting $113.7\times$ speedup exceeds the original 10 to $100\times$ runtime goal and should therefore be read as a deployment style marginal runtime result after training and calibration, not as a comparison of the full research cost of developing the model.

The component logs explain where the remaining runtime goes without needing a second table: mask generation takes about 0.30 s, torch priors about 0.04 s, and final neural prediction plus default .npz export about 0.54 s. The vectorized mask path was checked

Table 4.14: Full pipeline runtime comparison for one 513×513 Barcelona CKM sample. Both rows are measured on the same Ryzen 5 5600H / RTX 3050 Ti Laptop machine. The surrogate row includes geometric mask generation, prior computation, and prediction of channel attenuation, delay spread, and angular spread, including the default numerical .npz export. Checkpoint loading, one time CUDA warm up, and optional PNG exports are excluded.

Method	Timed path	Resource note	Time per sample	Dataset-scale extrapolation
MATLAB ray tracing	Pixel level receivers outside buildings; PL, DS and AS parsed from paths	Ryzen 5 5600H CPU; GPU off, PCT unavailable	99.89 s	448.96 h (≈ 18.7 days), $1\times$
HARP-NET CKM generator	Generated LoS/NLoS masks, final calibrated priors, PL/DS/AS prediction, and default .npz export	RTX 3050 Ti Laptop GPU; mixed fp16, $b = 1$	0.88 s	3.95 h, $113.7\times$ faster

against the original GT ray-caster with zero LoS/NLoS differences on the Barcelona samples and on 100 exported CKM samples; mixed precision changed RMSE by only 0.0014 dB, 0.0028 ns, and 0.0025° relative to fp32.

4.4 Qualitative Analysis

The qualitative evidence supports the quantitative story without needing the full diagnostic gallery in the main body. The early channel attenuation histograms showed NLoS mode collapse: visually plausible maps could follow the radial LoS trend while missing the blocked link tail. The selected final model removes the worst of that collapse, reduces RMSE relative to the frozen priors across the split and LoS/NLoS checks, and applies residual corrections whose sign follows the prior error maps. The diagnostic appendix keeps the histogram and spread quantisation checks; the wider gallery and split bar plots remain in the generated figure archive. The multilinear priors are already strong in RMSE because much of the CKM structure is explained by range, height, LoS state and coarse morphology. The main role of the residual model is therefore local: it sharpens corrections around building borders and other spatial transitions that a linear prior cannot represent. Structural metrics such as SSIM were not computed, but they may be more sensitive than RMSE to this type of spatial improvement.

The main lesson is that LoS path loss in this dataset has a strong radial ring structure, so a calibrated two ray prior is more efficient than asking a CNN to learn the same pattern from pixels. The remaining qualitative failures are concentrated in sparse NLoS and high spread cases, which is why future spread work should include an explicit high tail or outlier diagnostic.

4.5 Limitations

The dataset is produced by ray tracing, not by field measurements. The results therefore measure agreement with a simulator and not direct over the air deployment performance. The channel attenuation target is stored as uint8, which imposes approximately 1 dB quantization. The spread targets are also stored as integer maps: delay spread as uint16 in ns and angular spread as uint8 in degrees. A more suitable dataset for the final modelling stage would preserve the continuous ray traced targets, for example as float32 or as documented scaled integers, before any display-oriented quantisation. This should not be confused with a quantized model output: HARP-NET CKM emits continuous PL/DS/AS values. In that light, the final channel attenuation MAE of 0.9298 dB on labels stored at

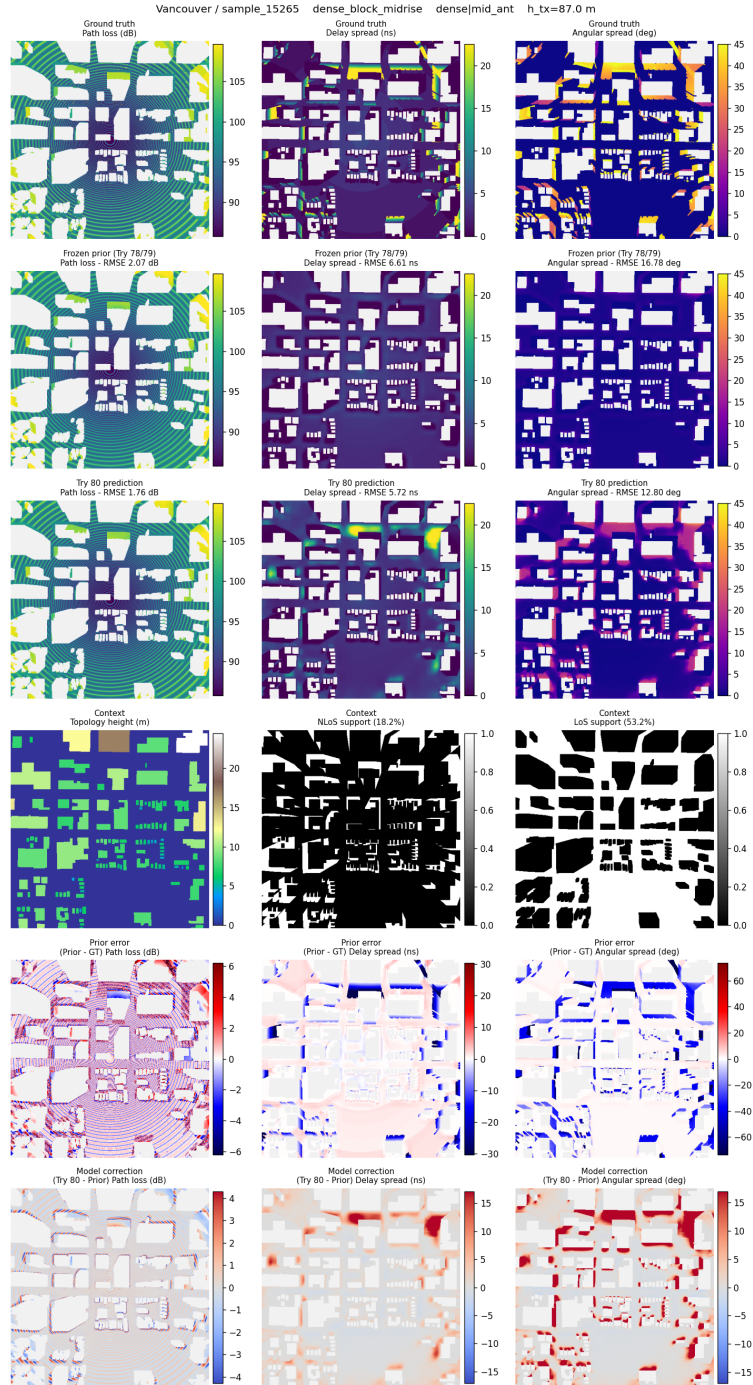


Figure 4.1: Representative final model inspection panel for a dense, mid altitude, held out test sample in Vancouver. This panel is kept here because it shows the final prior, residual correction and prediction side by side.

approximately 1 dB resolution is already close to the channel attenuation label step, even though the RMSE remains 1.6519 dB because it is dominated by the harder tail errors.

The effect can be written explicitly. Let y^* be the continuous ray traced value and $y_q = Q_\Delta(y^*) = y^* + \epsilon_q$ the stored value after rounding to step Δ . Under the usual high resolution quantisation approximation, $\epsilon_q \sim \mathcal{U}(-\Delta/2, \Delta/2)$, so

$$\sigma_q = \sqrt{\mathbb{E}[\epsilon_q^2]} = \frac{\Delta}{\sqrt{12}}. \quad (4.2)$$

With $\Delta = 1$ in the physical units used here, the hidden uncertainty is 0.289 dB for channel attenuation, 0.289 ns for delay spread, and 0.289° for angular spread. This is not the dominant error in the reported final model, but it is an irreducible resolution limit of the saved labels and becomes important once residual errors approach the sub unit regime.

Table 4.15: Approximate quantisation RMSE implied by the stored integer target precision in the current CKM HDF5 file. The table reports the actual storage used in this thesis.

Target	Stored type	Current integer step	Current floor
Attenuation	uint8	1 dB	0.289 dB
Delay spread	uint16	1 ns	0.289 ns
Angular spread	uint8	1°	0.289°

This also explains why getting very close to the best possible error is mathematically harder than the headline RMSE suggests. If model error and quantisation error are approximately independent, the observed squared error decomposes as

$$\mathbb{E}[(\hat{y} - y_q)^2] \approx \mathbb{E}[(\hat{y} - y^*)^2] + \sigma_q^2. \quad (4.3)$$

Near σ_q , a real reduction in model error produces only a small change in total RMSE because of the square-root compression. In addition, many different continuous ray tracing values collapse to the same stored integer, so the model cannot learn sub-bin structure from the labels even if the input geometry contains enough information. The heavy tailed NLoS distributions add a second difficulty: MSE is dominated by rare large misses, so once the common LoS pixels are near the quantisation scale, further improvements depend on resolving sparse hard NLoS cases rather than on smoothing the whole map.

The LoS/NLoS mask used in these experiments should be interpreted as a geometric binary visibility mask, not as an augmented electromagnetic propagation label. In the CKM setup the transmitter position is fixed at the map centre and the transmitter height is known for each sample, so the same binary mask can be derived by line of sight tracing through the building topology. It is therefore a deterministic geometric feature and branch key, not extra measured radio information unavailable at deployment.

The selected final model is evaluated on one city holdout split and one training seed. This demonstrates generalisation to unseen cities, but not full robustness: future work should repeat selection across seeds and add field measurements, especially for delay spread tails where quantised labels make sparse high delay artefacts difficult to interpret.

Finally, the dataset fixes frequency and simulator assumptions. Deployment would need conditioning on carrier frequency, bandwidth, antenna pattern, and environment class; the city holdout protocol tests urban layouts, not this broader domain adaptation.

Sustainability Analysis and Ethical Implications

The sustainability profile of this thesis is defined by a tradeoff: the exploratory computation needed to discover a reliable model versus the much lower marginal cost of reusing it for dense CKM generation. Since the result is a software pipeline rather than a manufactured device, the analysis focuses on energy use, reuse of trained artifacts, reproducibility, and the risks of using simulated predictions in planning decisions.

5.1 Sustainability matrix

Table 5.1: *Sustainability matrix summary for the thesis.*

Perspective	Development of BT	Project execution	Risks and limitations
Environmental	GPU training, laptop use, storage, and writing.	Reduced need for repeated ray tracing simulations once the surrogate is trained.	Extra retraining, larger models, or poor reuse of cached priors could increase footprint.
Economic	Student and supervision time, plus shared HPC use.	Lower marginal planning cost if the surrogate replaces repeated simulations.	Real deployment needs field validation and maintenance.
Social	Skills gained in wireless modelling, ML, HPC workflows and scientific reporting.	Faster elevated base station or UAV coverage planning for temporary, rural or emergency networks.	Bias toward represented city morphologies and risk of overclaiming simulated results.

5.2 Environmental impact

The main environmental cost during development was computation. The final HARP-NET CKM line used a four GPU RTX 2080 Ti node for approximately 12 h; the distribution first and prior calibration phase used additional GPU/CPU time. Across the full exploratory project, the total exceeded roughly 400 GPU hours. This is a one time research overhead and includes many failed or diagnostic experiments, not only the selected final run.

Several decisions reduced avoidable impact: reusing saved histories and frozen calibration artifacts, evaluating city holdout metrics from stored checkpoints, and keeping the final priors frozen instead of recomputing them blindly inside every neural experiment. The final design is also relatively efficient at inference. On the Barcelona benchmark, the local MATLAB ray tracing path took 99.89 s per full resolution sample, while the complete generator path with masks, priors, neural prediction and export took 0.88 s. The sustainability advantage therefore depends on reuse: if the same calibrated generator is queried many times, the marginal cost is much lower than repeated ray tracing; if the model is retrained from scratch for every new scenario, that advantage weakens.

The life cycle interpretation is therefore important. During the research phase, the footprint is dominated by exploration: many failed or partially successful attempts were necessary to discover that the final path should combine strong priors with a bounded

neural residual model. During the use phase, the footprint is dominated by inference and by occasional recalibration. The project is most sustainable when the trained model, frozen priors, cached masks and exported checkpoints are reused across many planning queries. It would be less sustainable if the same design were scaled by simply increasing model size or retraining from zero for every city.

The final generator also improves reproducibility, which has an environmental side. A documented .npz output, explicit mask policy, and fixed hardware/runtime benchmark make it easier to compare changes without rerunning expensive ray tracing baselines. Reproducible experiments are not only a scientific requirement; they reduce wasted computation caused by ambiguous settings, hidden preprocessing changes or repeated debugging runs.

For the development footprint, a rough quantitative estimate can be recovered from the experiment history. The final HARP-NET CKM line used four RTX 2080 Ti GPUs for approximately 12 h, about 18 kWh under a 1.5 kW four GPU node approximation. The distribution first and prior calibration phase added roughly another 18 h of four GPU time, about 27 kWh, plus lighter CPU and DirectML calibration runs. Across the full exploratory thesis line, including failed and diagnostic experiments, the total exceeded roughly 400 GPU hours. With the same power approximation this is about 150 kWh, or approximately 30 kgCO₂e if an illustrative 0.2 kgCO₂e/kWh grid factor is used. This is not a full life cycle assessment and does not include embodied carbon of shared GPUs or data centre infrastructure, but it gives an order of magnitude for the academic research phase.

Because this is a software thesis, raw material use is indirect: the laptop, shared GPU node, storage and network infrastructure already existed. No dedicated device or project specific material was purchased. Supplier environmental reports for the hardware were not audited, which is a limitation of the analysis. At the end of life, the project mainly generates digital artefacts that can be archived or deleted; no project specific physical waste is expected. If the work were repeated, the footprint could be reduced by freezing the experiment plan earlier, running smaller pilot subsets before full training, reusing checkpoints more aggressively and scheduling heavy jobs when cleaner electricity is available.

For the use phase, the older detailed estimate remains useful because it connects the benchmark with energy rather than only speed. Extrapolating the Barcelona runtime to the 16 180 CKM samples gives 448.96 h for the local ray tracing path and 3.95 h for the surrogate. Under a simple 100 W same laptop full load assumption, this corresponds to about 44.9 kWh versus 0.39 kWh. The assumptions are deliberately simple, but the conclusion is robust in direction: the environmental case depends on amortizing the research footprint through many reused predictions, not on claiming that model training itself is free.

5.3 Economic and social impact

Economically, this work is not yet a product ready to take to market. Its value is as a technical prototype showing that, under the fixed 7.125 GHz CKM contract used here, dense ray traced targets can be approximated much faster than rerunning the simulator for every candidate deployment once the target city representation and calibration

protocol are fixed. The main development cost was human time: literature review, dataset handling, training/debugging, analysis, supervision and writing. The compute cost is small compared with this time cost in an academic project, but it is still relevant because the full exploration required many GPU hours. In a planning workflow, a mature version of this surrogate could reduce repeated simulation runs and enable faster scenario studies over elevated transmitter heights or city layouts.

Table 5.2 gives an indicative professional cost view of the BT development. It is not a billed project budget, but it answers the guide’s request to quantify human and material resources, including working hours, hourly costs, computation and operating assumptions. The approximate calendar was literature review and proposal in the first stage, dataset handling and baseline implementation next, experiment design and debugging in the central stage, and writing, figures and review in the final stage, with supervision distributed throughout. No subcontracting, new computer purchase, material purchase or separate tax item is included; electricity is included in the shared compute estimate. Cost was reduced by using open source software, shared university compute, cached outputs and reusable scripts instead of buying dedicated hardware or repeating ray tracing unnecessarily.

Table 5.2: *Indicative economic cost of the BT development.*

Task or resource	Quantity	Unit cost	Cost	Notes
Literature review and proposal	60 h	12 EUR/h	720 EUR	Initial scope and state of the art
Dataset and baseline implementation	90 h	12 EUR/h	1 080 EUR	HDF5, U-Net/cGAN, masking
Experiment design and debugging	180 h	12 EUR/h	2 160 EUR	Priors, experts, evaluation
Writing and figures	120 h	12 EUR/h	1 440 EUR	Thesis text and diagrams
Supervision meetings and supervisors’ dataset simulation time	50 h	40 EUR/h	2 000 EUR	Advisor time
GPU compute, HARP-NET CKM	48 GPU hours	0.25 EUR/GPU hour	12 EUR	4× RTX 2080 Ti, 12 h
GPU compute, distribution and prior phase	72 GPU hours	0.25 EUR/GPU hour	18 EUR	Distribution first models plus prior calibration
Exploratory compute	≈400 GPU hours	0.25 EUR/GPU hour	≈100 EUR	Research iterations
Electricity	N/A	included in GPU hours	0 EUR	Covered by shared compute rate
Storage and code hosting	N/A	free	0 EUR	GitHub and local storage
Total indicative cost			≈7 530 EUR	

The most realistic economic path is a planning tool rather than a standalone consumer product. A mature version could be licensed as software for network planning teams,

integrated into a digital twin workflow, or offered as a service for rapid elevated base station or UAV coverage studies. Operators could use faster CKM generation to compare transmitter heights above roughly 10 m, test temporary base station locations, identify weak streets or squares before deployment, and prioritize the cases where full ray tracing or field validation should be run. Better radio maps would not create capacity by themselves, but they would make planning iterations cheaper and could reduce the risk of sending equipment or measurement teams to poor candidate sites. Patentability would require a novelty analysis beyond this thesis, because CKMs, radio map prediction and hybrid physics neural models already exist in the literature; the more plausible near term value is know how, implementation, calibration data and integration into operator planning workflows.

The economic limitations are therefore not only computational. The project would also need packaging, documentation, support, validation datasets, update procedures and periodic recalibration if it were maintained as planning software. The cost of dismantling is small because the project is software, but archival responsibility remains: code, checkpoints, data provenance and model metadata should be preserved so that future users do not have to rerun the same experiments only to recover the assumptions. The main viability risks are that operators may need other frequencies, antenna patterns, map formats or field validated accuracy levels, which would increase calibration cost. Other projects could still benefit from the reusable priors, mask generation code, evaluation scripts and documented city holdout protocol.

Socially, faster CKM prediction can support temporary coverage planning for public events, rural extensions, emergency response or infrastructure restoration. The direct users would be network planners; the indirect beneficiaries would be users receiving more reliable wireless coverage. The main social risk is overconfidence: a simulated city holdout result should not be treated as certified field performance. Errors should continue to be reported by LoS/NLoS state, transmitter height bin and morphology group rather than hidden behind one global RMSE.

The social value is strongest when the tool is used as decision support rather than as an automatic decision maker. A fast surrogate can help planners compare candidate elevated transmitter heights, identify weak coverage zones, and prioritize detailed simulation or measurement budgets. It should not replace domain review: the model has been validated on simulated CKM cities and inherits the assumptions of the ray tracer, the topology representation, and the ground mask definition.

From the development perspective, the work also had a personal and professional training impact. It required propagation theory, machine learning, HPC workflows, reproducible evaluation, scientific writing and continuous critical review of failed approaches. Inclusive language is used where applicable, and the thesis does not assign technical roles, expected users or beneficiaries by gender. From the project execution perspective, usability matters because the tool would likely be used by engineers rather than by the general public: input metadata, mask policy, antenna height, checkpoint version and limitations should be visible enough that users with different levels of ML expertise can understand what the generated map does and does not certify.

The current sector context is a growing use of digital twins, simulation assisted planning and machine learning surrogates for network design. The same problem could also be

addressed with full ray tracing, measurement campaigns, empirical propagation formulas or commercial planning tools; those options shift the balance between cost, accuracy, transparency and accessibility. The main barriers to this prototype are technical: users need access to map inputs, metadata and basic interpretation of uncertainty and masks. A future interface should provide accessible plots, documented defaults and clear warnings so that age, disability, cultural background or limited ML expertise do not become hidden barriers. A dependency risk also exists: if planners rely only on the surrogate and stop running measurements or high fidelity simulation, users could become vulnerable to systematic model errors.

There is also a fairness dimension in the morphology split. The model is tested on unseen cities, but the training distribution is still finite and urban morphologies are not equally represented. Dense high rise, unusual street canyons, informal settlements, vegetation heavy areas or non European building patterns may behave differently from the dominant CKM regimes. For that reason, future deployment should report uncertainty or at least split diagnostics by morphology and height, instead of using the global average as the only quality indicator.

5.4 Ethical implications and SDGs

The exact need addressed is faster dense CKM generation for UAV and elevated base station planning under the thesis dataset contract. This need was defined by the BT proposal, the supervisors' research context and the broader network planning problem of replacing repeated simulation with documented surrogates where this is justified. Consequences beyond the intended use include misplaced confidence in coverage maps, pressure to skip measurements, or inappropriate use outside the frequency, city and antenna assumptions evaluated here.

The main ethical obligation is honesty about validity. The dataset is generated by ray tracing, so the thesis demonstrates simulator level generalization to unseen cities, not real world deployment performance. A future operational system would require field measurements, calibration and human review, especially before public service or emergency use.

A second ethical issue is transparency. The final predictor is not a pure black box CNN: the attenuation and spread priors expose the geometric and morphological assumptions before the residual network is applied. This makes the system easier to audit, because a user can inspect whether an error comes from the LoS/NLoS mask, the prior, or the learned residual correction. That audit path is particularly relevant for emergency or infrastructure planning, where a wrong map can lead to misplaced confidence in predicted coverage.

The work does not process personal data, but operational radio planning can interact with sensitive contexts such as emergency response, public events or critical infrastructure. The generator should therefore be accompanied by clear metadata: input source, resolution, antenna height, mask generation mode, checkpoint version and known limitations. Keeping that metadata with the prediction is part of the ethical responsibility of making a fast tool: speed is useful only if the output remains traceable.

The project also aligns with the professional ethics expected in engineering and research work by documenting assumptions, avoiding data leakage between cities, reporting

limitations, and keeping the evaluation contract explicit. In UPC terms, the relevant principles are those of the Code of Ethics and the Code of Research Integrity: transparency, honesty in reporting results, responsible use of resources, and respect for the people who may be affected by technical decisions. A future operational version should include human review before coverage maps affect public service, emergency response or critical infrastructure decisions.

The project aligns most directly with SDG 9, because it develops a faster tool for wireless infrastructure planning; SDG 11, because it can support resilient urban and temporary connectivity; and SDG 13, because a validated surrogate can reduce repeated simulation energy. These contributions are indirect: the thesis provides a reproducible method and generator, not an operational network deployment by itself.

In the guide’s language, these SDG links are indirect but specific: SDG 9 is supported through a faster infrastructure planning method, SDG 11 through resilient urban and temporary connectivity planning, and SDG 13 through lower repeated simulation energy when the surrogate is reused responsibly. The work does not by itself deploy infrastructure or guarantee better coverage; it provides a reproducible method that can help future planning tools prioritize where slower simulation or real measurement is most needed.

Overall, the sustainability conclusion is conditional but positive. The research process was computation heavy, but it produced a reusable generator that can reduce many repeated full resolution simulation calls once validated for a target domain. The strongest sustainable use case is therefore not “train a larger model”, but “reuse HARP-NET CKM as a documented surrogate to screen planning options and prioritize slower simulation or real measurement”.

Conclusions and Future Work

6.1 Conclusions

This thesis demonstrates that dense Channel Knowledge Maps for UAV enabled networks can be predicted with varying UAV transmitter height, provided that the problem is treated as a structured propagation task rather than as generic image to image translation. The final system predicts channel attenuation, delay spread, and angular spread on 513×513 city maps under a strict city holdout protocol, with the UAV located at the map centre and its height changing from sample to sample. The work therefore answers the original practical question under the CKM simulation contract: a single surrogate can produce dense CKM targets for unseen cities while conditioning on continuous UAV transmitter height.

The central lesson is that the strongest model is not a pure black box neural network. It is a hybrid system in which calibrated physical and statistical priors explain the stable part of the channel, and a neural residual model corrects what remains. This division of labour is what made the final result reliable: the priors capture radial propagation, visibility, height effects, building layout, and spread scale statistics; the learned model then focuses on local residual structure and distribution mismatch.

The finished pipeline meets the original numerical targets. The standalone path loss prior achieved 1.9246 dB RMSE on the held out validation+test split, and 1.9383 dB when reused as the frozen prior on the final 14-city unseen test split. The spread priors reached 28.10 ns delay spread RMSE and 13.74° angular spread RMSE on the same final test split. HARP-NET CKM improved those anchors to 1.6519 dB, 26.5570 ns, and 11.3854° , respectively. These numbers are meaningful because the baseline being improved is already strong and train calibrated, not a weak baseline.

6.1.1 What the final result means

The clearest contribution is a working CKM predictor for variable height UAV maps. The final model receives geometry, masks, frozen priors, and a continuous height condition, and produces dense channel attenuation, delay spread, and angular spread maps jointly. The height dependence is physically important: the same city footprint changes its visibility structure, two ray interference pattern, and spread regimes as the transmitter altitude changes. The final architecture handles this by combining height aware priors with a sinusoidal height embedding injected through FiLM conditioning.

The channel attenuation result is the strongest success case. Most LoS path loss structure is radial, height dependent, and coherent enough that an enhanced two ray prior can explain much of it before any neural correction is applied. The residual model still matters, but its role is narrower and more realistic: it improves local blocked streets, transition regions, and geometry to radio misalignments that the analytic prior cannot fully capture.

Reducing the test RMSE from 1.9383 dB to 1.6519 dB after such a strong prior is therefore a substantial result.

Delay spread and angular spread are harder to judge by a single RMSE. Their maps contain sparse high value regions, spike like behaviour, and long tailed outliers. Qualitatively, the final predictions usually recover the main spatial structure: low spread and high spread regions appear in the right general parts of the city, and the LoS/NLoS geometry is respected. The remaining mistakes are mostly local outlier failures. Some groups of pixels with unusually high DS or AS are missed, while some false positive high spread patches are predicted where the target remains moderate. This explains why the spread result is less uniform than the channel attenuation result, while still being useful: the model improves the global structure and many local regimes, but rare spread hot spots remain the hardest cases. These RMSE gains should not be read as universal typical error gains: delay spread MAE increases, and angular spread MAE is nearly unchanged, so the spread claim remains an RMSE and structure claim rather than a calibrated error claim. The most important methodological conclusion is that CKM prediction improves when the model head follows the target distribution. Earlier point regression models could produce plausible-looking maps while missing NLoS attenuation modes or suppressing spread tails. The distribution first stage fixed that diagnosis by using GMM style outputs and histogram aware evaluation. The final model keeps that lesson in its head design: it emits mixture parameters around calibrated anchors and exports their local weighted point prediction, while the final reported claim remains the deterministic RMSE improvement selected after the RMSE/MAE dominant fine tuning stage.

6.1.2 Answers to the research questions

RQ1. Can a single deep learning surrogate predict per pixel channel attenuation, delay spread, and angular spread on 513×513 channel knowledge maps, under strict city holdout evaluation, with errors compatible with the thesis targets?

Yes, within the fixed CKM contract used in this thesis. The final model predicts PL, DS, and AS jointly on unseen cities and clears the original targets: 1.6519 dB channel attenuation RMSE, 26.5570 ns delay spread RMSE, and 11.3854° angular spread RMSE. The result should be understood under one carrier frequency, one centred UAV, valid ground receivers only, and city level train/validation/test separation.

RQ2. Which part of the prediction should be left to a calibrated analytic prior and which part should be left to the neural network, so that the hybrid model is both accurate and interpretable? The calibrated priors should handle the most predictable propagation structure: two ray LoS path loss, height-binned LoS calibration, morphology aware NLoS corrections, and log domain ridge models for DS and AS. These priors are strong enough to be useful on their own. The neural model is then valuable because it learns bounded residual corrections around them, reducing the frozen-prior test errors from 1.9383 dB, 28.1023 ns, and 13.7416° to 1.6519 dB, 26.5570 ns, and 11.3854° .

RQ3. How does the continuous UAV transmitter height (12 m to 478 m) affect the prediction error, and is the same architecture able to cover the full height range without per height retraining? Height affects the difficulty of the prediction through visibility, blocking and spread tail behaviour, so it must be represented explicitly. In the final model, height enters twice: through height aware calibrated priors and through a

sinusoidal FiLM conditioning path inside the neural residual model. Low altitude cases contain more blocked links, stronger transition zones and harder spread tails, while high altitude samples are easier even though they are less represented in the training data. This indicates that the height trend is physical, not only a sampling artifact.

6.1.3 Main contributions

The first contribution is the final variable height CKM predictor itself: HARP-NET CKM, a single final CKM model that jointly predicts channel attenuation, delay spread, and angular spread for unseen city layouts while conditioning on continuous UAV transmitter height.

The second contribution is the calibrated prior family. The path loss prior combines free space/radial structure, coherent two ray LoS behaviour, height dependent calibration, NLoS morphology features, and fallback regimes. The spread priors extend the same philosophy to DS and AS through log domain ridge calibration, topology/visibility grouping, height regimes, and robust fallbacks. Together, these priors put a large amount of physical and empirical knowledge into a reproducible estimator that is already competitive before the neural residual is trained.

The third contribution is the distribution first modelling lesson. The thesis shows that target histograms, LoS/NLoS marginals, and spike plus tail spread behaviour are not diagnostic extras; they determine which output head is appropriate. This is why GMM style residual prediction and histogram aware checks became part of the final design.

The fourth contribution is the evaluation contract: city holdout splitting, valid ground receiver masking, LoS/NLoS reporting, topology diagnostics, and transmitter height stratified reporting. This contract is essential because random splits or unmasked building pixels can make the same model appear stronger without improving the deployment-relevant task.

6.1.4 Practical implications and scope

The practical implication is that future CKM systems should be compared under the same contract before architectural claims are made. A new backbone is useful only if it improves held out city metrics after applying the same receiver mask, height conditioning, LoS/NLoS reporting, topology analysis, and prior baselines. Otherwise, a lower error may simply come from solving an easier problem.

The result is strongest as a reproducible surrogate for the CKM simulation contract used here. It is not a universal substitute for ray tracing reference generation across all frequencies, antenna patterns, or cities. It does show, however, that once the dataset contract is fixed, repeated simulator calls can be compressed into a fast and interpretable predictor. In the Barcelona runtime benchmark, pixel level MATLAB ray tracing took 99.89 s per sample, while the complete final generator path took 0.88 s, a 113.7 $\times$ reduction.

6.1.5 Negative results

The negative results are useful mainly as design evidence, so the detailed attempt history is kept in Appendix A rather than repeated in the main conclusion. The short lesson is that pure image translation, larger CNN backbones, topology experts, multi scale outputs, and uncertainty heads each exposed a different failure mode, but none removed the need for strong frozen priors and distribution aware residual learning.

This is why the thesis ends with HARP-NET CKM: the final design is not just a larger neural network, but a bounded residual predictor built around the parts of the channel that can already be explained physically or statistically.

6.2 Future Directions

Short-medium term. The next step is to strengthen the current residual pipeline without changing the evaluation contract, then extend it toward frequency and radio configuration conditioning. Useful additions include transmitter-oriented mixup, explicit corridor losses, and controlled ablations of each residual head and loss weight. The current dataset fixes the carrier and antenna assumptions, while a more general outdoor CKM model should learn how attenuation and spreads vary with frequency, bandwidth, antenna pattern, and possibly polarization. A natural dataset for this stage would simulate the same cities across several radio configurations so that the model can separate geometry effects from configuration effects.

A nearer term training direction is adaptation to new or under represented topology types. The per topology analysis suggests that dense high-rise, compact mid-rise, open low-rise, mixed urban layouts, and future morphology classes do not fail in exactly the same way. A practical continuation would therefore be to fine tune HARP-NET CKM on newly introduced urban typologies, or to train lightweight specialist heads on top of the shared backbone when only limited examples are available. This would test whether the model can recover local structure in topology regimes beyond the original training distribution, rather than only specialising to already defined classes.

For delay and angular spreads, another short to medium term direction is to optimize not only pixel-wise RMSE but also spatial structure. Spread maps can look physically coherent even when small shifts across quantized target boundaries are penalized harshly by RMSE. Future fine tuning should therefore test structural objectives, for example gradient consistency, region wise correlation, SSIM like terms, LoS/NLoS boundary consistency, or topology aware ranking losses, so that the model preserves the correct spread patterns rather than only minimizing the average numeric error.

Medium to long term. The final extension is field calibration. The attenuation and spread priors are interpretable, which makes them suitable for simulator to reality correction using a limited number of real measurements. In the medium to long term, this could lead to a CKM predictor that starts from ray-traced or prior generated maps, adapts to a new city with sparse measurements, and still reports uncertainty by LoS/NLoS regime, transmitter height bin, and topology class. Those safeguards should remain part of the method before considering operational use in emergency coverage, UAV network planning, or rapid deployment studies.

Bibliography

- [1] 3GPP. *Uncrewed Aerial System (UAS) Support in 3GPP; Stage 1*. Tech. rep. TS 22.125. Specification details page, accessed 21 May 2026. 3rd Generation Partnership Project, 2026. URL: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3545>.
- [2] Yong Zeng and Xiaoli Xu. "Toward Environment-Aware 6G Communications via Channel Knowledge Map". In: *IEEE Wireless Communications* 28.3 (2021), pp. 84–91. DOI: [10.1109/MWC.001.2000327](https://doi.org/10.1109/MWC.001.2000327).
- [3] Yong Zeng et al. "A Tutorial on Environment-Aware Communications via Channel Knowledge Map for 6G". In: *IEEE Communications Surveys & Tutorials* 26.3 (2024), pp. 1478–1519. DOI: [10.1109/COMST.2024.3364508](https://doi.org/10.1109/COMST.2024.3364508).
- [4] Xiaoli Xu and Yong Zeng. "How Much Data Is Needed for Channel Knowledge Map Construction?" In: *IEEE Transactions on Wireless Communications* 23.10 (2024), pp. 13011–13021. DOI: [10.1109/TWC.2024.3397964](https://doi.org/10.1109/TWC.2024.3397964).
- [5] Ç. Yapar et al. *Dataset of Pathloss and ToA Radio Maps With Localization Application*. IEEE DataPort dataset. 2022. DOI: [10.21227/0GTX-6V30](https://doi.org/10.21227/0GTX-6V30). URL: <https://doi.org/10.21227/0GTX-6V30>.
- [6] Çağkan Yapar et al. "The First Pathloss Radio Map Prediction Challenge". In: *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2023, pp. 1–2. DOI: [10.1109/ICASSP49357.2023.10433928](https://doi.org/10.1109/ICASSP49357.2023.10433928). URL: <https://doi.org/10.1109/ICASSP49357.2023.10433928>.
- [7] Fabian Jaensch, Giuseppe Caire, and Begüm Demir. *Radio Map Estimation: An Open Dataset with Directive Transmitter Antennas and Initial Experiments*. arXiv:2402.00878. 2024. DOI: [10.48550/arXiv.2402.00878](https://doi.org/10.48550/arXiv.2402.00878). URL: <https://arxiv.org/abs/2402.00878>.
- [8] Zijian Wu et al. "CKMImageNet: A Dataset for AI-Based Channel Knowledge Map Toward Environment-Aware Communication and Sensing". In: *IEEE Transactions on Communications* 73.12 (2025), pp. 14430–14443. DOI: [10.1109/TCOMM.2025.3615778](https://doi.org/10.1109/TCOMM.2025.3615778). URL: <https://doi.org/10.1109/TCOMM.2025.3615778>.
- [9] Theodore S. Rappaport. *Wireless Communications: Principles and Practice*. 2nd ed. Prentice Hall, 2002. URL: <https://www.pearson.com/en-us/subject-catalog/p/wireless-communications-principles-and-practice/P200000003199>.
- [10] Chia-Chuan Chiu et al. "Channel Modeling of Air-to-Ground Signal Measurement with Two-Ray Ground-Reflection Model for UAV Communication Systems". In: *2021 30th Wireless and Optical Communications Conference (WOCC)*. 2021, pp. 251–256. DOI: [10.1109/WOCC53213.2021.9603250](https://doi.org/10.1109/WOCC53213.2021.9603250).
- [11] COST Action 231. *Digital Mobile Radio Towards Future Generation Systems: Final Report*. Tech. rep. Luxembourg: Office for Official Publications of the European Communities, 1999. URL: <https://op.europa.eu/en/publication-detail/-/publication/f2f42003-4028-4496-af95-beaa38fd475f>.

- [12] 3GPP. *Study on Channel Model for Frequencies from 0.5 to 100 GHz*. Tech. rep. TR 38.901. Specification archive, accessed 15 May 2026. 3rd Generation Partnership Project, 2026. URL: https://www.3gpp.org/ftp/Specs/archive/38_series/38.901/.
- [13] A. Al-Hourani, S. Kandeepan, and S. Lardner. “Optimal LAP Altitude for Maximum Coverage”. In: *IEEE Wireless Communications Letters* 3.6 (2014), pp. 569–572. DOI: [10.1109/LWC.2014.2342736](https://doi.org/10.1109/LWC.2014.2342736).
- [14] W. Khawaja et al. “A Survey of Air-to-Ground Propagation Channel Modeling for Unmanned Aerial Vehicles”. In: *IEEE Communications Surveys & Tutorials* 21.3 (2019), pp. 2361–2391. DOI: [10.1109/COMST.2019.2915069](https://doi.org/10.1109/COMST.2019.2915069).
- [15] Abdul Saboor and Evgenii Vinogradov. *Millimeter-Wave UAV Channel Model with Height-Dependent Path Loss and Shadowing in Urban Scenarios*. arXiv:2511.10763. 2025. DOI: [10.48550/arXiv.2511.10763](https://doi.org/10.48550/arXiv.2511.10763). URL: <https://arxiv.org/abs/2511.10763>.
- [16] Evgenii Vinogradov et al. “Spatially Consistent Air-to-Ground Channel Modeling and Simulation via 3D Shadow Projections”. In: *2026 International Conference on Computing, Networking and Communications*. 2026. DOI: [10.1109/ICNC68183.2026.11416865](https://doi.org/10.1109/ICNC68183.2026.11416865).
- [17] P. Kyösti et al. *WINNER II Channel Models*. Tech. rep. IST-4-027756 WINNER II Deliverable, 2007. URL: <https://www.cept.org/files/8339/winner2%20-%20final%20report.pdf>.
- [18] Phillip Isola et al. “Image-to-Image Translation with Conditional Adversarial Networks”. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2017, pp. 1125–1134. URL: <https://arxiv.org/abs/1611.07004>.
- [19] Ron Levie et al. “RadioUNet: Fast Radio Map Estimation With Convolutional Neural Networks”. In: *IEEE Transactions on Wireless Communications* 20.6 (2021), pp. 4001–4015. DOI: [10.1109/TWC.2021.3054977](https://doi.org/10.1109/TWC.2021.3054977). URL: <https://arxiv.org/abs/1911.09002>.
- [20] Ziyue Yang et al. *Group Equivariant Convolutional Networks for Pathloss Estimation*. arXiv:2511.17841. 2025. DOI: [10.48550/arXiv.2511.17841](https://doi.org/10.48550/arXiv.2511.17841). URL: <https://arxiv.org/abs/2511.17841>.
- [21] Ju-Hyung Lee et al. “PMNet: Large-Scale Channel Prediction System for ICASSP 2023 First Pathloss Radio Map Prediction Challenge”. In: *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2023, pp. 1–2. DOI: [10.1109/ICASSP49357.2023.10095257](https://doi.org/10.1109/ICASSP49357.2023.10095257). URL: <https://doi.org/10.1109/ICASSP49357.2023.10095257>.
- [22] Yuxuan Li et al. “RMTransformer: Accurate Radio Map Construction and Coverage Prediction”. In: *2025 IEEE 101st Vehicular Technology Conference (VTC2025-Spring)*. 2025, pp. 1–5. DOI: [10.1109/VTC2025-Spring65109.2025.11174709](https://doi.org/10.1109/VTC2025-Spring65109.2025.11174709). URL: <https://doi.org/10.1109/VTC2025-Spring65109.2025.11174709>.
- [23] Mingran Sun et al. *WiCo-PG: Wireless Channel Foundation Model for Pathloss Map Generation via Synesthesia of Machines*. arXiv:2511.15030. 2025. DOI: [10.48550/arXiv.2511.15030](https://doi.org/10.48550/arXiv.2511.15030). URL: <https://arxiv.org/abs/2511.15030>.

- [24] Dong Yang et al. *FM-RME: Foundation Model Empowered Radio Map Estimation*. arXiv:2602.22231. 2026. DOI: [10.48550/arXiv.2602.22231](https://doi.org/10.48550/arXiv.2602.22231). URL: <https://arxiv.org/abs/2602.22231>.
- [25] Zhiyuan Liu et al. "A Fine-Grained 3D Radio Map Construction Paradigm With Ultra-Low Sampling Rates by Large Generative Models". In: *IEEE Journal on Selected Areas in Communications* 44 (2026), pp. 4397–4413. DOI: [10.1109/JSAC.2026.3677149](https://doi.org/10.1109/JSAC.2026.3677149). URL: <https://doi.org/10.1109/JSAC.2026.3677149>.
- [26] Wangqian Chen and Juntong Chen. "Diffraction and Scattering Aware Radio Map and Environment Reconstruction Using Geometry Model-Assisted Deep Learning". In: *IEEE Transactions on Wireless Communications* 23.12 (2024), pp. 19804–19819. DOI: [10.1109/TWC.2024.3487225](https://doi.org/10.1109/TWC.2024.3487225). URL: <https://arxiv.org/abs/2403.00229>.
- [27] Mukaram Shahid et al. "ReVeal: A Physics-Informed Neural Network for High-Fidelity Radio Environment Mapping". In: *2025 IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN)*. 2025, pp. 1–10. DOI: [10.1109/DYSPAN64764.2025.11115911](https://doi.org/10.1109/DYSPAN64764.2025.11115911). URL: <https://doi.org/10.1109/DYSPAN64764.2025.11115911>.
- [28] Borui Huang et al. "Transformer-Based Air-to-Ground mmWave Channel Characteristics Prediction for 6G UAV Communications". In: *Sensors* 25.12 (2025), p. 3731. DOI: [10.3390/s25123731](https://doi.org/10.3390/s25123731). URL: <https://www.mdpi.com/1424-8220/25/12/3731>.
- [29] Alex Kendall and Yarin Gal. "What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision?" In: *Advances in Neural Information Processing Systems*. 2017. URL: <https://arxiv.org/abs/1703.04977>.
- [30] Taha Saleh et al. "Probabilistic Path Loss Predictors for mmWave Networks". In: *Proceedings of the IEEE 93rd Vehicular Technology Conference*. 2021, pp. 1–6. DOI: [10.1109/VTC2021-Spring51267.2021.9448967](https://doi.org/10.1109/VTC2021-Spring51267.2021.9448967). URL: <https://doi.org/10.1109/VTC2021-Spring51267.2021.9448967>.
- [31] Dolores Garcia Marti et al. "A Mixture Density Channel Model for Deep Learning-Based Wireless Physical Layer Design". In: *Proceedings of the 23rd International ACM Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems*. ACM, 2020, pp. 53–62. DOI: [10.1145/3416010.3423229](https://doi.org/10.1145/3416010.3423229). URL: <https://dSPACE.networks.imdea.org/handle/20.500.12761/886>.
- [32] Youngmin Lee, Xiaomin Ma, and Andrew S. I. D. Lang. "Modeling of Time-varying Wireless Communication Channel with Fading and Shadowing". In: *2024 IEEE 100th Vehicular Technology Conference (VTC2024-Fall)*. 2024, pp. 1–6. DOI: [10.1109/VTC2024-Fall63153.2024.10757531](https://doi.org/10.1109/VTC2024-Fall63153.2024.10757531). URL: <https://doi.org/10.1109/VTC2024-Fall63153.2024.10757531>.
- [33] Xiucheng Wang et al. "RadioDiff: An Effective Generative Diffusion Model for Sampling-Free Dynamic Radio Map Construction". In: *IEEE Transactions on Cognitive Communications and Networking* 11.2 (2025), pp. 738–750. DOI: [10.1109/TCCN.2024.3504489](https://doi.org/10.1109/TCCN.2024.3504489). URL: <https://arxiv.org/abs/2408.08593>.
- [34] Ethan Perez et al. "FiLM: Visual Reasoning with a General Conditioning Layer". In: *Proceedings of the AAAI Conference on Artificial Intelligence*. 2018. DOI: [10.1609/aaai.v32i1.11671](https://doi.org/10.1609/aaai.v32i1.11671). URL: <https://arxiv.org/abs/1709.07871>.

- [35] Pavel Izmailov et al. “Averaging Weights Leads to Wider Optima and Better Generalization”. In: *Proceedings of the 34th Conference on Uncertainty in Artificial Intelligence*. 2018. URL: <https://arxiv.org/abs/1803.05407>.
- [36] Ali Saeizadeh et al. *AIRMap: AI-Generated Radio Maps for Wireless Digital Twins*. arXiv:2511.05522. 2025. DOI: [10.48550/arXiv.2511.05522](https://arxiv.org/abs/2511.05522). URL: <https://arxiv.org/abs/2511.05522>.
- [37] Yuan Gao et al. “Effective Outdoor Pathloss Prediction: A Multi-Layer Segmentation Approach With Weighting Map”. In: *IEEE Transactions on Vehicular Technology* (2026), pp. 1–6. DOI: [10.1109/TVT.2026.3658966](https://doi.org/10.1109/TVT.2026.3658966). URL: <https://doi.org/10.1109/TVT.2026.3658966>.
- [38] Ferdaous Tarhouni, Muneer AlZubi, and Mohamed-Slim Alouini. “Machine Learning-based Path Loss Prediction in Suburban Environment in the Sub-6 GHz Band”. In: *IEEE Transactions on Vehicular Technology* (2025), pp. 1–11. DOI: [10.1109/TVT.2025.3633604](https://doi.org/10.1109/TVT.2025.3633604). URL: <https://doi.org/10.1109/TVT.2025.3633604>.
- [39] Zhijie Zhong et al. “PathFinder: Advancing Path Loss Prediction for Single-to-Multi-Transmitter Scenario”. In: *Pattern Recognition* 179 (2026), p. 113725. DOI: [10.1016/j.patcog.2026.113725](https://doi.org/10.1016/j.patcog.2026.113725). URL: <https://doi.org/10.1016/j.patcog.2026.113725>.
- [40] Stefanos Bakirtzis et al. “The First Indoor Pathloss Radio Map Prediction Challenge”. In: *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2025, pp. 1–2. DOI: [10.1109/ICASSP49660.2025.10889381](https://doi.org/10.1109/ICASSP49660.2025.10889381). URL: <https://doi.org/10.1109/ICASSP49660.2025.10889381>.
- [41] ICASSP 2025 Indoor Radio Map Challenge. *The First Indoor Pathloss Radio Map Prediction Challenge: Results*. 2025. URL: <https://indoorradiomapchallenge.github.io/results.html>.
- [42] Wenlihan Lu et al. “SIP2Net: Situational-Aware Indoor Pathloss-Map Prediction Network for Radio Map Generation”. In: *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2025, pp. 1–2. DOI: [10.1109/ICASSP49660.2025.10890319](https://doi.org/10.1109/ICASSP49660.2025.10890319).
- [43] Bin Feng et al. “IPP-Net: A Generalizable Deep Neural Network Model for Indoor Pathloss Radio Map Prediction”. In: *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2025, pp. 1–2. DOI: [10.1109/ICASSP49660.2025.10890663](https://arxiv.org/abs/2501.06414). URL: <https://arxiv.org/abs/2501.06414>.
- [44] Xin Li et al. “TransPathNet: A Novel Two-Stage Framework for Indoor Radio Map Prediction”. In: *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2025, pp. 1–2. DOI: [10.1109/ICASSP49660.2025.10887939](https://arxiv.org/abs/2501.16023). URL: <https://arxiv.org/abs/2501.16023>.
- [45] Zeyao Sun et al. “RadioPiT: Radio Map Generation with Pixel Transformer Driven by Ultra-Sparse Real-World Data”. In: *2025 30th Asia-Pacific Conference on Communications (APCC)*. 2025, pp. 1–6. DOI: [10.23919/APCC64555.2025.11279807](https://doi.org/10.23919/APCC64555.2025.11279807). URL: <https://doi.org/10.23919/APCC64555.2025.11279807>.
- [46] Ju-Hyung Lee et al. “PMNet: Robust Pathloss Map Prediction via Supervised Learning”. In: *Proc. IEEE Global Communications Conference (GLOBECOM)*. 2023. DOI: [10.1109/GLOBECOM54140.2023.10437562](https://arxiv.org/abs/2211.10527). URL: <https://arxiv.org/abs/2211.10527>.

- [47] Guanshu Yang et al. “Machine-Learning-Based Prediction Methods for Path Loss and Delay Spread in Air-to-Ground Millimetre-Wave Channels”. In: *IET Microwaves, Antennas & Propagation* 13.8 (2019), pp. 1113–1121. DOI: [10.1049/iet-map.2018.6187](https://doi.org/10.1049/iet-map.2018.6187). URL: <https://onlinelibrary.wiley.com/doi/10.1049/iet-map.2018.6187>.
- [48] Andrea Goldsmith. *Wireless Communications*. Cambridge University Press, 2005. DOI: [10.1017/CB09780511841224](https://doi.org/10.1017/CB09780511841224).
- [49] J. Walfisch and H. L. Bertoni. “A Theoretical Model of UHF Propagation in Urban Environments”. In: *IEEE Transactions on Antennas and Propagation* 36.12 (1988), pp. 1788–1796. DOI: [10.1109/8.14401](https://doi.org/10.1109/8.14401).
- [50] F. Ikegami et al. “Propagation Factors Controlling Mean Field Strength on Urban Streets”. In: *IEEE Transactions on Antennas and Propagation* 32.8 (1984), pp. 822–829. DOI: [10.1109/TAP.1984.1143419](https://doi.org/10.1109/TAP.1984.1143419).
- [51] Rong-Terng Juang. “Explainable Deep-Learning-Based Path Loss Prediction from Path Profiles in Urban Environments”. In: *Applied Sciences* 11.15 (2021), p. 6690. DOI: [10.3390/app11156690](https://doi.org/10.3390/app11156690).
- [52] Xuesong Cai et al. “An Empirical Air-to-Ground Channel Model Based on Passive Measurements in LTE”. In: *IEEE Transactions on Vehicular Technology* 68.2 (2019), pp. 1140–1154. DOI: [10.1109/TVT.2018.2886961](https://doi.org/10.1109/TVT.2018.2886961). URL: <https://doi.org/10.1109/TVT.2018.2886961>.
- [53] T. Izydorczyk et al. “Angular Distribution of Cellular Signals for UAVs in Urban and Rural Scenarios”. In: *Proc. EuCAP*. 2019. URL: <https://ieeexplore.ieee.org/document/8739910>.
- [54] Guillem Moreno Garcia. *TFGAllProgress_Tries_and_Attempts: Development History and Experimental Attempts*. Public GitHub repository. Snapshot commit 7c13efd51708. 2026. URL: https://github.com/unworthyzeus/TFGAllProgress_Tries_and_Attempts (visited on 05/06/2026).
- [55] Hang Mi et al. “Measurement-Based Prediction of mmWave Channel Parameters Using Deep Learning and Point Cloud”. In: *IEEE Open Journal of Vehicular Technology* 5 (2024), pp. 1059–1072. DOI: [10.1109/OJVT.2024.3436857](https://doi.org/10.1109/OJVT.2024.3436857). URL: <https://doi.org/10.1109/OJVT.2024.3436857>.
- [56] Guillem Moreno Garcia. *CKMGenerator: Generator of Path Loss, Delay Spread and Angular Spread*. Public GitHub repository. Snapshot commit 8b6c5fd26c30. 2026. URL: <https://github.com/unworthyzeus/CKMGenerator> (visited on 05/06/2026).
- [57] Guillem Moreno Garcia. *Final_Code_TFG: Final Training and Evaluation Code*. Public GitHub repository. Snapshot commit 63d12a3d1a05. 2026. URL: https://github.com/unworthyzeus/Final_Code_TFG (visited on 05/06/2026).
- [58] Guillem Moreno Garcia. *TFG_ActualText_LaTeX: Thesis Source*. Public GitHub repository. Snapshot tag final-thesis-2026-05-15. 2026. URL: https://github.com/unworthyzeus/TFG_ActualText_LaTeX (visited on 05/06/2026).

Detailed Attempt Log

This appendix records the detailed experimental sequence used to build the final thesis narrative. It is intentionally more operational than the main chapters: the purpose is to show how the final design emerged from earlier failures, not to re-argue every result.

A.1 How to read the attempt numbers

The attempt numbers are internal experiment identifiers. They are kept here because they make the development path auditable, but they should not be read as eighty independent thesis contributions. Many intermediate runs tested one small change inside a larger phase. The table below maps each number to the public technical role it plays in the thesis narrative.

The main chapters use only the attempts that carry a methodological or quantitative claim; this appendix keeps the full numbering context. The chronology is separated from the final formulas: Chapter 3 contains both the methodology and the mathematical audit trail, while Chapter 4 contains the quantitative results.

A.2 Try-number map

Table A.1: *Try-number map, early image-regression phase.*

Try	Public definition
1	Initial U-Net / pix2pix-style cGAN proof of concept for aligned CKM image to image regression.
2	Linear-scale path-loss training variant: physical normalization, dual dB/linear reporting, and optional distance-from-centre channel.
3	Hybrid path-loss branch on the early U-Net/cGAN base: coarse path-loss head, confidence head, saturation masking, and fused evaluation/export.
4	Differentiable fused-map path-loss objective in dB, turning the hybrid branch into the optimized prediction rather than only an export.
5	TinyGAN / BatchNorm / low-resolution discriminator stabilization run for the early cGAN family.
6	Second early cGAN stabilization run, keeping the TinyGAN-style low-resolution discriminator while checking repeatability.
7	Additional cluster replication of the low-resolution-discriminator cGAN baseline before changing the physical inputs.
8	Final early image to image baseline before explicit height and visibility conditioning became central.
9	First explicit UAV-height conditioning run, introducing FiLM-style height modulation beside the older image to image baseline.
10	LoS/NLoS split path-loss run, comparing visibility-specific training with and without height conditioning.
11	Follow-up height/visibility conditioning pivot, used to separate geometry information from pure adversarial image translation.
12	Late geometry-conditioning continuation before committing to FiLM-specialized LoS/NLoS branches.
13	FiLM-only cGAN variant, isolating height modulation without the full visibility-specialized split.
14	Separate LoS-FiLM and NLoS-FiLM branches; last geometry-conditioning run before artifact-control work.
15	Start of artifact-control phase: GroupNorm, distance features, weaker GAN pressure, and smoothing-oriented variants.
16	Artifact-control continuation focused on keeping the U-Net/cGAN structure while tuning normalization and distance features.
17	Training-stability artifact run before the decoder was changed, mainly testing whether artifacts came from optimization rather than architecture.
18	Late artifact-control branch with blended outputs and cluster experiment bookkeeping; still pre-decoder-fix.
19	Final artifact-control run before replacing the decoder path with explicit bilinear and multiscale tests.
20	Bilinear decoder change; reduced checkerboard artifacts.
21	Multiscale supervision change; improved large scale field learning.
22	Combined bilinear decoder and multiscale supervision; strongest clean pre-prior baseline.

Table A.2: *Try-number map, spread branches, masks, and first priors.*

Try	Public definition
23	Delay/angular multiscale branch, transferring the Try 21/22 multiscale idea to spread-related targets.
24	Intermediate spread-branch continuation before attention was added; retained as a small internal pivot.
25	Bottleneck-attention spread/path-loss branch, testing whether global context in the compressed representation helped.
26	Delay/angular gradient target/loss run, making spread prediction more sensitive to spatial change rather than only absolute value.
27	Intermediate spread/prior exploratory run before topology attention; retained as a small internal pivot.
28	Topology-attention variant, using environment class information more directly in the spread/path-loss branch.
29	Radial path-loss objective variant, testing whether distance-structured supervision reduced physically implausible maps.
30	Spread-priority objective, deliberately weighting delay/angular spread behaviour more heavily than the path-loss baseline.
31	Prior-residual path-loss branch, early move from direct image regression toward learning corrections over a physical estimate.
32	Support-amplitude spread decomposition, last spread/prior exploratory run before the ground-mask contract changed.
33	Ground-only receiver mask becomes mandatory; building pixels are excluded from loss and metrics.
34	Restart under the corrected ground-mask evaluation contract, used to make post-mask scores comparable.
35	Spread-plus-building-mask run, applying the corrected receiver mask to spread-oriented learning.
36	Second spread-building-mask run, checking whether the masked spread branch was stable across reruns.
37	Building-mask path-loss rerun, returning the corrected receiver contract to path-loss prediction.
38	Hybrid two ray prior input run, injecting a physical propagation prior as an input channel rather than only a loss/metric.
39	Building-mask continuation before Try 40; small internal pivot for masked evaluation and spread/path-loss balance.
40	Spread-building-mask consolidation, final masked-evaluation run before explicit prior-residual learning.

Table A.3: *Try-number map, prior-residual and PMNet-style phase.*

Try	Public definition
41	Explicit physical-prior plus learned-residual formulation; raw prior and train only calibrated prior were diagnosed.
42	PMNet-style residual regressor; improved over prior-only calibration but NLoS remained dominant.
43	PMNet no-prior direct-control run, testing how much of the result came from the learned network without a physical residual anchor.
44	No-prior PMNet-v3-style control; showed that LoS could be learned but NLoS remained weak.
45	PMNet residual run tied to the formal prior formulas and calibration notes; bridge between simple prior residuals and MoE routing.
46	LoS/NLoS mixture-of-experts prior run, splitting the residual learner by visibility regime.
47	U-Net prior-residual MoE with obstruction-feature precomputation and calibration, replacing the PMNet-only route.
48	Warm-started Try 42 branch with widened 96-channel PMNet and stage-2 refinement chains.
49	PMNet stage-1 w112 prior branch with MAE-dominant ablation and tail-refiner stage 2; reduced average error but left hard NLoS tails.
50	Prior-research/NLoS sandbox with formula-only, hybrid, HGBoost, tabular, and MLP-delta variants; confirmed scalar corrections over LoS-shaped priors were insufficient.

Table A.4: *Try-number map, PMHHNet, regularization, and distribution diagnosis.*

Try	Public definition
51	Literature-backed PMNet prior stage-1, using a wider w112 branch and a stage-2 tail refiner.
52	Paper-routed city-type-aware PMNet branch: 112 channels, three city-type NLoS experts, automatic city-type routing, and a Try 52 teacher refiner.
53	Literature/cyclic PMNet branch with stage-2 teacher refinement and an added stage-3 NLoS global-context cyclic refiner.
54	First full partitioned-expert PMHHNet branch: six topology experts, topology classifier, continuous height conditioning, and no-data head.
55	Same partitioned PMHHNet layout as Try 54, but with the objective changed toward final map RMSE plus auxiliary no-data loss and EMA checkpoint smoothing as a training parameter.
56	Try 26 U-Net family converted into six topology experts, with topology-mask input, no-data output, and stronger configurable dropout.
57	Partitioned PMHHNet final-map-RMSE branch deployed across the six topology experts after the Try 54 and Try 55 objective change.
58	Try 26 topology expert generator profile: topology mask, no-data BCE head, stronger dropout, AdamW and EMA training defaults, and rewind to best.
59	Second Try 26 topology-expert generator-only pass under the Try 59 branch, later kept as a separate spread/prior source rather than merged with Try 58.
60	No-prior ablation of the Try 57 family: direct path-loss prediction from topology, LoS, distance, and height without formula/confidence prior channels.
61	No-prior NLoS-focused objective: strong LoS/NLoS reweighting, explicit NLoS term, composite checkpoint, and open-sparse-vertical split into LoS/NLoS experts.
62	Paper-like reset after Try 61: restored formula prior, added obstruction proxy channels, removed manual NLoS/no-data losses, and used a stage-2 refiner.
63	Coarse-to-fine Try 62 follow-up: cheap 128×128 stage 1 plus full 513×513 stage 2, with obstruction proxies still enabled.
64	Coarse-to-fine Try 63 follow-up with obstruction proxies disabled, isolating resolution/refinement from explicit blocker features.
65	Grokking-style stress test: single-stage 513×513 training, no early stopping, no rewind, no stage 2, high constant learning rate/weight decay, and very long horizon.
66	Consolidated PMHHNet reference: continuous multi-altitude prediction, sinusoidal height embedding into FiLM, physical channels, and topology experts; LoS was reasonable, but NLoS remained the bottleneck.
67	Try 66-family geometry-assisted pivot with knife-edge/physical-channel ideas; start of the late stabilization and diagnostic phase.
68	Try 66 rerun with PMHHNet stem, high-frequency-path fix, and CutMix removed when height FiLM was active.
69	Knife-edge and LoS/NLoS residual diagnostic with dense/mixed/open macro experts plus legacy sparse expert variants.
70	Try 68 PMHHNet with multi-scale quad residual heads at 513, 257, 129, and 65 grids.
71	Heteroscedastic uncertainty diagnostic; identified hard pixels but did not solve the map error.
72	Try 70-era open-sparse-lowrise expert continuation, checking whether the multi-scale PMHHNet refinements transferred to the hardest low-rise sparse regime.
73	Open-sparse-lowrise plus open-sparse-vertical continuation, broadening the Try 72 late-PMHHNet diagnostic to vertical sparse cases.
74	Band 45 to 55 all-city LoS/NLoS histogram experts, used to expose distribution mismatch and high-tail underprediction.
75	All-city small-LoS expert and decisive histogram diagnosis: NLoS path loss and spread tails required distribution aware modelling.

Table A.5: *Attempt map, final distribution, prior, and joint model phase.*

Internal line	Public definition
76	Distribution first GMM path-loss architecture: sample level mixture estimation plus pixel assignment.
77	Distribution first spread architecture: delay/angular spread GMM and spike-plus-tail modelling.
Calibrated path loss prior	Enhanced two ray LoS branch plus a legacy mixed branch linear calibration table with 18 visibility/morphology/height regimes, from which the final assembly uses the NLoS pixels.
Calibrated spread priors	Log-domain ridge calibration for delay and angular spread, with 36 exact regimes per metric and fallbacks.
HARP-NET CKM	Frozen path loss and spread priors plus a shared neural residual model selected through the wide residual model, relaxed residual bounds, a large capacity stage, and the final residual fine tuning stage.

A.3 Expanded chronological timeline

A.3.1 Early image to image baselines

Tries 1 to 8 treated CKM prediction as aligned image to image regression on the same 513×513 grid used by the targets. This was a natural first formulation because the input maps and output maps are spatially registered. The early U-Net and pix2pix-style cGAN runs established the data pipeline and showed that dense convolutional prediction was feasible at map level.

The phase ended because adversarial image realism was the wrong proxy for radio map accuracy. A visually plausible shadow could still be tens of decibels wrong, and the metric rewards calibrated physical values under a ground mask, not texture realism.

A.3.2 Geometry conditioning and artifact control

Tries 9 to 14 introduced explicit UAV-height conditioning and LoS/NLoS-aware diagnosis. Height could not remain hidden in the sample identity because it changes path length and reflection geometry. Likewise, LoS and NLoS pixels follow different propagation laws, so mixing them into one global number made failure analysis unreliable.

Tries 15 to 22 then attacked visible artifacts and missing large scale structure. GroupNorm was better suited to the small full-resolution batches than BatchNorm, distance channels exposed radial geometry directly, Try 20 isolated bilinear upsampling, Try 21 isolated multiscale supervision, and Try 22 combined both. In the early contract, Try 22 was the strongest clean pre-prior path-loss baseline at about 16.72 dB. The phase ended because the maps became cleaner, but the model still had to rediscover physics from pixels.

A.3.3 Masking and evaluation contract

Tries 33 to 40 are important mainly because they changed the metric contract. Pixels inside buildings are not valid receiver locations, so after Try 33 the project excluded $topology_map \neq 0$ pixels from training loss, validation/test metrics, and error-map visualization. This made the task stricter and made older scores non-comparable except as historical trend markers.

The lesson that survived into the final thesis is simple: every model, prior, and residual head must be evaluated on the same valid-receiver mask. This is why the main chapters treat old direct-regression rows as chronology rather than as a homogeneous leaderboard.

A.3.4 Prior-guided residual learning

Tries 41 to 50 moved from direct prediction toward residual learning over a physical estimate. Try 41 made the residual formulation explicit. The raw formula prior was poor, around 67.23 dB, but train only regime-aware calibration improved it to about 24.16 dB. This showed that physical structure mattered, but also that a formula alone could not replace learning.

Try 42 moved to a PMNet-style residual regressor. Try 44 showed that a no-prior PMNet-v3-style control could learn much of the smooth LoS component but still left NLoS weak. Try 49 added prior confidence and robust loss logic, reaching about 18.87 dB in its stage-1 branch while still leaving hard NLoS tails. Try 50 then served as negative evidence: richer scalar NLoS corrections and tabular/MLP-delta variants did not solve blocked-link error. The phase ended because the remaining error was distributional, not merely a missing scalar correction.

A.3.5 PMHHNet and expert-routing phase

Tries 51 to 66 expanded the residual idea into PMHHNet and topology-expert training. The important knobs were physical channels, height FiLM, high-frequency paths, six topology experts, no-data handling, obstruction features, coarse-to-fine refinement, and controlled no-prior ablations. The rows in the map above separate these attempts because they were not interchangeable: for example, Try 52 was a paper-routed city-type-aware PMNet branch, while Try 63 was a coarse-to-fine Try 62 follow-up with obstruction proxies still enabled.

Try 66 is the reference point from this phase. It reached 9.41 dB overall, with LoS near 3.75 dB, but NLoS remained near 31.48 dB. The project was no longer failing uniformly. The phase ended because backbone engineering improved the easy regime, but the hard NLoS tail still dominated the physical error.

A.3.6 Regularization and distribution diagnosis

Tries 67 to 75 tested whether late training refinements could rescue the PMHHNet family. Geometry assisted channels, high frequency path fixes, multi scale quad heads, uncertainty heads, checkpoint smoothing parameters such as EMA or SWA, and histogram focused experts all helped diagnosis. They made the failure mode clearer, but they did not change the central fact that point regression could collapse hard tails.

Try 74 and Try 75 were the turning point because histogram analysis made the distribution mismatch visible. The target was not simply a smooth surface with local noise. NLoS path loss and the spread targets had tails and regime structure that needed either distribution aware modelling or a much stronger prior. This is why the next phase turned away from larger encoders as the main answer.

A.3.7 Distribution first modelling

The distribution first path-loss GMM line reframed NLoS path loss as a distributional problem. Instead of forcing one value everywhere, Stage A estimated sample level mixture structure and Stage B placed components spatially. This produced much stronger NLoS behaviour, with the DirectML summary around 3.34 dB pixel-weighted RMSE. The key lesson was that a model can approach the hard regime by estimating the right distribution

before assigning pixels.

The distribution first spread expert line transferred the same philosophy to delay and angular spread. It was not the final deployed spread baseline, but it explained why spread targets should not be treated as ordinary smooth images. Together, the two distribution first lines are the conceptual bridge to the mixture based HARP-NET CKM residual head retained in the final attempt.

A.3.8 Frozen calibrated priors

The calibrated path-loss prior showed that LoS path loss was more deterministic than the neural history suggested. The dominant residual over FSPL was a radial ring field, well explained by the enhanced two ray model with fitted effective reflection parameters. Its stored mixed-branch linear calibration table contains 18 visibility, morphology, and height regimes. In the final frozen path loss assembly, the LoS pixels come from the enhanced two ray branch, while the NLoS pixels come from that calibrated branch.

The calibrated spread-prior module applied the same estimator discipline to delay spread and angular spread: use log-domain ridge calibration over geometry, LoS/NLoS state, height, topology, and local morphology. Its final taxonomy is 36 exact regimes per metric, 72 exact regimes total, and 114 keys when fallback entries are included. These priors moved the final system away from “deep learning versus physics” and toward “frozen calibrated estimator plus learned residual.”

A.3.9 HARP-NET CKM

The final attempt is HARP-NET CKM. It starts from the frozen path loss prior and spread priors, then learns bounded residual corrections for path loss, delay spread, and angular spread with one shared multi task network. The GMM head remains part of the residual output design even when the selected final model is chosen by deterministic validation RMSE.

The first training runs for this final line gave only small improvements over the frozen priors, which was itself useful information: the priors were already strong and the residual model was initially too constrained. The final training path therefore moved through a wide residual model, relaxed residual bounds, a large-capacity residual model, and the final residual fine tuning stage. The last stage kept the large architecture and relaxed caps, but used an RMSE-dominant fine-tune with light anchor and prior-guard terms. This is the selected final model reported in Chapter 4.

Representative HARP-NET CKM Inspection Panels

This appendix keeps three representative inspection panels: one open, one mixed, and one dense urban case. The full panel manifest and additional figures remain in the generated figure archive, but the submitted thesis keeps only these examples to avoid turning the appendix into a gallery.

Each panel uses the tall 6×3 layout. The six inspection stages are rows and the three targets are columns. The rows are:

1. **Ground truth:** the ray-traced reference field.
2. **Frozen calibrated prior:** the calibrated analytic prior that HARP-NET CKM receives as an input channel and anchors to.
3. **HARP-NET CKM prediction:** the residual-over-prior output of the joint model, in the same colour scale as columns 1 and 2.
4. **Context:** topology height for the path-loss row, NLoS support for the delay spread row, and LoS support for the angular spread row.
5. **Prior error:** the signed difference Prior – GT. Red indicates a prior overshoot, blue indicates a prior undershoot, and white indicates where the analytic prior is already accurate.
6. **Model correction:** the signed residual Model – Prior that the neural head applies on top of the prior. If the model has learned a useful residual, row 6 should *anti-correlate* with row 5: where the prior is too high, the model should push down, and vice versa.

Rows 5 and 6 are therefore the main *diagnostic* rows: the thesis claim is that HARP-NET CKM is a bounded residual corrector, and the visible agreement (in sign and location) between the prior-error map and the model-correction map is the direct qualitative evidence for that claim; in the PNGs, colourbars use 1st to 99th percentile clipping, while RMSE is computed on all valid pixels.

Table B.1: HARP-NET CKM appendix panels kept in the submitted thesis.

Expert	City	Sample	Height bin
open/low	Segovia	sample_13213	low
mixed/mid	Nice	sample_10828	mid
dense/high	Tunis	sample_14694	high

CHAPTER B. REPRESENTATIVE HARP-NET CKM INSPECTION PANELS

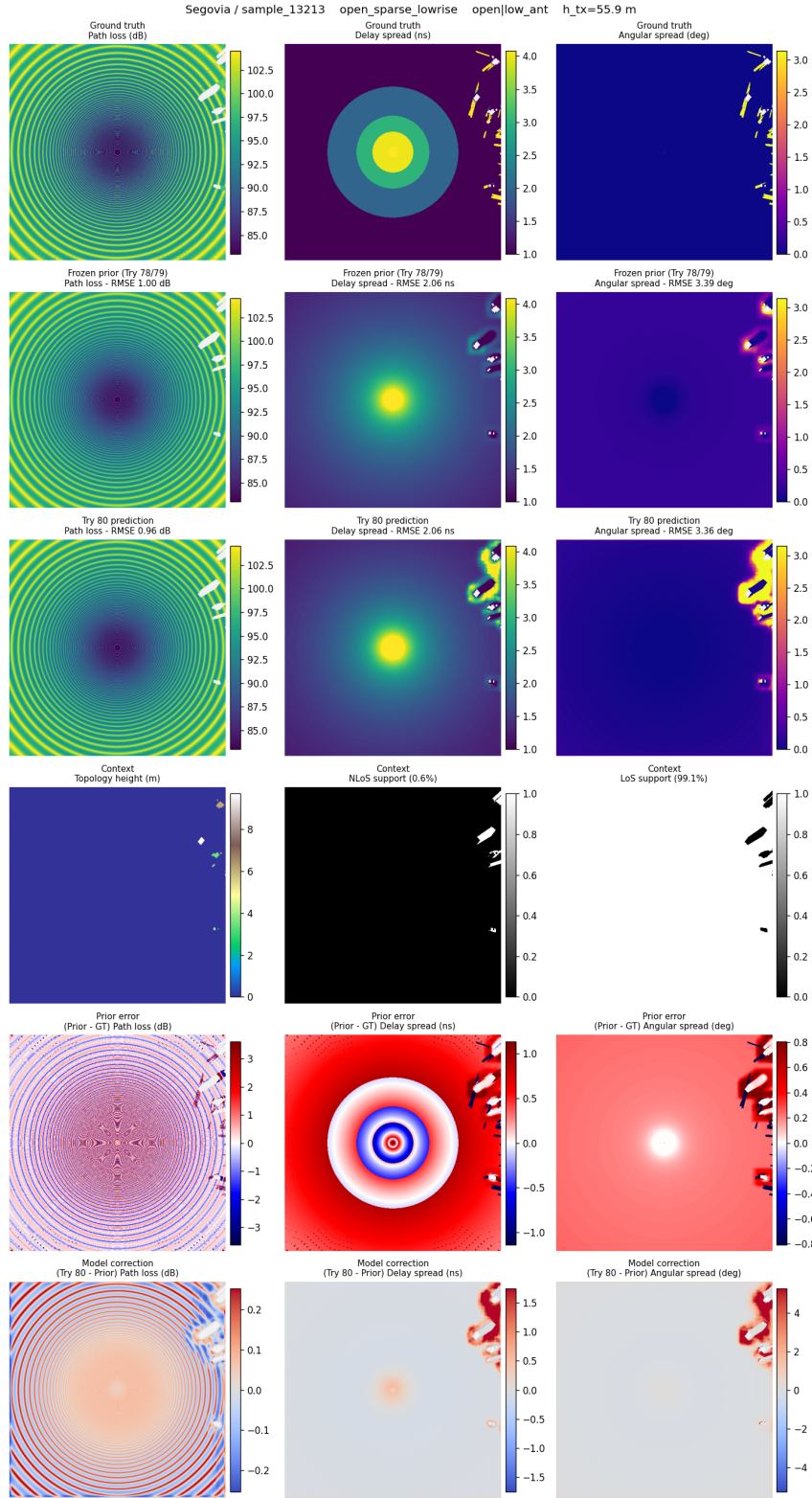


Figure B.1: HARP-NET CKM 6×3 inspection panel for an open, low-antenna validation sample in Segovia.

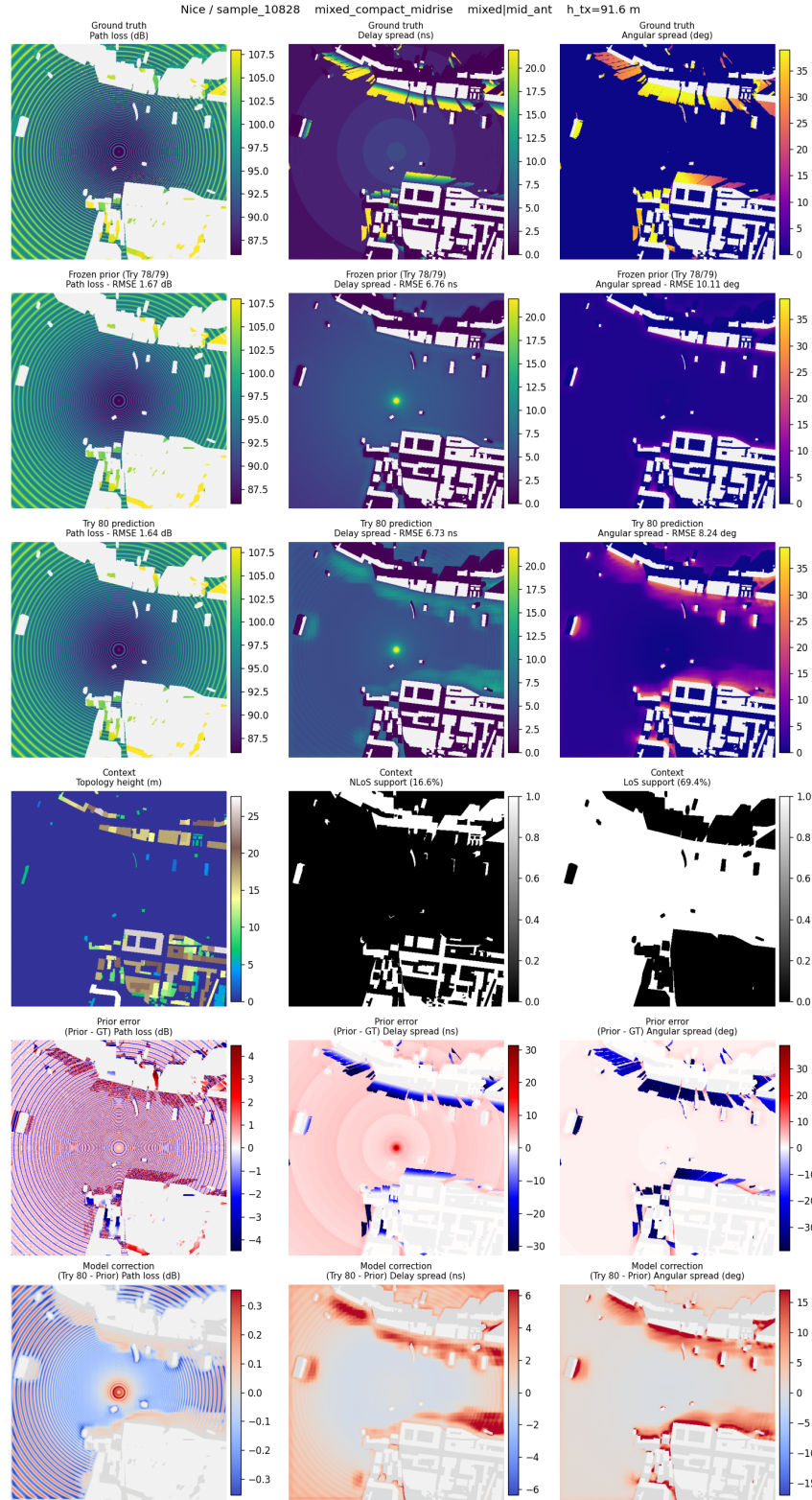


Figure B.2: HARP-NET CKM 6×3 inspection panel for a mixed, mid-antenna validation sample in Nice.

CHAPTER B. REPRESENTATIVE HARP-NET CKM INSPECTION PANELS

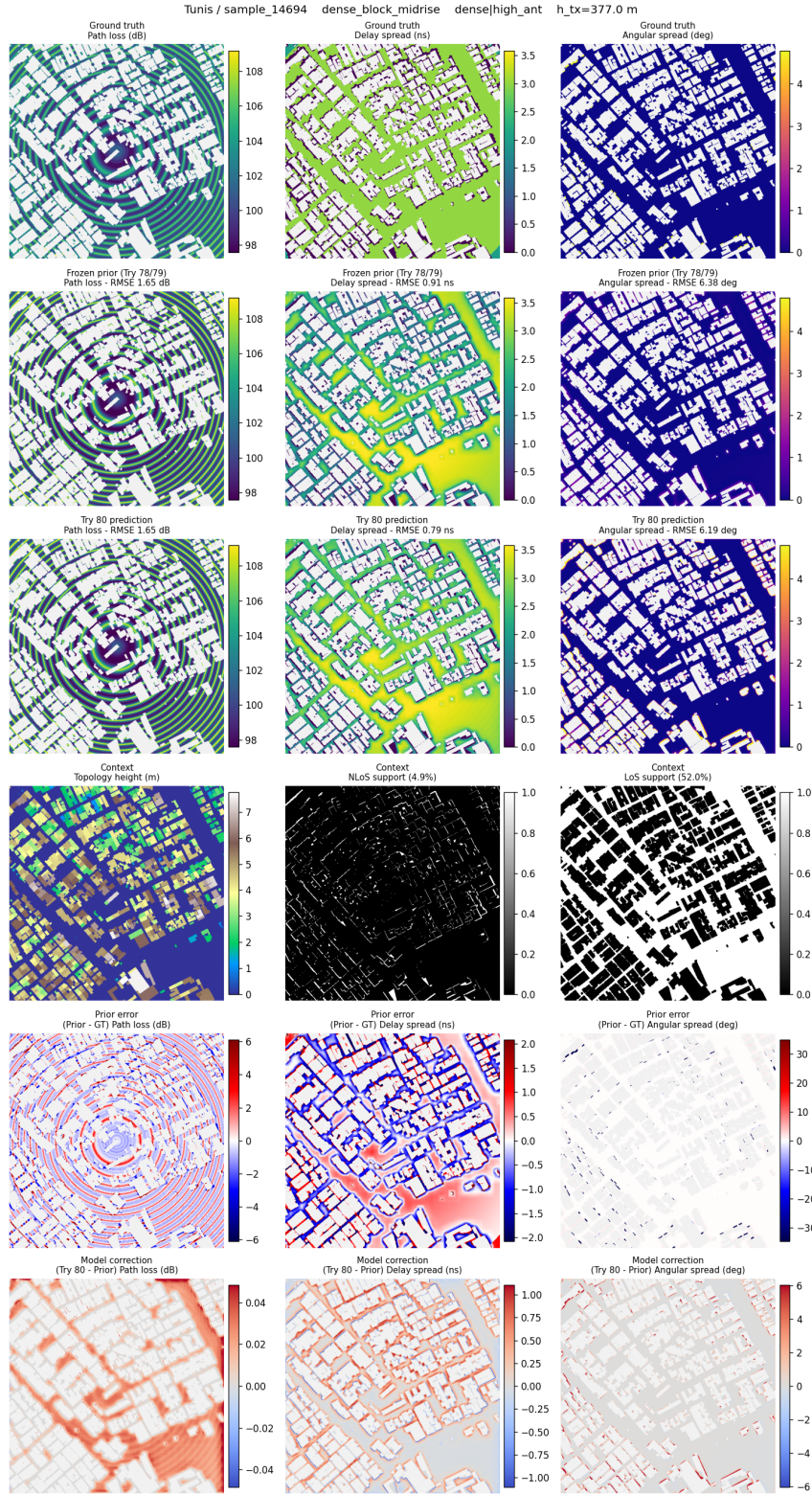


Figure B.3: HARP-NET CKM 6×3 inspection panel for a dense, high-antenna validation sample in Tunis.

CKM Generator Interface and Reproducibility Tool

This appendix documents the standalone CKM Generator delivered with the final HARP-NET CKM pipeline. The tool was built to make the trained model usable outside the training scripts: it accepts topology inputs, prepares them with the same geometric conventions used in the thesis, computes or imports LoS/NLoS masks, evaluates the frozen calibrated priors, optionally runs the final neural model, and exports the numerical arrays used for reproducible analysis. It can also export visual figures when those optional inspection switches are enabled.

The generator is intentionally available through two front ends. The Streamlit application is designed for inspection, rapid manual testing, and demonstrations with uploaded files. The command-line interface uses the same backend and is intended for reproducible scripted runs, validation, and larger batches.

C.1 Public source record and checkpoint artifact

The reproducibility record is split across public repositories. The standalone generator is maintained in CKMGenerator; the final training and evaluation code is collected in Final_Code_TFG; the broader experimental history is preserved in TFGAllProgress_Tries_and_Attempts; and the thesis source is kept in TFG_ActualText_LaTeX [56, 57, 54, 58].

The final checkpoint loaded by the generator is `models/best_model.pt`. Since this file is larger than GitHub’s regular single-file limit for ordinary Git history, it is treated as a large model artifact rather than normal source code. The generator therefore accepts either the local default checkpoint path or an explicit `–checkpoint` path, while the public source repository remains small enough to clone and inspect normally.

In the public repository, this checkpoint is distributed through Git LFS. After cloning, users should run `git lfs pull` if `models/best_model.pt` has not been materialized automatically. The published checkpoint is 322,553,518 bytes, with SHA256:

```
0C245DE7D7D090D12563D17DC5C3D2E1DE9ED96C32DEBC8461E15CAC7A7C9E25
```

C.2 Interface overview

Figure C.1 shows the main controls exposed by the Streamlit interface. The left sidebar controls the input source, antenna height, runtime device, mask policy, model execution, export format, image scaling, pixel spacing, HDF5 filtering, and output folder.

After generation, the application presents the fitted topology next to the LoS and NLoS masks actually used by the priors and model. This is important because the masks are not only visual outputs: they are part of the final model input tensor and determine the receiver-valid support used by the priors. Figure C.2 shows this inspection view.

When the final model is enabled, the generator produces the three target maps in their native units. Figure C.3 shows the joint prediction panel saved by the tool.

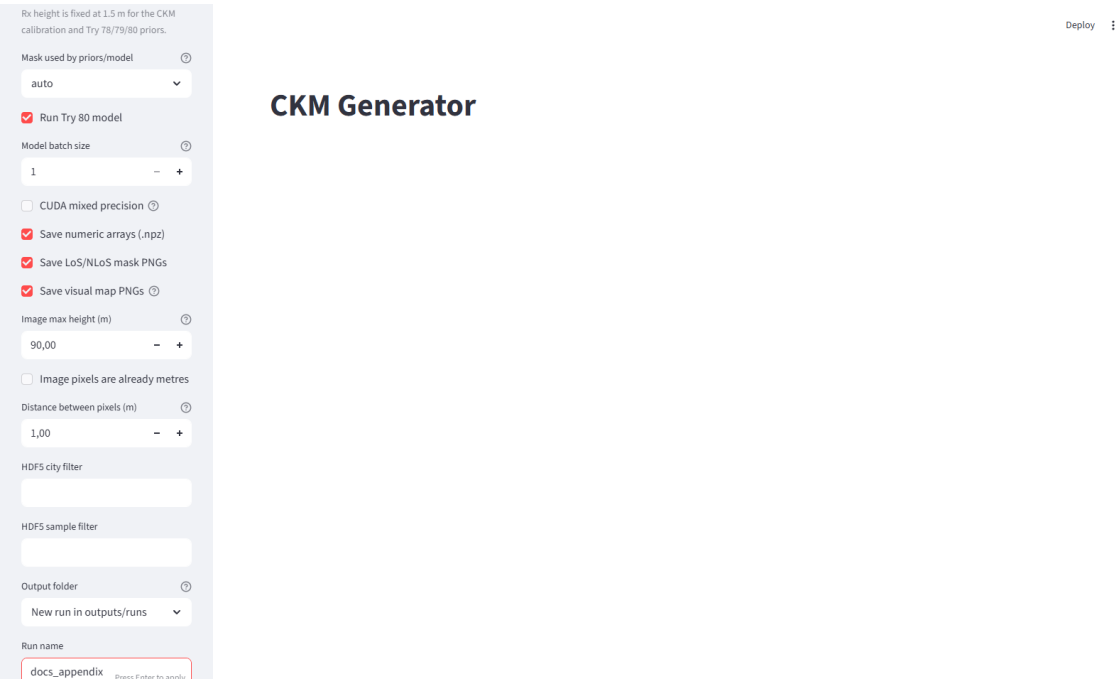


Figure C.1: CKM Generator Streamlit interface. The sidebar exposes the same settings that are available through the command-line interface.

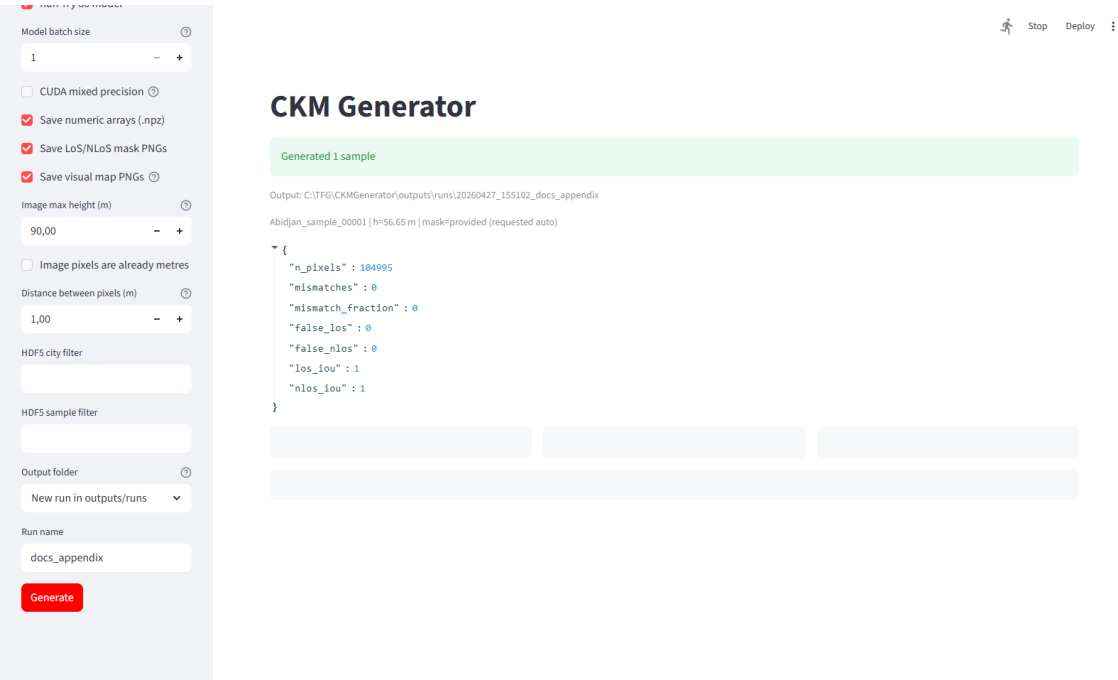


Figure C.2: Generated topology and masks for an HDF5 sample. The view reports the resolved mask source and displays topology, LoS mask, and NLoS mask side by side.

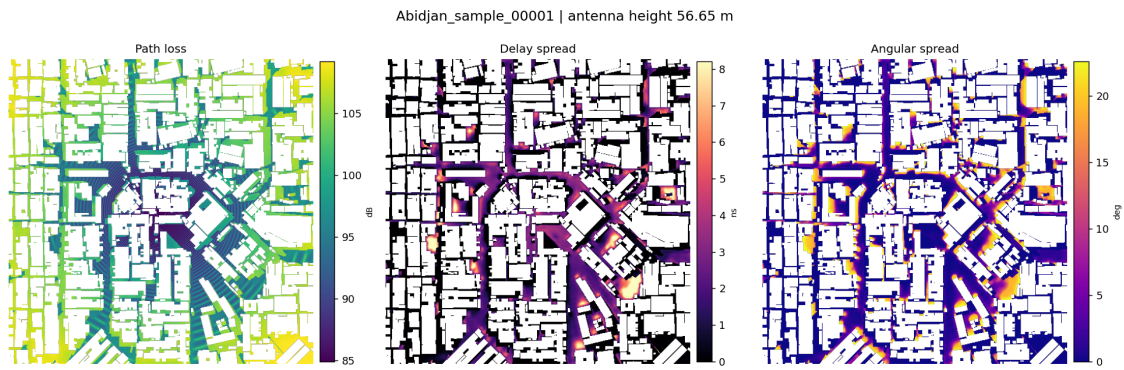


Figure C.3: Joint prediction figure exported by the CKM Generator, with path loss in dB, delay spread in ns, and angular spread in degrees.

C.3 How the tool is run

The dependencies are installed in the Python environment that will run the tool:

```
cd C:\TFG\CKMGenerator
C:\Python310\python.exe -m pip install -r C:\TFG\CKMGenerator\requirements.txt
```

The runtime check is executed before inference:

```
C:\Python310\python.exe C:\TFG\CKMGenerator\ckm_doctor.py
```

The doctor command checks PyTorch, CUDA, DirectML, the checkpoint, available memory, and whether the selected runtime can execute the final model. In auto mode, the runtime priority is CUDA, then DirectML, then CPU.

The Streamlit application is started with:

```
C:\Python310\python.exe -m streamlit run C:\TFG\CKMGenerator\ckm_app.py
```

The equivalent terminal command for an HDF5 sample is:

```
C:\Python310\python.exe C:\TFG\CKMGenerator\ckm_generate.py ^
--input C:\TFG\CKMGenerator\inputs\samples\Abidjan_sample_00001.h5 ^
--run-name abidjan_check ^
--mixed-precision
```

For an image-only topology, the transmitter height must be supplied explicitly unless it is stored in a structured input file:

```
C:\Python310\python.exe C:\TFG\CKMGenerator\ckm_generate.py ^
--input C:\path\topology_map.png ^
--height 56.65 ^
--topology-max-m 120 ^
--meters-per-pixel 1.0 ^
--mask-source auto ^
--mixed-precision
```

C.4 Accepted inputs

The generator accepts three families of input files:

- Raster topology images: .png, .jpg, .jpeg, .tif, .tiff, and .bmp.
- Numeric arrays: .npy, .npz, .csv, .txt, and .json.
- HDF5 files: .h5, .hdf5, and .hdf.

For HDF5 and NPZ inputs, topology arrays are detected using the keys `topology_map`, `topology`, `building_height`, `building_heights`, `height_map`, or `terrain`. The optional LoS and NLoS masks can be read from `los_mask`, `LoS_mask`, `los`, `LoS`, `nlos_mask`, `nLoS_mask`, `NLoS_mask`,

nlos, or NLoS. The optional antenna height can be read from `uav_height`, `antenna_height`, `antenna_height_m`, `height`, `height_m`, or `h_tx`.

For image or array topology files, sidcar masks are detected when they share the same base name:

```
sample_topology.png
sample_los_mask.png
sample_nlos_mask.png
```

The exported manual-input folder also uses a sibling `masks_png` directory, which is checked automatically. The valid transmitter-height range is 12 m to 478 m. Receiver height is fixed at 1.5 m, as in the CKM calibration and in the calibrated-prior formulas.

C.5 Configurable controls

The Streamlit interface and command-line interface expose the same operational choices: input path or upload, antenna height, runtime device, mask source, whether to run the neural residual model, batch size, mixed precision, numeric array export, optional PNG exports, topology scaling, pixel spacing, HDF5 filters and output folder. The thesis keeps this as a reproducibility summary; the public generator repository contains the full option help.

C.6 Image scaling and raw numeric data

The `Image max height (m)` setting applies only to raster images. It exists because PNG, JPG, TIFF and BMP files normally store integer intensities rather than physical building heights. When image pixels are not already metres, the conversion is

$$h_m(x) = \frac{I(x)}{I_{\max}} h_{\max}, \quad (\text{C.1})$$

where $I_{\max} = 255$ for 8-bit images and $I_{\max} = 65535$ for 16-bit images. The value $h_{\max}$ is the selected image maximum height in metres.

If `Image pixels are already metres` is enabled, this normalisation is skipped and the image pixel values are used directly as metre values.

The maximum-height setting is ignored for HDF5, NPZ, NPY, CSV, TXT and JSON inputs. These formats already contain numeric values, so rescaling them with an image-intensity maximum would destroy the physical values. For raw numeric inputs, the loaded values are treated directly as topology heights.

C.7 Model-grid fitting and interpolation

The final model was trained and calibrated on a fixed 513×513 grid with 1 m pixels. Every input is therefore converted to that representation before priors or neural inference are computed.

The `Distance between pixels (m)` or `-meters-per-pixel` value describes the physical spacing of the input grid. The loader first converts the input to a 1 m/pixel grid:

$$H_{\text{out}} = \text{round}(H_{\text{in}} s), \quad W_{\text{out}} = \text{round}(W_{\text{in}} s), \quad (\text{C.2})$$

where s is the input spacing in metres per pixel. A value of $s = 1$ leaves the physical sampling unchanged, $s < 1$ downsamples a finer grid, and $s > 1$ upsamples a coarser grid.

Topology arrays are resampled with bilinear interpolation. Mask arrays are resampled with nearest-neighbour interpolation and then binarised again. This prevents LoS/NLoS masks from becoming fractional probability maps.

After resampling, the array is fitted to the 513×513 model grid:

- If it is larger than 513×513 , the centred 513×513 crop is used and the outer area is ignored.
- If it is smaller than 513×513 , it is centred and padded.
- Topology padding uses a 1 m building height. The outside area is therefore non-ground and receiver-invalid, so the model does not predict there.
- Mask padding uses 0, meaning no LoS or NLoS receiver support outside the supplied data.

The receiver-valid ground mask is derived after this fitting step:

$$m_{\text{ground}}(x) = \mathbf{1}\{h_{\text{topology}}(x) \leq 10^{-6}\}. \quad (\text{C.3})$$

Only zero-height pixels are treated as receiver locations. Building pixels and padded outside pixels are excluded from both LoS and NLoS support.

C.8 Mask policies

The model consumes one LoS mask and one NLoS mask as input channels. They are also used by the physical priors. The available policies are:

- **auto**: use a provided mask when present; otherwise generate a ray-cast mask from topology and antenna height.
- **provided**: require a provided mask and fail if none exists.
- **generated**: ignore provided masks and generate masks from topology and antenna height.

The original CKM HDF5 data normally stores the LoS mask. The NLoS mask is derived rather than read as an independent ground truth label:

$$m_{\text{LoS}} = m_{\text{LoS,raw}} m_{\text{ground}}, \quad m_{\text{NLoS}} = (1 - m_{\text{LoS,raw}}) m_{\text{ground}}. \quad (\text{C.4})$$

The multiplication by m_{ground} is essential; otherwise building pixels would be labelled as NLoS receiver pixels.

For topology-only inputs, generated LoS/NLoS masks are an approximate geometric fallback. They allow the final model to run outside the original CKM HDF5 dataset, but they are not expected to be bit-identical to CKM reference masks.

C.9 Outputs

Each run writes to a timestamped folder:

`outputs/runs/<timestamp>_<run-name>/`

or to the explicit output folder selected in the app or passed with `-out`. File names include the sample id and antenna height:

`<sample>_h<height>m`

The output tree can contain:

```
masks/
*_los_mask.png
*_nlos_mask.png
*_generated_los_mask.png
*_generated_nlos_mask.png
priors/
*_prior_path_loss.png
*_prior_path_loss_los.png
*_prior_path_loss_nlos.png
```



```

*_prior_delay_spread.png
*_prior_angular_spread.png
predictions/
*_pred_path_loss.png
*_pred_delay_spread.png
*_pred_angular_spread.png
*_pred_joint.png
arrays/
*.npz
metadata/
*.json

```

Figure C.4 shows the concrete artifacts produced by the example HDF5 run used for the appendix screenshots. The tree view is an expanded visual output example: the current fast default writes the numerical .npz archive and metadata, while PNG exports are optional.

Generated CKM output folder

A complete run stores visual PNGs, numeric arrays, and JSON metadata.

```

outputs/runs/20260427_155102_docs_appendix/
arrays/
  Abidjan_sample_00001_h56p65m.npz          3882.7 KB
metadata/
  Abidjan_sample_00001_h56p65m.json          2.8 KB
masks/
  Abidjan_sample_00001_h56p65m_generated_los_mask.png    7.9 KB
  Abidjan_sample_00001_h56p65m_generated_nlos_mask.png   12.3 KB
  Abidjan_sample_00001_h56p65m_los_mask.png              7.9 KB
  Abidjan_sample_00001_h56p65m_nlos_mask.png             12.3 KB
priors/
  Abidjan_sample_00001_h56p65m_prior_angular_spread.png 214.6 KB
  Abidjan_sample_00001_h56p65m_prior_delay_spread.png   199.4 KB
  Abidjan_sample_00001_h56p65m_prior_path_loss.png      269.9 KB
  Abidjan_sample_00001_h56p65m_prior_path_loss_los.png  322.5 KB
  Abidjan_sample_00001_h56p65m_prior_path_loss_nlos.png 195.7 KB
predictions/
  Abidjan_sample_00001_h56p65m_pred_angular_spread.png  221.3 KB
  Abidjan_sample_00001_h56p65m_pred_delay_spread.png    229.1 KB
  Abidjan_sample_00001_h56p65m_pred_joint.png           689.4 KB
  Abidjan_sample_00001_h56p65m_pred_path_loss.png       276.5 KB

```

Figure C.4: Generated output folder for one CKM Generator run. The run contains binary masks, prior maps, prediction maps, a numerical archive, and a JSON metadata file; the PNG folders are optional visual exports.

PNG files are visual inspection products. The authoritative numerical output is the NumPy archive in arrays/. It is uncompressed by default for faster batch export and compressed only when `Compress numeric arrays` or `-compress-arrays` is enabled. It stores all arrays on the final 513×513 grid. Important keys include: `topology`, `ground_mask`, `los_mask`, `nlos_mask`, `generated_los_mask`, `generated_nlos_mask`, `reference_los_mask` when available, `prior_path_loss`, `prior_path_loss_los`, `prior_path_loss_nlos`, `prior_delay_spread`, `prior_angular_spread`, `pred_path_loss`, `pred_delay_spread`, and `pred_angular_spread`.

The JSON metadata stores provenance and bookkeeping: sample id, antenna height, source file, checkpoint path, requested and resolved mask source, topology class, antenna bin, mask comparison statistics when a reference mask exists, the selected save options, save timings, and the paths of all generated files.

C.10 Performance controls

CUDA is the preferred runtime when available. The app and CLI can also use CPU or DirectML. The model batch size can increase throughput on GPUs with enough VRAM, while keeping batch size 1 remains the safest option on smaller cards.

The benchmarked fast path is the same one reported in Chapter 4: vectorized LoS/NLoS mask generation, torch prior evaluation, mixed-precision neural inference, and default numerical .npz export after the checkpoint has already been loaded. The appendix keeps only the operational interpretation: mask generation and prior evaluation are implementation optimizations, not changes in the mathematical definition of the generator. The vectorized mask backend was checked against the original GT ray-caster with zero LoS/NLoS differences on the Barcelona benchmark and on 100 exported CKM samples, while the torch prior backend changed RMSE only in the fourth decimal place relative to the NumPy prior implementation.

The mixed-precision numerical difference remains negligible. On the full Barcelona subset, the RMSE between mixed precision and fp32 predictions was 0.0014 dB for path loss, 0.0028 ns for delay spread, and 0.0025° for angular spread, which is far below the validation RMSE scale of the model. Its benefit is speed: with the torch prior backend and provided masks, the same 50-sample validation path took 0.72 s per map in fp32 and 0.62 s per map with CUDA mixed precision.

For batch export, keeping `Save visual map PNGs`, `Save LoS/NLoS mask PNGs`, and `Compress numeric arrays` disabled avoids Matplotlib rendering and compression overhead while preserving the masks, priors, predictions, and metadata in the numerical archive. Disabling `Run final residual model` is useful when the goal is only to validate topologies, masks, and prior generation.

Declaration on Generative AI Use

Generative AI tools were used during the preparation and software development of this thesis. The tools used were OpenAI GPT based assistants and OpenAI Codex. They were used as assistant tools for writing support, literature and terminology checks, LaTeX editing, consistency audits, Python code generation, debugging, and refactoring.

In the software work, these tools were useful for translating author defined ideas into executable code. For example, they helped convert prototype logic into scripts, port calibrated priors from NumPy prototypes to PyTorch implementations, wire evaluation utilities, inspect errors, and keep thesis and paper sources consistent. They also supported drafting and revising text, but all references and claims were checked by the author.

The generative AI tools did not originate the scientific direction, research questions, model ideas, experiment choices, final interpretation, reported numbers, plots, or conclusions of the project. The project design, validation protocol, final source selection and responsibility for the submitted document remain with the author. Numerical results were obtained from the project scripts, stored artifacts and compiled thesis sources, not from generative model outputs.

The CKM dataset itself was processed through local scripts and stored experiment outputs rather than pasted into prompts. Code changes and final text were reviewed by the author before inclusion.